Part II

Monetary Policy and Economy

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Chapter 3

The Role of Money and Monetary Policy in the Economy

3.1. Introduction

Precisely how monetary policy plays a role in the economy has always been the subject of public debate. Some support a monetary policy focus on price stability. Indonesia's bitter experience during the 1960s showed that using monetary policy to finance lighthouse projects created hyperinflation, which culminated in an economic crisis and the failure of the economic development program during the Sukarno era. Meanwhile, another section of the public believes that monetary policy should play a role in stimulating output growth. Some hold this view due to political interests, while many others consider that demand for monetary stimuli during an economic downswing or even recession should support the fiscal stimuli of the Government. On a practical level at central banks, the core of the debate has become the essence of how monetary policy should offset the trade-off between price stability, in the form of low and stable inflation, and fostering economic growth.

The issue concerning the role of money and monetary policy in the economy has been the focus of studies for as long as monetary theory has been developed. The most fundamental debate concerns the neutrality of money in the economy, meaning that money only affects prices but is neutral in terms of real output.[1] The crux of the debate begins with the proposition of Keynes (1936) in The General Theory, which aimed to attack classical thinking, namely Say's law – "supply

[1]Another proposal is the Classical Dichotomy, which separates the treatment of real economic theory on relative prices from monetary theory on prices. Therefore, in the analysis of factors influencing prices, the role of theory on relative prices can be debated (Johnson, 1967). Starting with the classical argument of economic flows, Lange (1942), that Say's law – "supply creates its own demand" – logically breaks all the arguments in monetary theory because of the combination with Walras' law – "the total supply of goods and money to the market must be equal to the total demand of goods and money" – which implies that excess demand of money is zero and realized in the absolute price (intermediate).

Central Bank Policy: Theory and Practice, 49–79

doi:10.1108/978-1-78973-751-620191006

creates its own demand." Departing from the classical view of economic equilibrium (full employment) and, therefore, money only affects prices and is neutral in its effect on real output, the proposition of Keynes is based on the assumption that nominal wages are rigid, so if there is a change in the stock of money, real wages would also change and influence real output. In other words, the proposition stresses that monetary economics will reach equilibrium through several possibilities, where full employment is just one extreme condition. Such a concrete proposition has become the overarching focus of subsequent debates between Keynesian and Classical economists, and later with Monetarists, Neoclassical, and Neo-Keynesian economists.

Consensus from the protracted debate between the Classical and Keynesian views concluded a synthesis that money only affects prices and is neutral in its effect on real output in the long run but can influence real output in the near term. Nonetheless, one core aspect – whether money is important or not – has not yet been satisfactorily addressed. In the evolution of monetary theory to the present day, opposing economists have developed approaches (economic models) assuming certain theories, which ultimately conclude the neutrality of money in the economy, or vice versa. Metzler (1951) as well as Gurley and Shaw (1960), for instance, showed that money affects real interest rates and output through government debt and open market operations by the central bank. Departing from the Classical and Keynesian economic views, Stiglitz & Greenwald (2003) developed a new paradigm of monetary theory based on the role of the stock of credit, not money, in an imperfect financial market due to asymmetric information between the banks and borrowers. According to such conditions, bank lending, in terms of quantity or interest rates, affects monetary aggregates and the real economy.

The five sections of this chapter discuss the role of monetary policy in a closed economy. Following the introduction, the differences between theoretical views concerning the neutrality of money are reviewed in terms of general equilibrium, imperfect information and imperfect markets as well as the view of monetarists. The subsequent section presents the empirical evidence concerning the role of money and monetary policy in the economy, including the results of empirical studies in Indonesia. In the penultimate section, as a new theoretical paradigm, the views of Joseph E. Stiglitz are presented, wrapped up with some concluding remarks.

3.2. Theoretical Review

Two issues relating to the structure of Classical theory, which treats relative prices as a factor affected by real demand and real supply, and the price of money as a factor dependent upon the stock of money, were the primary focus of the work of Don Patinkin in the 1950s (Johnson, 1967). In this case, the neutrality of money relates to conditions where an increase in the stock of money would raise all prices by the same magnitude, thus the relative price does not depend on the stock of money. If the effect of money is neutral, an increase in the stock of money would edge up prices without changing the relative price and interest rate and, therefore, output. In the terminology of Pigou, money is merely a veil of real activity underlying the workings of a system.

Patinkin's argument on the neutrality of money, formulated as a short-term macroeconomic system with a structure à la Keynesian but based on the assumption of classical behavior, produced a classical paradigm that changes in the stock of money would not change real equilibrium in the economic system, as reflected by changes in the relative price and interest rate. The firm conclusion linked to the argument built on the assumptions required to realize money neutrality, namely wage and price flexibility, inelastic expectations, no money illusion, no distribution effect, homogenous securities, and no government debt or open market operations.

The assumptions became a target of criticism from Metzler (1951) as well as Gurley and Shaw (1960), who described conditions where the effect of money was not neutral. Through government debt or open market operations by the central bank, Metzler argued that the relationship between worth and savings used in the Pigou effect analysis by Keynes had theoretical implications where changes in the stock of money could affect the interest rate and, therefore, output growth. Assuming that the real value of government bonds is fixed and the interest on government debt is paid as public income, Metzler showed that rising prices due to monetary expansion through open market operations caused the public stock of real assets to decrease and the propensity to save to increase, thereby lowering the equilibrium interest rate. A lower interest rate due to the distribution effect, assuming that private sector performance is affected by government real debt performance, would stimulate output growth.

The seminal contribution of Gurley and Shaw (1960) to the debate on the neutrality of money was not only based on the assumptions of rigidity, the distribution effect, the money illusion, and expectations, but also based on analyses of structural and financial sector intermediation linkages with economic growth and monetary policy, as well as the differing liquidity characteristics of various assets. They differentiated two important concepts of money, namely inside money created by the monetary authority and banking industry in relation to private sector lending as well as outside money, in the form of currency outside of the private sector and not related to loans/claims, for instance banknotes and coins. Money supply that consists of inside and outside money implies that changes in the stock of money not only raise or lower general prices but also the relative price and, therefore, output. This is because if the public holds both types of money, then a change in the stock of one type without a corresponding change in the other would trigger a change in the assets and liabilities ratio, which would, in turn, change real variables in the economic system.

In its evolution, the debate on money neutrality has splintered into two concepts, namely neutrality and superneutrality. In terms of general equilibrium analysis, the neutrality of money occurs if (Handa, 2000): (1) all prices rise proportionally; (2) the real value of the endowment does not change; (3) interest rates are paid on all components of money; and (4) no further price changes are anticipated. In this case, money is considered neutral if a once-for-all change in money supply does not affect the values of real economic variables, including output. Meanwhile, superneutrality occurs if a continuous change in money supply (a change in money supply growth) has no real effect. In terms of general equilibrium analysis, the superneutrality of money, in addition to the influence of the first three factors of neutrality, is also affected by errors in the formation of inflation expectations.

In this regard, the theoretical arguments generally agree that the neutrality of money may not occur, at least in the near term. In this case, money supply is non-neutral if economic imbalances are present. In fact, if there is an increase in the once-for-all of money supply, the non-neutrality of money on real variables can occur during an adjustment phase from one balance to another in the economic system. In this case, an increase in total money supply would not necessarily precipitate a proportional increase in the absolute prices of goods and nominal wages, thus the real price of goods and real endowment value would change and trigger a real change in the economy.

In other words, increases in total money supply and inflation have a real effect, at least in the near term, when rigidity and imbalances occur in the economy. In economic theory, a debate emerged on whether the imbalances are merely temporary, followed by quick return to equilibrium, such that the consequences can be ignored. In general, Neoclassical and Modern Classical economists assume that economic dynamics quickly return to balance. The analysis has only focused on equilibrium conditions and, therefore, money is neutral on real economic variables. In contrast, Keynesian and Neo-Keynesian economists generally believe that protracted economic imbalances can persist, therefore money could affect real variables in the economy.

3.2.1. The Role of Money in the General Equilibrium Analysis

Money plays a role in the economy in line with its primary functions, namely as a medium of exchange, as a store of value, and as a unit of account. How money affects real decisions and dynamic balance in an economy remains a topic of theoretical and empirical studies. In this respect, a general approach was developed to explain how money affects real decisions and dynamic balance in an economy by linking the presence of money based on certain assumptions in a general equilibrium analysis of an economic system. One important implication when evaluating different approaches to monetary economic theory is to observe how certain assumptions underlie the economic models developed, or on ways to introduce the role of money into an economic model.

General equilibrium analysis in a closed economy developed by classical economists is based on equilibrium analysis in the real sector and on the money market. The formula is recursive, namely that balance in the real sector will affect money market equilibrium but not vice versa. Analysis of the real sector consists of two core equations, namely aggregate supply that reflects real output on the production side as a function of capital and human capital, and aggregate demand that explains real output on the expenditure side as a function of consumption, investment and government spending. Balance in the real sector will determine real output, real wages, labor absorption, capital formation, consumption, and real investment. Meanwhile, analysis of the money market consists of balancing the supply of money by the central bank and demand as a function of real interest rates and real output. With the recursive formula, equilibrium analysis of the two economic sectors has shown that real output will determine the real interest rate. In other words, total money supply will only affect prices and remain neutral in terms of real output and other real economic variables.

According to Keynesian economists, however, general equilibrium theory assumes nominal wage rigidity in the economic system. The assumption was born out of observations during economic recessions after the industrial revolution in Europe at the beginning of the nineteenth century, which demonstrated that labor contract negotiations did not always produce the same level of real wages per the view of classical economists. Therefore, with real wages affected by prices, the equation structure in the economic system becomes simultaneous. Equilibrium analysis in the real sector and money market will simultaneously determine not only the total money supply but also prices and, thereafter, labor absorption, real output, real interest rate, capital formation, as well as real consumption and investment. In other words, nominal wage rigidity implies that the effect of money is not neutral in the economy, meaning that total money supply not only affects prices, but also real output and other real economic variables.

General equilibrium analysis concerning the neutrality of money in an economy will depend on the assumptions built into the economic system, including: wage and price flexibility, elasticity of expectations, the money illusion, distribution effect, homogenous securities as well as government debt or open market operations. Of the various approaches available, two are the focus of attention in this section, namely the overlapping generations (OLG) model based on the distribution effect of money between residents, and the money-in-the-utility (MIU) function that can demonstrate whether the money illusion is present in terms of public proclivity to hold currency in the analysis of money neutrality in an economy. Meanwhile, money neutrality analysis based on the effect of information availability on expectations as well as market imperfections and price rigidity will be discussed in the subsequent sections.

(a) OLG Model: "Non-superneutrality of Money"

A general equilibrium analysis approach that shows the non-neutrality of money based on the distribution effect in an economy is the OLG model. Built based on the consumption-loan model of Samuelson (1958), the OLG model structure appears simple: namely that time is discrete, individuals live in two periods, N_t individuals are born in period t, and the population grows at a rate, n, which can be normalized to $N_t = (1+n)^t$. Individuals born in period t are labeled young, and in period $t+1$ they become old. Each individual is equipped with one unit (consumable) in their youth but not in old age. Individuals have access to storage technology, thus each unit stored in period t will produce $1+r$ units in period $t+1$. If productive units are stored, then $r>0$, but if the stored unit is broken, then $r=-1$. The utility function of the individual born in period t is $u(c_{1t}, c_{2t+1})$.

For instance, if, during period zero, the government gives money totaling H to the old individual and the money can be exchanged for goods in period t with a price P_t, then the maximization problem for the individual born in period t, $t \geq 0$ is

$$Max\ \ u\left(c_{1t}, c_{2t+1}\right) \tag{1}$$

Subject to: $P_t\ (1 - c_{1t}) = M^d_t$ and $P_{t+1}c_{2t+1} = M^d_{t,}$ where M^d_t is individual demand for money.

Considering the lack of natural uncertainty in the model, the condition of perfect foresight is assumed, in cases where the expected and actual prices in period t+1 are the same. The first-order condition for maximization is

$$\frac{M_t^d}{P_t} = L\left(\frac{P_t}{P_{t+1}}\right) \tag{2}$$

which is a function of demand for money and simultaneously a function of savings. The rate at which money is acquired is therefore, P_t/P_{t+1}, or equivalent to $1 + g_t$, where g_t is the rate of deflation.

Per general equilibrium, Walras' law, which equates demand of the young individual based on the function of demand for money (2) with the inelastic supply of the old individual based on the total money saved, H, is as follows:

$$(1-n)^t\ M_t^d = H \tag{3}$$

Using (3) and (2) in the periods t and t+1 produces

$$\left(1+g_t\right)^{-1}\left(1+n\right) = \frac{L\left(1+g_t\right)}{L\left(1+g_{t+1}\right)} \tag{4}$$

In a steady state, g is constant so, according to Equation (4), $g = n$. Deflation must be the same as the rate of population growth, implying that prices experience a change at a rate where the supply of money expands at the same pace as demand for money, namely equivalent to population growth.

The previous analysis requires constant nominal money stock growth. If nominal money stock grows at a rate, μ, then inflation will be $\mu - n$ in a steady state. The influence of money growth on the allocation of resources depends on how the additional money is calculated in the economy. In the case where the additional money is transferred in a lump sum to the old individual (consumables), the maximization problem must be modified in the constraint function of Equation (1) as follows:

$$P_t\left(1-c_{1t}\right) = M_t^d \quad \text{and} \qquad P_{t+1}c_{2t+1} = \Delta M_t + M_t^d \tag{5}$$

Therefore, considering that ΔM_t is not affected by individual behavior, the solution for the first-order condition is

$$\frac{M_t^d}{P_t} = L\left(\frac{P_t}{P_{t+1}}, \frac{\Delta M_t}{P_{t+1}}\right) \tag{6}$$

Demand for money is a function of the rate at which money can be accumulated and transferred in the second period, which is exogenous.

Meanwhile, the nominal stock of money grows, as an aggregate, at a rate, μ, thus:

$$H_{t+1} = H_t + \Delta H_t = (1+\mu) H_t \tag{7}$$

Additional new money will be distributed equally to the old individual, $\Delta H_t = N_t \Delta M_t$, therefore, equilibrium on the money market and goods market in period t requires that:

$$N_t M_t = H_t \tag{8}$$

In a steady state, the real cash balance per capita is constant, thus $H_t / P_t N_t$ is also constant, implying that deflation must occur at a rate, g, where $1 + g = (1 + n)(1 + \mu)$, where small values of n and μ are equivalent to $n - \mu$.

Considering the impact on inflation and, therefore, the rate at which money can be accumulated, money growth also influences the allocation of resources in the economy. Therefore, money is not "super-neutral" (non-superneutrality of money). Differing from the concept of money neutrality, which emphasizes the neutral impact of money on changes in the levels of real variables, the super-neutral concept stresses the influence in the form of changes in growth. In other words, through the distribution effect of money between residents (in this case, between the younger and older generations), the OLG model shows that money supply growth will influence changes in prices (inflation), real output, and other real economic variables.

(b) MIU Function Model: Superneutrality of Money

General equilibrium analysis, by including the public preference to hold currency in the MIU model, demonstrates the neutral effect of money in the economy. The role of money was first introduced to the model based on the optimal behavior of economic agents by Sidrauski (1967), namely by assuming the direct presence of money in the utility function of each respective economic agent. This is based on the idea that introducing the role of money in the form of a demand function for money in the economy is inadequate, thus requiring that economic agents be shown to prefer holding currency. By including the preference to hold currency as a viable commodity in the utility function of economic agents (consumers), the MIU model can prove whether money influences the decisions of producers when determining the use of production factors (capital) and real output. This approach represents an advancement on the Neoclassical model (the Ramsey non-monetary model) and is recognized in the MIU analysis framework.

In the model of Sidrauski, each respective individual lives indefinitely, while the population, N, grows at a rate, n. The maximization problem is therefore expressed as follows:

$$Max \;\; V_s = \int_s^{\infty} u(c_t, m_t)\exp[-\theta(t-s)]d_t, \qquad u_c, u_m > 0, u_{cc}, u_{mm} > 0 \tag{9}$$

where c and m are consumption (C) and real cash balance per capita. An individual can hold wealth in the form of cash (M) or capital (K). The budget constraint function is as follows:

$$C + \frac{dK}{dt} + \frac{dM \,/\, dt}{P} = wN + rK + \chi \tag{10}$$

where P is the price, w and r are the level of wages and real interest rate, respectively. X is the transfer from the government. With total household net worth defined as $A = K + M/P$ and transformation of the constraint function in the form of per capita (divided by N), the following equation is produced:

$$\frac{da}{dt} = [(r-n)a + w + x] - [c + (\pi - r)m] \tag{11}$$

where a is total net worth per capita and π is inflation. The equation shows that the magnitude of change in net worth per capita is the difference between income and consumption.

Meanwhile, the non-Ponzi-game conditions are:

$$\lim_{t\to\infty} a_t \exp-\left[\int_0^t (r_v - n)\,dv = 0\right] \tag{12}$$

The design problem above requires an optimal control solution, where λ_t, as a co-state variable, is linked to constraint function above. The Hamiltonian function can be expressed as follows:

$$H = \{u(c,m) + \lambda[(r-n)a + w + x - c - (\pi + r)m]\}\exp(-\theta t). \tag{13}$$

The first-order condition are:

$$\begin{aligned} &u_c(c,m) = \lambda \\ &u_m(c,m) = \lambda(\pi + r) \\ &\frac{d\lambda}{dt} - \theta\lambda = -(r-n)\lambda \\ &\lim_{t\to\infty} a_t \lambda_t \exp(-\theta t) = 0 \end{aligned} \tag{14}$$

The two first-order conditions imply that the marginal rate of substitution between consumption and real cash balance is the same as the nominal interest rate.

Steady-state analysis involves additional assumptions, per the decentralized Ramsey model, concerning conditions where corporate technology is a constant return to scale, market production factors are competitive, and there is the same lump-sum transfer with seigniorage from issuing currency, namely:[2]

$$\begin{aligned} r &= f'(k) \\ w &= f(k) - kf'(k) \\ x &= \frac{dM/dt}{PN} = \left(\frac{dM}{M}\right)\left(\frac{M}{PN}\right) = \mu m \end{aligned} \tag{15}$$

where μ is the rate of money growth.

Per steady state conditions, $da/dt = dm/dt = d\lambda/dt = 0$, where $dm/dt = 0$, implying that $\pi = \mu - n$.

Steady state conditions for capital stock, consumption, and real cash balance are as follows:

$$\begin{aligned} f'(k^*) &= \theta + n \\ c^* &= f(k^*) - nk^* \\ u_m(c^*, m^*) &= (\theta + \mu)u_c(c^*, m^*) \end{aligned} \tag{16}$$

The salient implication of such conditions is that money growth does not affect the real interest rate or wages as well as capital growth and real consumption, thus, per the steady state, money is superneutral (superneutrality of money). This differs from the common perception that by including the preference of economic agents to hold money, money will influence real output and other real economic variables. The previous analysis showed that money supply growth only affects inflation, in addition to the marginal utility of holding money and the real value of government transfers.

3.2.2. Relationship between Money—Output and Imperfect Information

To explain the empirical factors of money neutrality to real output, Neoclassical economists have changed strategy through a different modeling approach and by expanding the assumptions to include imperfect information. This was first achieved by Lucas in 1973. In this case, assuming rational economic behavior in terms of optimizing decisionmaking, or rational expectations, Lucas showed how an equilibrium model using a decentralized market and imperfect information could explain the impact of nominal shocks on output.

In the Lucas Island model, the economy consists of separate competitive markets spread over various islands. The shocks are distinguished into two categories, namely aggregate shocks that affect nominal money and idiosyncratic shocks that

[2]Fundamentally, understanding seigniorage refers to state receipts, including government seigniorage and central bank seigniorage, which stem from issuing currency.

have a specific impact at the sectoral level. When determining the decisions of market participants, namely suppliers (producers-workers), in response to special shocks, considering the lack of sufficient information to differentiate, the respective responses of each market player will be different. Therefore the impact of the shocks, including nominal shocks, will influence output.

For instance, the supply function for each respective market $i = 1, \ldots, n$ is as follows:

$$y_t^i = b(p_t^i - E[p_t \mid I_t^i]) \tag{17}$$

where y^i_t and p^i_t are, respectively, the (log) output and output price on market i in period t. I^i_t is the information set available on market i during period t, which contains complete information on economic conditions, including specific prices, P^i_t. The supply function implies that an increase of supply would depend on the perception of the supplier in terms of the relative prices of the goods produced to general prices, P_t, which is known indirectly through an assessment based on information regarding distribution.

Based on the preliminary information that the distribution of p_t is normal with a prior mean and variance, observations will be made based on the following relationship:

$$p_t^i(z^i) = p_t + z_t^i \tag{18}$$

where z^i are special shocks, or relative demand-side shocks, with a total value for all markets of zero. The characteristic of relative shocks, z^i, is white noise, distributed normally with variance σ^2_z. Through Bayes rule, the supplier forms a posterior mean from the general price as follows:

$$\theta = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_z^2}$$

In this case, the posterior mean represents the weighted average of the prior mean and special price on market i, with the weight influenced by the relative variance of the preliminary price distribution and z.

If a price change is signaled due to relative demand-side shocks, limited information will cause supplier errors in their perception of which shocks require a response. If σ^2_z is small, however, the increase in P^i_t would be perceived more as a signal of higher general prices, rather than relative prices, $p_t^i - E[p_t \mid I_t^i]$, which would cause a small, or no, change in the output produced.

From the three equations, therefore, the following can be expressed:

$$y_t^i = b(1-\theta)(p_t^i - E[p_t \mid I_t]) = \beta(p_t^i - E[p_t \mid I_t]) \tag{19}$$

where $\beta = \dfrac{b\sigma_z^2}{\sigma_p^2 + \sigma_z^2}$, and aggregating among markets, we have

$$y_t = \beta(p_t - E[p_t \mid I_t]) \tag{20}$$

Equation (20) is known as the Lucas (aggregate) supply function, as explained by the Phillips curve under balanced conditions, namely that output is a positive function of unexpected increases (shocks) in general prices.

The aggregate demand is specified simply based on the Classical quantity theory of money, assuming a constant income velocity, equal to zero, as follows (as a logarithm):

$$m_t = p_t + y_t \tag{21}$$

The aggregate supply and demand functions produce conditions where $E[p_t \mid I_t]) = E[m_t \mid t]$, and can substitute the function of output and prices on total money supply (actual and anticipated), namely:

$$\begin{aligned} y_t &= \left(\frac{\beta}{1+\beta}\right)(m_t - E[m_t \mid I_t]) \\ p_t &= \left(\frac{1}{1+\beta}\right)(\beta E[m_t \mid I_t] - m_t). \end{aligned} \tag{22}$$

From the equation above, it can be concluded that output only responds to unanticipated money changes or money disturbances.

The analysis explains the empirical fact of a correlation between money and real output as a short-term phenomenon because of imperfect information among economic agents due to unanticipated changes in money supply. Therefore, Neoclassical economics recommends formulating monetary policy based on rules rather than discretion, thus clarifying and providing assurance when anchoring economic expectations. According to this perspective, money neutrality requires monetary policy to focus on price stability, while real output must be driven by productivity gains and technological advancements. Through its evolution, Neoclassical analysis in the 1980s, which referred to equilibrium, was directed toward developing the real business cycle theory, where long-term economic fluctuations are attributable more to real shocks, such as government spending (Barro, 1986) as well as technology and productivity (Prescott, 1986), rather than nominal shocks, such as money and the erroneous perceptions of economic players.

3.2.3. The Money—Output Correlation against Imperfect Market Competition and Price Rigidity

Rebutting the Neoclassical analysis of money neutrality under conditions of imperfect information, Neo-Keynesian analysis points to economic models that evidence a correlation between money and real output based on rational expectations, if nominal (and real) price and wage rigidity exists. This model supports the Keynesian viewpoint that monetary policy, through aggregate demand, affects output, and the long-term neutrality of money but not the short-run non-neutrality of money. Fischer (1977) and Taylor (1980) provided important contributions by developing a wage-price mechanism that assumes nominal rigidity and rational expectations and has implications on the role of money and monetary policy in the economy.

Fischer (1977) introduced his theoretical model, which can be expressed simply as follows:

$$y = (m - p) + u$$

$$y = -(w - p) \tag{23}$$

$$w = E(p \mid -1)$$

The first equation shows aggregate demand, with a unit elasticity of money held by the public to output, y, while u is the non-policy demand disturbance. The second equation shows aggregate supply, resolved from the maximization problem of corporate profit, which is assumed to have a unit elasticity to wages, w. The third equation determines wages, showing that nominal wages are set at the beginning of the period based on the information available (including the previous values of m and u) to achieve real wages and the job availability constant expected.

Assuming rational expectations, the reduced form solution can be expressed as follows:

$$y = (1/2)\left[\left(m - E(m \mid -1)\right) + \left(u - E(u \mid -1)\right)\right] \tag{24}$$

An important implication is that demand and money shocks can only affect output if the shocks are not anticipated. On one hand, the reduction equation is consistent with the Lucas model, but the channel used is the same as Keynes, namely that prices are flexible but nominal wages do not change, thus demand shocks raise prices, lower real wages and boost output.

Considering that wages are only adjusted in one period, in the model, policymakers do not have any additional information than those setting wages and can, therefore, react flexibly, thus minimizing the impact of policy to stabilize output. Fischer's model was subsequently modified to include staggered wage setting over two periods, which produced the same general conclusions, namely that output performance depended on unanticipated demand and money shocks.

Taylor (1979, 1980) developed a staggered wage-setting model over two periods, very similar to Fischer's model but with an important difference, namely that not only could wages be set earlier but the same nominal value could be set for two periods. The simple version of the model can be written as follows:

$$y = (m - p)$$

$$p = 1/2\left(w - w(-1)\right) \tag{25}$$

$$w=(1/2)\left[p+E\left(p(+1)\,|\,-1\right)\right]+(1/2)a\left[E(y\,|\,-1)+E\left(y(+1)\,|\,-1\right)\right]$$

Differing from Fischer's model, Taylor's aggregate demand equation disregards other shocks except nominal money shocks. The second equation is a price-setting equation, representing the weighted average wage for two periods. In this case, each month, half of the labor force selects a wage for the two periods, namely the current period and subsequent period. Therefore, during a given period, half of the labor force will be paid a total of, w, while the other half will be paid $w(-1)$. The final equation shows that if the labor force sets nominal wages for the two periods, it will depend on current prices and output as well as expected prices and output in the upcoming period. In this regard, money developments are only assumed to be known after wages have been set and, thus, cannot be used as an information set to determine wages.

The equation to illustrate wage behavior is as follows:

$$w=(1/2)\left[w(-1)+E\left(w(+1)\,|\,-1\right)\right]+a\left[E(y\,|\,-1)+E\left(y(+1)\,|\,-1\right)\right] \qquad (26)$$

In this case, in addition to workers taking into account relative wages, with $w(-1)$ and $E(w(+1)|-1)$, the wages are paid to half of the labor force in the current period and subsequent period.

If money performance follows a stochastic process in the form of a random walk, the solution in the assumption of rational expectations implies the unanticipated dynamic influence of changes in money as follows:

$$w=kw(-1)+(1-k)m(-1)$$

$$p=1/2\left(w-w(-1)\right) \qquad (27)$$

$$y=(m-p)$$

where $k = (1 - a^{1/2})/(1 + a^{1/2}) < 1$, and k is a measure of real money persistence, while, a, reflects the effect of labor market conditions on wages. A smaller value of a will produce a value of k closer to 1, which indicates a longer persistence of real money influence.

Considering that wages are determined prior to knowledge of money developments, such developments will not affect wages and prices in the current period and, therefore, will influence real output. Subsequently, in line with the corrections to wages and prices that occur, real output will gradually return to equilibrium following an exponential decline.

3.2.4. The Role of Money in the Monetarist Analysis Framework: "Money Matters" and the Keynesian-Neoclassical Synthesis

The "money matters" proposition, namely that money influences the economy, was explicitly proposed by Milton Friedman and other Monetarists. Friedman

(1958) argued that changes in money supply influence nominal spending and output (not real output) in the long term. This was subsequently proved using nearly 100 years of US monetary history in Friedman and Schwartz (1963). The empirical evidence also demonstrated a lag concerning the influence of money supply on nominal output, implying the non-neutrality of money in terms of short-term real output in an economy adjusting to a new equilibrium. In addition, Friedman developed a doctrine based on empirical and theoretical studies that the demand for money function, including the relationship with income (income velocity), is stable; thus, strengthening the case for non-neutrality of money in terms of real output in the long term. That doctrine was widely accepted during the early 1960s and contributed to the emergence of the Keynesian-Neoclassical synthesis through investment savings-liquidity money (IS-LM) model framework development to determine aggregate demand. The synthesis emphasizes the need for monetary policy to focus on price stability in the long term, despite the need to also consider the effect on real output in the near term.

Another contribution of Friedman, in terms of explaining the linkages between money supply and real economic variables, was through development of the natural rate hypothesis when analyzing the Phillips curve. As elaborated previously, the presence of the Phillips curve under conditions of economic equilibrium showed that output represents a positive function of unanticipated price shocks. In this case, Friedman argued that the natural rate of employment, or output under full-employment conditions, is independent of the anticipated or desired inflation rate, thus output and unemployment fluctuations indeed relate to actual inflation deviation from inflation expectations. This is known as Friedman's expectations-augmented Phillips curve.

Friedman's proposition concerning the real effect of money was developed more specifically by monetarists through the St. Louis model as an empirical procedure to assess the linkages between national income and total money supply. The model was formulated as a reduced form of the short-term macromodel equation as follows:

$$Y_t = \alpha_0 + \sum_i a_i M_{t-1} + \sum_j b_j G_{t-j} + \sum_s c_s Z_{t-s} + \mu_t \tag{28}$$

where Y is nominal national income; M is the vector of the nominal value of monetary aggregates in the current period and previous period; G is the vector of the fiscal variable in the current and previous periods; Z is the vector of other explanatory variables; and μ is the disturbance term.

An assessment conducted by Anderson and Jordan (1968) showed that monetary aggregates have a strong, positive, and rapid effect on nominal national income, which is more significant than the impact of fiscal policy. In general, these findings are in line with Friedman but the lagged effect of monetary policy was much shorter compared to the findings of Friedman and Schwartz (1963). The lag implies adequate space to utilize monetary policy for the stabilization of short-term real output, as proffered by the Keynesian school. The absence of lag would imply that total money supply affects nominal output immediately and, therefore, money only affects rising prices and not real output, as found by the Classical school of thought.

Consensus between the two economic schools of thought was achieved through the Keynesian-Neoclassical synthesis, which showed that monetary policy can influence real output in the near term but is neutral concerning the impact on the economy in the long term. The Keynesian-Neoclassical synthesis was the subject of a study conducted by Gali (1992), seeking to empirically validate the IS−LMPhillips curve model. To that end, the fundamental structure of the economy was represented through three equilibrium conditions, namely the goods market (IS), money market (LM), and price adjustment process to long-term equilibrium (Phillips curve):

$$\begin{aligned} y &= \alpha + \mu_S - \sigma(i - E\Delta p_{+1}) + \mu_{IS} \\ m - p &= \varphi y - \lambda i + \mu_{MD} \\ \Delta m &= \mu_{MS} \\ \Delta p &= \Delta p_{-1} + \beta(y - \mu_S) \end{aligned} \tag{29}$$

where (y, i, p, and m) represent, respectively, real output, nominal interest rate, prices, and total money supply, while (μ…) indicate four different types of shock in the economy, namely: aggregate supply, aggregate demand, money demand, and money supply.

An evaluation of that equation system for the US data from 1955 to 1987 identified the dynamic response to four economic shocks, which were generally consistent with the qualitative predictions of the IS−LM−Phillips curve analysis framework in line with the Keynesian-Neoclassical synthesis. An aggregate demand shock (at least in the short-term) affects output and other real economic variables due to the lagged adjustment of nominal variables. Monetary shocks (including money demand and money supply shocks) are transmitted to the real sector through changes in the real interest rate. Meanwhile, real output and prices move in the same direction in response to an aggregate demand shock, but in opposing directions in response to an aggregate supply shock.

3.3. Empirical Evidence and Related Issues

3.3.1. Monetary Policy Modeling and Variable Selection Issues

How money plays a role in the economy is, ultimately, an empirical question. In this case, based on the empirical literature, consensus is generally reached among economists that money neutrality only occurs in the long term, while in the near term, monetary disturbances can have a significant impact on real variables, such as output. Furthermore, several approaches have been developed to assess the effect of money and monetary policy on real economic activities. The methodology used has evolved over time because of advancements in time-series econometric techniques as well as changes in the theoretical models used. The two main objectives of the testing are to prove whether changes in monetary policy affect real economic performance and to seek empirical justification and consistency for the various theories that have been developed. The seminal studies in monetary economics include Friedman and Schwartz (1963), Sims (1972, 1980, 1992), Eichenbaum (1992) as well as Leeper, Sims, and Zha (1996).

The study conducted by Friedman and Schwartz (1963), based on the US macroeconomic data covering a period of around 100 years, was based entirely on the timing pattern of the money supply growth and economic growth cycles, which supports the causal relationship between money supply and output fluctuations. From the findings, the data fully supported the conclusion that total money supply growth is a positive reference (with a long lead) of the economic cycle. In this case, faster growth of money supply would be accompanied by a corresponding increase in output above the normal trend and, vice versa, slower money supply growth would trigger a decline in output. Furthermore, variations in money supply would be accompanied by variations in real economic activity, with a long and diverse lead.

Criticisms of that conclusion quickly emerged relating to the view that the evidence, based on precise movements and simple correlations, could not show the actual causality of the influence of money, considering a comparison of reference points of the economic cycle tended to ignore the information contained in time-series behavior of the variables of money supply, output and interest rates, thus making it difficult to assess the actual effect of monetary policy on output. Tobin (1970) opined that the correlation between money and output, fundamentally, is the result of reverse causality, namely that output influences money supply. Meanwhile, King and Plosser (1984) stressed the presence of inside money, a component of money supply that represents banking sector liabilities as the main factor explaining the correlation between money and output. Furthermore, it was concluded that the correlation between money supply (M1 and M2) and output emerges as a result of an endogenous response from the banking sector to economic shocks that do not correspond to the application of monetary policy.

Efforts by Friedman and other monetarists seeking empirical evidence concerning the links between money and economic activity were not only based on developing graphical analyses and simple correlations. Friedman and Meiselman (1963) developed an econometric modeling approach, which was later developed and became known as the St. Louis model, to analyze the effect of monetary policy or fiscal policy on nominal income. Although more significant and stable empirical evidence was found concerning the effect of monetary policy, rather than fiscal policy, on output, the specification of the approach was criticized, particularly due to endogenous money, which could impact assessment consistency when merely using a simple regression technique.

The issue of endogenous money was overcome by an important contribution from Sims (1972), namely by developing the Granger Causality approach for money and nominal income. Sims' early study concluded that past money behavior (M1 and narrow money) helped explain future nominal income. In a subsequent study, Sims (1980), using the Vector Autoregression (VAR) approach, where the production index was used as a measure of real output and the nominal interest rate as another variable that impacts the economic system, showed that the role of money in terms of explaining variations in output declined significantly. This implies that, in the near term, monetary policy could affect real output, but the effect fades in the longer term. Nevertheless, Sims conceded that the approach was very sensitive to which variable specifications should be considered. Relating to the trend of time-series variables, Eichenbaum and Singleton (1986) showed that using the first difference produced a conclusion regarding the effect of money, which was less significant than the specification of level. Departing from that study, Stock and Watson (1989) demonstrated that by systematically specifying the trend, the effect of money could explain output in upcoming periods.

Applying the VAR approach to seeking empirical evidence concerning the linkages between money and output has received more attention since a large-scale VAR modeling system was development, not just bivariate and multivariate systems. Sims (1992), by calculating several variables in the system, namely production, the consumer price index (CPI), short-term interest rates as a measure of policy impact, money supply, exchange rates, and the commodity price index, concluded that the real output response to the interest rate was in line with the advanced country studies, namely that monetary policy affects output, with a hump-shaped distribution. The conclusion implies that in the near term, monetary policy can influence real economic activity and, in the long term, the effect tends to fade.

Another focus of the VAR approach (four variables) was developed by Eichenbaum (1992), namely by evaluating the effect of monetary policy using several alternative measures of monetary policy, such as money supply or policy rate shocks, as well as by analyzing how those alternatives produced puzzling results. If M1 shocks are used as a measure of monetary policy, then a positive M1 shock would be accompanied by an increase in the policy rate and lower output. This output puzzle challenges the standard model and shows that expansive monetary policy would push up output and that money demand is inversely proportional to the interest rate.

In contrast, if interest rate shocks are used as a measure of monetary policy, then a positive policy rate shock (contractionary shock) would be accompanied by an output contraction and rising prices. The output puzzle would not occur in this case, rather the opposite would appear, known as the prize puzzle, where contractionary monetary policy would edge up prices. In general, current consensus lies in the fact that the variables used in the VAR modeling system do not represent the availability of information required by the monetary authority to set the policy rate. For instance, based on the prediction of heightened inflationary pressures in upcoming periods, the monetary authority would opt to raise the policy rate. If the policy response is inappropriate or poorly timed, then the higher policy rate would be accompanied by higher prices.

3.3.2. Consensus Regarding the Role of Money and Monetary Policy

The general conclusion of the various studies previously conducted is that the role of monetary policy shocks in terms of explaining output follows a hump-shaped distribution. An important question, therefore, is how far the role of monetary policy shocks can explain fluctuations in the economic cycle. Using monthly US economic data from the beginning of the 1960s to the beginning of 1996, Leeper, Sims, and Zha (1996) concluded that the role of monetary policy shocks was comparatively unimportant. Nonetheless, that conclusion demands further scrutiny considering that during the observation period, the Federal Reserve applied a different set of monetary policy operating procedures. This was evident in subsequent VAR models for two different sub-samples (1965.01–1979.09 and 1983.01–1994.12), which produced opposing conclusions concerning the role of monetary policy shocks to explain output fluctuations (Walsh, 2001). In this case, the role of monetary policy shocks was less important during the period when the Federal Reserve applied money supply as the operational target (1965.01–1979.09), but the reverse was true when the interest rate was used as the operational target (1983.01–1994.12), namely that monetary policy significantly influenced output fluctuations.

Congruent with the criticisms of VAR modeling, studies into the role of monetary policy shocks were also conducted using structural models, large and small. One pioneering approach, using the large-scale econometric model developed by the Federal Reserve, applied a structure that allowed for simulations based on various assumptions concerning the formation of expectations (Brayton & Tinsley, 1996). Based on that approach, it was concluded that monetary policy using the interest rate had a significant impact on output and the influence followed a hump-shaped distribution like that of the simple VAR approach. Nevertheless, differing from the VAR model, which achieved the maximum influence after around two years, the large-scale econometric model achieved maximum influence in a much shorter period, namely around one year.

Ultimately, although consensus was reached among economists concerning the role of money and monetary policy shocks in an economy, at least in the near term, no consensus has been reached regarding the transmission mechanism and role of a systematic monetary policy response. Several empirical questions arise, including whether money supply or the interest rate would more relevantly represent monetary policy? How long is the lag? And, how does the monetary policy transmission mechanism affect inflation, output and other real economic variables? This is primarily because of differing views concerning two aspects, namely what is the actual structure? And, to what extent is there interconnectedness between the variables in a policy regime? Operationally, the issues remain difficult to overcome through existing modeling approaches, namely VAR or structural econometric modeling.

3.3.3. Empirical Evidence in Indonesia

Several studies exploring the influence of monetary policy on economic activity in Indonesia have been conducted concerning the issue of monetary policy transmission mechanisms (Muelgeni, 2004; Warjiyo & Agung, 2002). Those issues will be explored in more depth in Chapter 5. Using a different framework, Solikin (2005) studied the influence of monetary policy variables on the final policy target, namely output, inflation and job opportunities. The study developed a structural cointegration vector autoregression (SVAR) approach. Inter-variable linkages and the magnitude of the monetary policy lag were observed using the impulse response function by identifying generalized functional innovation linkages. Nine variables were shown to influence the economic system, namely the policy rate (BI Certificates – SBI) and narrow money, as policy variables, output (GDP), inflation (CPI), job opportunities, exchange rate, London interbank offered rate (LIBOR), and Jakarta stock exchange (JSX).

Using quarterly data, with an observation period from 1980.01 to 2003.04, the study concluded at least four empirical findings as follows: *First,* the interest rate has an adverse impact on economic growth, is neutral in terms of employment opportunities, and has a positive impact on inflation. The positive impact on inflation is linked to inflation expectation formation (supply side) and the possible accumulation of narrow money due to principal and interest payments on BI Certificates (SBI) sold in the previous period (demand side). The impact on inflation is considered more direct. There are also other channels of influence. On the demand side, this can be indicated through the impact of a higher interest rate on M0-real money balance (negative: 3–6 quarters), and the impact of M0 on aggregate demand and inflation (positive: 3–6 quarters). Therefore, an interest rate hike would anchor inflation expectations on the supply side and subdue inflation on the demand side.

Second, the expansion of narrow money would positively impact output (only in the near term, not in the long term), and always impact positively on prices and employment opportunities. The indirect channels of influence could be explained through the impact of an increase to M0 on the interest rate (which is positive in the short term but negative thereafter), and subsequently, the impact of the interest rate on output (positive in the short term) and inflation (positive).

Third, the influence of both policy variables on the ultimate policy target tend to be similar, which is due to the ongoing role of both policy variables in terms of operational monetary control during the observation period, where the impact tended toward complementary. During the post-crisis period, the two indicators were mutually responsive. In this case, from the causality test of the impact of the policy variables on the final target, namely output, a policy response in the form of hiking the interest rate during conditions of excess liquidity would be comparatively strong handed in comparison to raising monetary aggregates during an excessive monetary contraction by increasing the interest rate. In contrast, the opposite is true when using prices as the ultimate target of monetary policy.

Fourth, by using the policy variables, SBI interest rates and the money base, complementarily, the lag in the impact of monetary policy on output and prices was identified as between one year, when the first extreme (peak) is reached, and two years, when the influence experiences a more persistent trend. Meanwhile, the impact on job availability gradually fades during the periods after the third year.[3]

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[3]The impact of monetary policy on job availability is difficult to identify due to, among others, the characteristics of the data, which contain conceptual and technical weaknesses.

Therefore, the average lag of monetary policy was identified at around 1.5 years, which is consistent with previous observations. One of the problems encountered, however, was that the influence of monetary policy on real economic activity and inflation, was transmitted through the various channels. Therefore, more rigorous and complete understanding of the monetary policy transmission mechanism in Indonesia is required to enrich the substance of the analysis.

In relation to the results, it is important to understand the possible emergence of economic phenomena that deviate from theoretical predictions, known as the economic puzzle. In general, several of the fundamental empirical findings of this study corroborated the inter-variable linkages consistent with theoretical predictions. In this respect, the specification of the modeling system successfully negated the liquidity puzzle. Using monetary aggregates (M0) to gauge the general monetary policy stance could also negate the price puzzle and exchange rate puzzle. Of the various economic puzzles observed, only two were interesting enough as puzzles, namely when using the SBI rate to assess the monetary policy stance. The *first* concerns the impact of SBI rate innovation on higher inflation. *Second* is the impact of SBI rate innovation on exchange rate depreciation.

Fundamentally, an alternative explanation of the two phenomena is interconnectedness between the variables, namely the interest rate, inflation, and exchange rate. This is possible because the exchange rate is also influenced by interest rate and inflation expectations at home and internationally. On the assumption that the influence of international interest rates and inflation is neutral (ceteris paribus), a domestic interest rate hike would (directly) raise inflation expectations and, therefore, exacerbate domestic inflation and exchange rate depreciation. That argument was supported by the findings of a previous study (Solikin & Sugema, 2004b), where a change in the interest rate was an important variable in determining inflation expectations in the manufacturing sector, primarily due to the component of output price setting on the supply side, including the cost of interest.[4]

Meanwhile, on the demand side, trends are inextricably linked to the use of SBI as the main instrument of monetary control through open market operations. On the other hand, however, is BI's commitment to pay interest on SBI sold in previous periods. Therefore, the impact of monetary policy shocks through hikes to SBI rates during the initial period would undermine the nominal growth

[4]According to Sims (1992), a corresponding increase in inflation and the interest rate (prize puzzle) could be explained by conditions where the monetary authority has sufficient information on possible future inflationary pressures. Therefore, seeking to dampen the inflationary pressures, contractionary monetary policy would be required through a higher interest rate. Under such conditions, prices would continue to rise after the monetary contraction, albeit not as significantly as if contractionary monetary policy had not been implemented. By arguing for such anticipatory measures, several research papers recommend using forward-looking variables, for instance international commodity prices, to overcome the prize puzzle.

In this research, considering that the monetary policy framework in Indonesia is primarily backward looking, the argument does not support the empirical findings.

of the monetary base because SBI sales would absorb money supply and, subsequently, lower inflation. Thereafter, assuming BI does not tighten monetary policy, the monetary base (and money supply) would accelerate as principal and interest payments are made on the BI Certificates (SBI). In addition to the resultant inflationary pressures, the consistent increase in the monetary base (and money supply) would also precipitate rupiah exchange rate depreciation.[5] That argument is also supported by the findings of a previous study (Mochtar, 2003), which demonstrated an increase in SBI demand as the SBI rate begins to rise.[6]

3.4. Market Imperfections and the New Paradigm of Monetary Economics: Credit Matters

The theory of monetary economics presented in the previous sections explained the role of monetary policy in an economy based on the transaction demand for money. On one hand, the Classical view regarding the quantity theory of money (MV = PT) and the Neoclassical view of the real business cycle à la Lucas (1972) or Kydland and Prescott (1977), where prices and wages are assumed to be fully flexible, concluded that money does not have a real impact. Furthermore, by also assuming rational expectations and full employment, money supply expansion would only be interpreted or anticipated by economic agents as an increase of inflation in upcoming periods; and, consequently, prices at that time would increase immediately, thus real income would not change. On the other hand, the Keynesian view, which assumes market failure and, hence, inflexible prices and wages (rigidity), suggests that money does have a real impact. With the presence of rigidity, money supply expansion would make economic agents feel an increase in real wages because prices would not yet have changed, thus driving up real economic activity as well.

Although the two schools of monetary economic theory can generally explain the effect of money on the economy, the issue of whether money matters has not been definitively addressed. Specifically, monetary economic theory based on the bank intermediation function has not been developed in response to how money created by the central bank can subsequently be circulated in the financial sector and finance various economic activities. Joseph E. Stiglitz proposed a new paradigm of monetary theory based on the demand and supply of credit (Stiglitz & Greenwald, 2003). In that context, banks have access to asymmetric information concerning borrower conditions, thus encountering default risks when disbursing loans. Therefore, understanding the behavior and ability of banks to manage risk is crucial when faced with imperfect information in terms of lending. Considering

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[5]Theoretically, the phenomenon of rising inflation due to tight monetary policy in the previous period is known as the "unpleasant monetarist arithmetic" (Ljungvist & Sargent, 2000; Sargent & Wallace, 1981).

[6]Mochtar (2003) showed that using government securities (T-Bills) as an instrument of open market operations had a different impact on inflation than raising the interest rate.

that credit performance affects monetary aggregates as well as output and other real variables, the banks' behavior will therefore determine institutional economics. This represents a departure from traditional monetary theory that considers the performance of banking institutions as a given or exogenous.

Several reasons have been proposed on why focusing on banking institutions is important, the most fundamental of which is the role of banks in terms of the intermediation function, namely by mobilizing deposits (demand deposits, savings deposits, and term deposits) from the public and subsequently disbursing the funds as loans to finance economic activity. In this case, monetary policy would have a direct effect on bank deposits and total money supply (M1 and M2) in the economy. Nevertheless, monetary policy affects not only credit performance but also bank lending behavior in the face of asymmetric information and credit risk. In other words, changes in monetary policy do not always feed through to proportional changes between bank deposit and loans and, therefore, the loan-to-deposit ratio (LDR) is not always 100%. Credit performance is affected more directly, compared to the influence of money supply on output and other real economic variables, therefore, monetary economics should be based on the banks' lending behavior.

Rapid growth of the financial sector has also had a profound effect on monetary policy. Development of the money market account (MMA), for instance, through the securitization of long-term government debt securities in the United States will continue to replace term deposits at banks, which means the effectiveness of monetary policy through the quantity of money (M0, M1, and M2) will fade and, therefore, monetary policy should be based on its effect through the interest rate. Meanwhile, globalization and financial sector integration will bring parity between domestic and international interest rates, thus leaving standard monetary policy theory impotent in the case of a small open economy. Nevertheless, this does not mean that efforts to stimulate the economy through monetary policy cannot be applied, due to many factors other than interest rates that affect lending, for example, through the reserve requirement, LDR, prudential principles for lending and by allocating subsidies to local banks. Such thinking demonstrates the importance of monetary policy based on analysis of the structure of the banking sector in terms of implementing the intermediation function in a country.

Does money matter? Stiglitz believes that money (monetary institutions and policy) matters; the only problem is that the influence of money is sometime too small to be detected.[7] Furthermore, the definition of money is becoming

[7]Following Keynes' three motives of demand for money, most money is not under the control of the central bank. The very definition of money itself has become increasingly blurred with the advent of financial engineering. Differences between the checking account (typically low interest) and the T-Bill account (market rates) have also become unclear. In reality, the use of money is not linked to the income generating motive but to transactions on financial markets (decoupling). Nonetheless, it is not money that is used for many financial transactions but credit. Under such

increasingly blurred and much falls outside the control of the central bank due to financial transaction and product innovation, including, more recently, card-based payment instruments and electronic money. With the emergence of such issues, monetary policy is still expected to influence the real sector due to: (1) market imperfections, for instance rigidity of nominal prices and wages; and (2) the redistributive effect of monetary policy through the financial sector. Therefore, the focus of alternative monetary theories should be directed toward credit availability, namely in terms of the supply and demand of credit, which is linked directly to imperfect financial markets and takes into consideration the distributive effect of credit on real sector activity. Therefore, per Stiglitz, the prevailing trend is credit matters rather than money matters.

3.4.1. Credit Availability and Bank Behavior: Credit Rationing Equilibrium

Credit availability in an economy is closely related to bank behavior. Fundamentally, banks are risk averse because banks are limited in terms of diversifying and distributing the risks. This is linked to the very essence of the function as a trusted intermediation institution, which, with relatively little capital, can mobilize the funds of the public and disburse them in the form of loans and other economic financing. The comparatively small amount of capital leads to solvency risk at the bank, which underlies the numerous regulations issued and enforced by the relevant authority to ensure the soundness of individual banks (microprudential policy). The banks also face liquidity risk because the depositors can withdraw their funds at any time, but the funds themselves have already been disbursed as loans. Lending also contains risks (non-performing loans – NPL) because the borrowers may be unable to repay the loan due to economic or other reasons.

With the risk-averse behavior of banks, any changes that occur in the economy will affect the availability of bank credit and economic activity in general. Similarly, the monetary policy response to influence the availability of credit (loanable funds) is also affected. During a recession, however, banks tend to apply stricter lending standards, which exacerbates economic conditions. Furthermore, monetary policy effectiveness in terms of influencing bank lending also fades. In this case, risk-averse behavior in the banking industry and imperfect information when lending can create a phenomenon known as credit rationing, as the price mechanism fails to bring about equilibrium in the market. Therefore, credit market equilibrium is reached in terms of total credit, where demand outstrips the supply of loanable funds and, subsequently, general economic equilibrium is

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conditions velocity is no longer constant or predictable as assumed. Meanwhile, even if the Neoclassical view always stresses that macroeconomic analysis must be based on microeconomics, the assumptions are still ad hoc, for example, money could suddenly appear in the utility or production function or be perceived as money donated to overcome the cash-in-advance (CIA) constraint.

reached at a level of real output that is below full employment and oversupply of labor or unemployment (Stiglitz & Weiss, 1981).[8]

Credit market equilibrium with credit rationing can be illustrated as follows. When lending, banks will consider the interest rates and risks. Despite monitoring and selecting prospective borrowers, the banks still have incomplete information concerning the borrowers' actual conditions due to asymmetric information. Therefore, the banks tend to include credit risk (NPL) in their decision to offer how much credit and at which rate to the borrower. On the other hand, however, the interest rate stipulated by the bank affects the repayment capacity of the borrower. With full knowledge of the business conditions rather than the information known to the bank, the borrower will consider a lower level of credit risk (or no credit risk) when applying for a bank loan. With the influence of credit risk when setting the interest rate, conditions where demand for credit is the same as supply may not be reached. Generally, credit rationing occurs, where credit market equilibrium is achieved in terms of interest rates and total credit, with demand outstripping the supply of loanable funds.

In addition to credit rationing, another lending phenomenon is known as adverse selection bias, which appears as a consequence of different borrowers having different repayment capacities. Considering the expected return of the bank depends on the repayment capacity of the borrower, the bank will strive to identify borrowers with the highest repayment capacity through various screening techniques. Nevertheless, considering how difficult it is to identify such borrowers, the banks screen potential borrowers referring to the interest rate the borrower could afford to pay (on average), the borrower who is willing to pay a higher interest rate may actually contain more risk. Adverse selection bias in terms of lending can manifest considering that the ability of the borrower to pay a high interest rate is possible due to the low probability of the borrower repaying the loan. Therefore, higher interest rates lead to a higher average degree of debtor risk, which erodes the profit obtained by the bank.

Consequently, increases in the expected return are slower than increases in lending rates and, under certain conditions, the expected return could decline, as illustrated in Fig. 3.1. In this case, the interest rate at which the expected return is maximized is known as the bank-optimal rate, where there is clearly excessive demand for loanable funds.

According to loanable fund theory, the interest rate is determined by the demand for and supply of loanable funds at a point where the demand for loanable funds curve intersects the supply curve of loanable funds (Fig. 3.2). During

[8]Traditional monetary theory states that monetary policy is an effective means by which to control economic activity. In this case, the government or central bank can control the money supply, which affects interest rates and, thus, aggregate demand. Consequently, the economy is also believed to tend towards equilibrium. Per the new monetary economics paradigm, however, the self-adjustment mechanism is not fully functional, thus making economic equilibrium difficult to achieve or prompting a protracted recession.

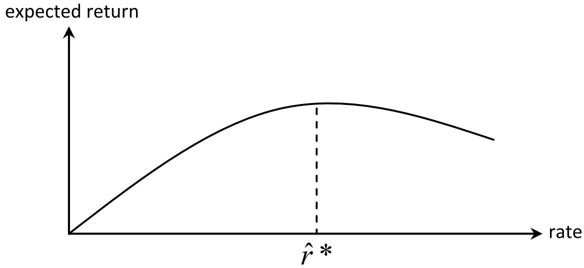

Fig. 3.1: Bank-optimal Rate.

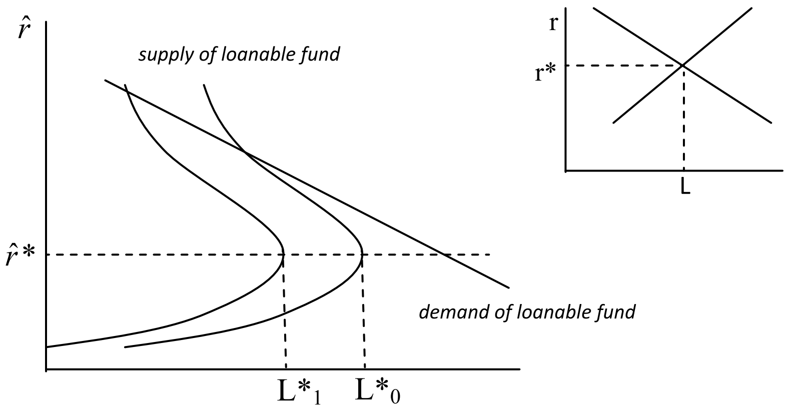

Fig. 3.2: Market Equilibrium Loanable Funds.

a recession, however, the demand curve (influenced by the demand for investment goods) shifts leftwards. Meanwhile, the supply curve (influenced by savings deposits) also moves to the left because savings will also decline in line with decrease of income. During a recession, the supply decline is more significant than the drop in demand (implying excessive demand). Under such conditions, the interest rate would not be determined by the intersect between the supply and demand curves because the borrower will face a higher real interest rate, which exacerbates matters.

After understanding the behavior of the equilibrium (optimal) interest rate and equilibrium in the loanable funds market, the next stage is to understand how determining credit market equilibrium is linked to credit rationing, as presented in Fig. 3.3. Considering that demand for funds depends on the interest rate set by the bank, r, while the supply of funds is influenced by the average expected return, P, using the conventional supply and demand diagram is not possible. Quadrants I and IV represent interest rate equilibrium as elaborated previously, while Quadrant III shows a positive correlation between the supply of loanable funds and the expected return. If a bank is free to compete for depositors, then P is the interest rate received by the depositors.

From Fig. 3.2, credit rationing equilibrium occurs when demand for loanable funds exceeds supply. When the bank sets the interest rate, the income per unit

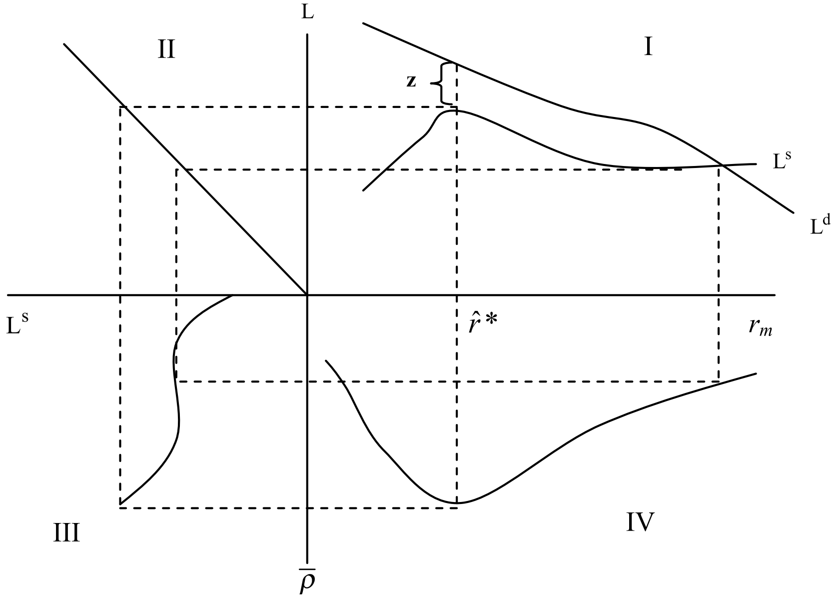

Fig. 3.3: Equilibrium Credit Rationing.

of currency of the funds borrowed will decline. In this case, the amount of excess demand for funds is estimated as Z. It is important to stress that there is a rate at which the demand for loanable funds is the same as the supply of loanable funds, namely r_m, but the rate is not the equilibrium interest rate. Therefore, the bank could increase its profits by setting $\hat{r}*$, not r_m. Through comparative statics, excess demand for loanable funds will decrease in line with an increase in the supply of loanable funds. Nonetheless, the interest rate determined by the bank will not change as credit rationing is maintained. Ultimately, congruent with the increase in the supply of funds, Z can decline and reach zero, prompting lower market rates.

3.4.2. A New Paradigm in Monetary Economics Theory

Stiglitz's criticism of traditional monetary theory was not due to a perceived lack of persuasiveness of the theoretical foundation but more because of the erroneous direction of the approach to analyzing monetary policy; with a focus on the incorrect policy variable, for instance money supply or the interest rate, which fundamentally have an unstable relationship with output and other real variables, particularly during a crisis.[9] Meanwhile, an analysis focused more on credit,

[9]Per traditional monetary economics theory, the emphasis of the role of money, as reflected by the presence of the LM curve (the transactions motive for money demand), received criticism due to the unstable nature of the relationship between money and output.

rather than money, and bank behavior, better represents how monetary policy influences the economy.

Stiglitz's proposition ushered in a new theory paradigm and monetary policy based on credit and bank behavior. The focus was not merely limited to how credit and the interest rate affect output and other real economic variables but also on how bank behavior transmits the effects of monetary policy in the economy. Furthermore, the analysis is based on the factual phenomenon that financial markets are always in an imperfect state (rather than general equilibrium) due to various factors, including asymmetric information, moral hazard, default risk, the influence of net worth, and so on.

Stiglitz and Greenwald (2003) provided a detailed elaboration of their new monetary economics paradigm. *First,* that the quantity of credit is clearer and directly affects (depending on the quantity if money) output and other real variables. The quantity of credit disbursed by the banking industry is clearly linked to the need for financing for production (capital and workforce) and spending (consumption, investment, exports, and imports) that creates output. On the other hand, the correlation between the quantity of money and output is becoming weaker and more difficult to predict. In addition to currency, the definition of money supply continues to develop in line with rapid financial innovation, therefore including products that are only traded on the financial markets and not just linked to transactions and economic financing. Fluctuations in the quantity of money are operational target always consistent with the quantity of credit. In banking terminology, such conditions are reflected in a LDR that is not always one or fixed over time.

Second, the interest rate is not a viable instrument or indicator to represent monetary policy. The spread between the funding rate (demand deposits, savings deposits, and term deposits) and the lending rate is not always stable because of several factors, including operating costs, profit margin, and the risk premium. Moreover, the interest rate only represents one factor in the decision of the bank to extend a loan. In general, credit is not allocated through an auction or market mechanisms but through a selection process by the bank using an assessment of net worth based on the relevant risk factors. In this case, the fundamental characteristic of the credit market is imperfect competition because the bank may not have complete and accurate information on the potential risks of the prospective borrower. As a result of asymmetric information and internal conditions, credit rationing has become a phenomenon in the banking industry, thus interest rates and the quantity of credit are determined during excess demand on the credit market. Such conditions represent a complete departure from traditional monetary theory, which typically assumes general equilibrium, where the interest rate will offset the demand and supply of credit.

Third, bank behavior and internal conditions also affect credit performance and, therefore, impact the economy. Internal conditions, including capital, liquidity, profitability, and NPLs, strongly influence bank lending ability. Similarly, when anticipating the economic outlook, the banking sector tends toward procyclical behavior, implying more lending during an economic upswing (boom) and extreme prudence during an economic downswing (recession). The banks' funding preference and strategy (retail a or wholesale) as well as credit strategy (retail or corporate) will also have an influence. Therefore, the interest rate and quantity of bank funds (demand deposits, savings deposits, and term deposits), which generally represent the interest rate and money supply in traditional monetary economics theory, cannot always explain credit developments and the influence on the economy. Understanding bank behavior and internal conditions are crucial in terms of economic and monetary policy theory.

Fourth, the availability of complete and accurate information on the general conditions of the economy, business dynamics, and internal conditions of the customer is also vital for the banking sector. Despite screening and monitoring prospective and existing borrowers, the banks will never be able to fully detect and anticipate potential business failure or the risk of loan default. The agency problem between bank and borrower could surface due to asymmetric information and other factors. The borrower's business could fail due to an inappropriate business strategy, moral hazard or general economic conditions. Consequently, the influence of information availability on bank lending behavior is an important element of the new paradigm of economic and monetary policy theory.

The new paradigm of monetary economics has several fundamental implications for central bank monetary policy. *First,* the monetary policy stance (tight/loose) can be measured more precisely based on credit availability, rather than total money supply or real interest rates. To that end, a methodology needs to be developed to measure the optimal level of credit availability in the economy, which should be linked to inflation, output, and other macroeconomic variables. The range and threshold would help determine how consistent the credit gap is with the goal of price stability as well as the trade-off with economic growth. The credit gap can be used as an important indicator of monetary policy in conjunction with the output gap (the gap between actual and potential output) and the interest rate gap (the real interest rate compared to the natural rate of interest) when formulating monetary policy.

Second, the interest rate can still be used as a monetary policy instrument by strengthening its effectiveness. The central bank implements its monetary policy stance by influencing interest rates (and liquidity) on the money market, which is subsequently transmitted to term deposit rates, lending rates and interest rates on other financial assets, thus affecting aggregate demand, output, and inflation. Nevertheless, the effectiveness of monetary policy transmission through the interest rate channel depends on banking and financial sector efficiency. Operating costs, profit margin, and the risk premium affect the magnitude of spread between lending and funding rates (demand deposits, savings deposits, and demand deposits) as well as money market rates. Therefore, market deepening efforts to boost efficiency are required to narrow the interest rate spread, including regulations concerning transparency and market conduct for banks and other financial institutions when setting interest rates.

Third, bank regulatory policy plays a salient role in terms of strengthening the use of traditional monetary instruments to influence lending. In addition to the interest rate and reserve requirement as monetary instruments, bank regulations on capital, liquidity, profitability, and loan loss provisions have a significant impact on credit. Furthermore, development of the credit information system is also important to reduce the problem of asymmetric information faced by banks when disbursing loans. It is important to stress that bank regulatory policy, in terms of monetary policy, must be based on bank behavior theory. The application of an inappropriate approach will be inefficient (pareto) and also counterproductive, which could actually exacerbate the risks faced by the banking industry. If banking supervision and monetary policy are conducted separately by two institutions, sound coordination is required between the two. Furthermore, bank regulatory policy and monetary policy are best applied not only paying due consideration to aggregate demand but also to aggregate supply.

Fourth, there is a greater need for policymakers to understand how monetary policy influences bank behavior. Banks tend to be procyclical, meaning they offer more credit during an economic upswing and are more prudent lenders during a downswing. Consequently, during a period of economic moderation, to prevent exacerbating conditions further, the monetary authority must respond with extra prudence by tightening the monetary policy stance to avoid the probability of bank default. Relaxing banking regulations, if possible, may be considered under such circumstances. Conversely, to control aggressive bank lending during an economic boom, the central bank may consider a combination of raising the interest rate and tightening bank regulatory policy. Under both sets of conditions, the central bank must meticulously monitor bank conditions and behavior.

Fifth, it is becoming increasingly important for the central bank to use micro, disaggregated, and granular analyses. Credit is heterogeneous, where excess funds at one bank cannot substitute shortfalls at another. The market and industry microstructure can cause frictions and disruptions to the circulation of funds and credit between banks and economic sectors. Therefore, policy that refers to aggregates tends to be misleading. The sectoral impact of monetary policy is not proportional, therefore, an overreliance on monetary policy may be distortive and, consequently, undermine effectiveness. Micro, disaggregated, and granular analyses are required to determine how far the traditional monetary instruments, such as the interest rate, need to be complemented by bank regulatory policy to reduce the distortions that emerge.

Finally, central bank or government intervention is possible if distortions cannot be corrected using traditional monetary instruments or bank regulatory policy. In general, loans are not disbursed through auctions and market allocation is not pareto-efficient, therefore intervention could raise prosperity to the pareto-optimal level. Under such conditions, the government could create a credit institution or scheme to complete the missing parts of the system, particularly in terms of the allocative and redistributive impact of credit on public welfare, for instance through interest rate subsidies or providing credit programs for micro, small, and medium enterprises (MSME). During crisis conditions, particularly during periods of financial system restructuring, the government could introduce measures to maintain the intermediation function by providing a blanket guarantee scheme to prevent bank runs and tax incentives for credit restructuring, or by showing willingness to purchase high-quality financial assets that the market cannot absorb.

3.5. Concluding Remarks

In this chapter, monetary policy and theory have been discussed in the context of a closed economy. From the theoretical studies and empirical evidence, the neutrality of money in an economy fundamentally depends on the assumptions used. The lengthy debate between Classical and Keynesian economists has produced a synthesis that money is neutral in the long run but not in the near term. Therefore, in the long term, total (growth of) money supply only influences prices (inflation) and not real output (economic growth), but affects both variables in the short term. Consequently, monetary policy should be directed toward achieving price stability, in other words low and stable inflation, while considering the impact on economic growth, particularly in the near term.

To what extent there is a trade-off between inflation and economic growth in the near term depends on the assumptions underlying monetary economics in a country's economy. In an economy with perfect markets (high economic flexibility) and rational economic agents (with complete information), per the assumptions of classical theory, the short-term trade-off between inflation and economic growth tends to be small. Nonetheless, those assumptions are not always present in an economy, as observed by Keynesian economists. Price and wage rigidity, incomplete information and irrational economic behavior are all present, which means a market is not always in equilibrium and, therefore, has implications on quantity and prices. Total money supply also has distributive and allocative effects through the impact of real interest rates on output, among others, due to holdings of government bonds and central bank monetary operations. The various factors explain why monetary policy not only affects prices but also output and other real economic variables.

The discussion on theory and empirical findings also showed that the real interest rate is a more appropriate monetary policy instrument than money supply in terms of its effect on the economy. The relationship between money supply and output is not stable and difficult to predict due to the innovation of products similar to money that may only be traded in the financial sector and do not relate directly to economic transactions. In fact, reverse causality has been found, meaning that economic growth has more of an impact on demand for money supply. Consequently, central bank monetary operations should be directed more toward influencing interest rates on the money market, which would be expected to affect interest rates in the financial sector and, thus, aggregate demand, inflation and economic growth. Therefore, how effectively the policy rate is transmitted to various interest rates in the financial sector and economic variables becomes significant. In this case, the assumptions debated between Classical and Keynesian economists, including the flexibility of economic sectors, the completeness of information, and the rationality of economic agents, also influence the effectiveness of the interest rate in the economy.

Nobel prize winner, Joseph Stiglitz, proposed a new paradigm of monetary economics theory, believing that the availability of bank credit has a direct effect on various economic activities. Monetary policy using credit availability, concerning quantity and the interest rate, is more effective in terms of influencing inflation and economic growth than monetary policy using money supply. Furthermore, imperfect markets and asymmetric information in the credit market cause the phenomenon of credit rationing, where credit quantity and interest rates are determined under conditions of excess demand. Therefore, credit availability is not only determined by the interest rate, but also by internal bank conditions and the information available concerning general economic conditions and the conditions of the customers. Consequently, the effectiveness of monetary policy transmitted through the interest rate channel should be strengthened through bank regulatory policy, the provision of general economic information and a debtor information system, as well as greater understanding of business microconditions and bank behavior.

Understanding the various aspects regarding the role of money and monetary policy presented in this chapter is crucial for the discussions in subsequent chapters. The mechanisms and effectiveness of monetary policy transmission, in terms of influencing the economy through the money supply channel, interest rate channel and bank credit channel will be discussed in Chapter 5. The strategic and operational frameworks of monetary policy as applied by the central bank will then be explored in Chapters 6 and 7. Those two chapters will lay the foundation for the discussions in Chapters 8 and 9 on the ITF commonly applied by central banks, including BI. Bank regulatory policy to influence credit performance as a macroprudential policy measure has become more widespread since the GFC in 2008, which will be discussed in Chapters 15, along with the linkages to monetary policy in a central bank policy mix.

Chapter 4

Exchange Rates and the Economy

4.1. Introduction

Monetary policy and theory in an open economy are dimensionally disparate from monetary policy and theory in a closed economy on two levels. *First,* the flow of foreign exchange from international trade and investment activities manifests on the balance of payments (BOP). Foreign exchange also affects total money supply in a country and is transmitted to monetary and real sector variables. *Second,* exchange rates also have an economic impact due to domestic and external factors. The exchange rate also influences the relative prices of goods and financial assets, thus affecting the economic decisions of economic players. Consequently, flows of foreign exchange and the exchange rate demand constant vigilance and represent important considerations underlying monetary policy formulation by the central bank.

Since the GFC of 2008/2009, external influences on domestic economic developments have increased and become harder to predict. Through the trade channel, global spillovers have occurred because the sluggish global economic recovery has undermined export performance and, therefore, the magnitude of foreign exchange flows from international trade. Meanwhile, through the financial channel, flows of foreign exchange from international trade have become more fluctuative, especially during the period of US monetary expansion from 2008 to 2013, which subsequently subsided but became less certain after the Federal Reserve announced its plan to normalize monetary policy in May 2013. Increasing exchange rate volatility and foreign exchange flows since the global crisis have become a complex challenge for central banks when formulating appropriate policy to mitigate the impact of global spillovers on domestic monetary and financial system stability.

The above description demonstrates the close interconnectedness between the exchange rate, foreign capital mobility and the central bank's monetary policy response in an open economy. In the literature for monetary economics and international finance, that interconnectedness has become known as the impossible trinity. More specifically, the theory states that exchange rate stability, foreign capital mobility and monetary policy autonomy to achieve domestic price stability are three goals that are impossible to attain simultaneously in an

Central Bank Policy: Theory and Practice, 81–113

doi:10.1108/978-1-78973-751-620191007

open economy.[1] Only two of the three goals are attainable at any one time. If an authority chooses to pursue exchange rate stability and foreign capital mobility, for instance, monetary policy autonomy to achieve domestic price stability would be difficult to attain because the significant influence of foreign capital flows on money supply and, subsequently, domestic prices. Meanwhile, if foreign capital mobility and monetary policy autonomy are desired to achieve domestic price stability, exchange rates would tend to fluctuate in line with the magnitude of foreign capital flows in order to maintain control over domestic money supply without affecting price stability.

The issue of the policy trilemma has dominated central bank discussions on monetary policy and theory in an open economy. This chapter will present in-depth discussions on exchange rate policy and theoretical aspects, while the discussion on foreign capital flows in terms of the policy trilemma in an open economy will be discussed in Chapter 14. The discussion in this chapter is arranged into five sections. After the introduction, exchange rate theory will be discussed in the second section based on an analytical framework in relation to exchange rates, foreign capital flow mobility, and the monetary policy response in an open economy. Various relevant theories, including the Mundell–Fleming model, Dornbusch's overshooting model, and exchange rate models with imperfect information and an imperfect foreign exchange market microstructure, shall be discussed. Then, the empirical findings of the impact of the exchange rate on inflation, growth, and other economic variables in various countries and Indonesia shall be presented in the third section . The fourth section will discuss the exchange rate system and exchange rate policy practiced in several countries, highlighting the debate on bipolarization of the exchange rate system, namely to apply a pegged or floating system. The causes and fallout of exchange rate crises, as well as the appropriate policy response to stabilize the exchange rate will also be discussed in the conclusion.

4.2. Exchange Rate Determination Theory

Numerous theories have been developed in the literature concerning international economics and finance to explain the exchange rate and relationships between variables in an open economy. Initially, exchange rate determination theory tended to emphasize international trade on the BOP. Nevertheless, as global financial markets became more developed and financial liberalization was pursued in various countries, exchange rate determination theory tended to evolve with a focus on capital

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[1] In this case, monetary policy autonomy is the central bank's ability to formulate monetary policy to achieve price stability, particularly in the face of external shocks such as the impact of economic developments, inflation, and external interest rates. The term should not be confused with the institutional independence of the central bank, namely to use monetary policy instruments (instrument independence) and remain free from intervention by a third party (institutional independence), as discussed in Chapter 12.

transactions on the BOP. Modern approaches view the exchange rate as an asset on the international financial market, which emphasize monetary aspects, currency substitution, and cross-border investment portfolio balance. Fig. 4.1 clearly illustrates the evolution of exchange rate determination theory.

The analyses contained in various theories are not only limited to exchange rate determination theory but also the impact of the exchange rate on various economic variables and the implications on the macroeconomic policy pursued. Although not all theories successfully explain the complex exchange rate puzzles and inter-variable relationships in an open economy, the various theories described below provide a framework to analyze the linkages between three important aspects of international finance and economics, namely exchange rate stability, cross-border capital mobility, and the macroeconomic policy response, particularly monetary policy in an open economy.

Exchange rate determination theory was initially developed through purchasing power parity (PPP), which is also recognized as exchange rate inflation theory. Fundamentally, PPP theory states the law of one price for various products traded internationally.[2] This theory is based on the concept of flows in terms of

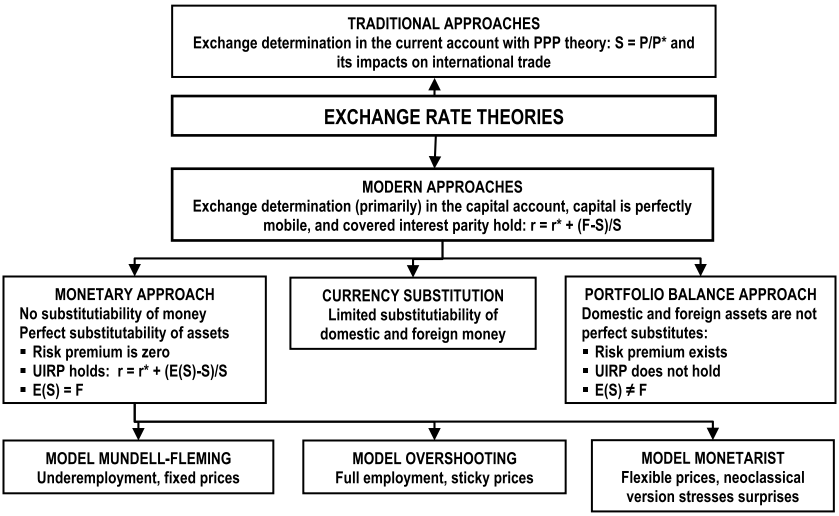


Fig. 4.1: Exchange Rate Determination Theory
Source: Gartner (1993, p. 28), modified.

[2]The law of one price states that if calculated in the same currency, freely traded commodities have the same price assuming perfect market mechanisms (namely small transaction costs, tax free, homogenous products, and no uncertainty). If the prices of homogenous products are different from one another, arbitrage will occur to take advantage of differing prices for the same asset until the prices are the same.

exchange rate determination from international trade activities, where the flow of demand for foreign exchange to pay for imports is the same as the flow of supply of foreign exchange proceeds from exports.[3] If there are no barriers to international trade or arbitrage taking advantage of foreign exchange trade, the prices of goods in each country will be the same when prevailing exchange rates are taken into consideration in absolute terms, namely $P = P^* + S$, where S is the nominal exchange rate, while P and P^* are the domestic and international prices of goods. In relative terms, however, if the methods and basket of goods used to calculate inflation are not too dissimilar, then according to PPP, changes in exchange rates will mirror the differences in inflation between countries, namely $\Delta S = \pi - \pi^*$, where π and π^* are domestic and international rates of inflation, as put forward by Cassel (1918).[4]

As trade has expanded to many countries, PPP theory has become the foundation for measuring weighted exchange rates in real terms, otherwise known as the real effective exchange rate (REER), namely REER $= \Sigma\ \omega_j \pi*_j / \pi$, where ω is the respective value of trade as a share of each trading partner. In general, REER is expressed as an index with a base year that reflects normal conditions of the country in question. The REER index can be used as a measure of currency misalignment from fundamental conditions. Furthermore, the REER index is also used to measure real exchange rate competitiveness, thus facilitating analysis of the impact on exports and imports.

Departing from PPP, which is based on the trade balance, interest rate parity (IRP) theory emphasizes the concept of capital flows between countries in the capital account. With the presence of the international asset market, IRP conditions are congruent with the law of one price of cross-border interest rates. Such conditions apply, assuming that the foreign exchange market is efficient in terms of transforming information into exchange rate movements and does not face significant transaction barriers to perfect market competition. Perfect capital mobility is also present with assets traded on relatively homogenous financial markets with perfect market substitutability.[5] Under such conditions, the returns on investment or interest rates in different countries will be the same after exchange rates are taken into consideration.

IRP theory can be expressed in the form of covered interest rate parity (CIRP) or uncovered interest rate parity (UIRP). Per UIRP, the return on investment or domestic interest rates will be the same as international interest rates after market

[3]PPP is considered a flow concept because it is based on the flow of goods and services in the trade balance to determine exchange rates.

[4]If speculative behavior is taken into consideration, Roll (1979) suggested an alternative PPP equation in the form of expectations, namely in absolute terms: $E(P) = E(P^*) + E(e)$, or in relative terms, where $\pi - \pi^* = E(e_t +_1 - e_t)$, where $E(.)$ is the expectations operator.

[5]Examples of capital market barriers include restrictions on cross-border capital mobility (such as foreign exchange controls), transactions costs, tax as well as disparate risks when investing in securities.

expectations of interest rate fluctuations are taken into consideration, namely $r = r^* + [E(S) - S]/S$, where $\{r,r^*\}$ = domestic and international interest rates and $\{E(S), S\}$ = expected and actual spot exchange rate. Meanwhile, CIRP indicates IRP for cross-border investment hedged against currency risk using forward transactions, thus $r = r^* + (F - S)/S$, where F = forward exchange rate. IRP may occur because arbitrage on international financial markets will prevent abnormal profits from investment transactions or borrowing on the domestic asset market compared to the international market. Therefore, interest rate disparity between countries will be lost due to capital mobility and arbitrage.

4.2.1. Mundell–Fleming Model: Policy Trilemma

Departing from the PPP and IRP theories that use partial modeling, various exchange rate determination theories were subsequently developed in the 1970s from the perspective of an open macroeconomic system. According to the IS-LM-Philips Curve, the relationships between domestic and international economic variables can be analyzed using the BOP, with the exchange rate represented as an asset price on the international financial market, through a monetary, money substitution, or portfolio balance approach (PBA). The monetary model, for instance, emphasizes the use of money only in the domestic economy, whereas the money substitution model assumes the internationalization of money and in the portfolio balance model there are is money along with other financial assets. In this case, there are differences in dealing with price behavior and capital mobility between countries. The neoclassical model of Mundell (1963) and Fleming (1962) was the first monetary model to determine exchange rates assuming fixed prices (thus possible to ignore) due to underemployment. In an open economy, this model assumes that the transportations costs between countries are very low, thus satisfying PPP conditions, while the risk premium and UIRP are effective and cross-border capital mobility is perfect.

Therefore, the Mundell–Fleming model applies the IS-LM approach for a closed economy and includes the relationship of an open economy with PPP and UIRP theories. The model's structure is based on analyzing the balance between three markets, namely the goods market, money market and foreign exchange market. An IS curve indicates balance on the goods market as follows:

$$Y = C + I + G + NX \tag{1}$$

where Y = gross domestic product (GDP), C = consumption that depends on the interest rate (r) GDP, I = investment that is also influenced by the interest rate (r) and GDP, G = government expenditure, NX = net exports that are influenced by international GDP (Y^*) and the real exchange rate $Q = E[P/S{\cdot}P^*]$ with S = nominal exchange rate and $(P{\cdot}P^*)$ = domestic and international prices.

Meanwhile, the LM curve indicates balance on the money market as follows:

$$M^d / P = L(Y,r) \quad \text{and} \quad M^d = M^S = M \tag{2}$$

where M^d = demand for money, M^s = supply of money, M = total money supply, and r = nominal interest rate. On the other hand, the foreign exchange (FE) curve indicates balance on the foreign exchange market, which illustrates balance on the BOP as follows:

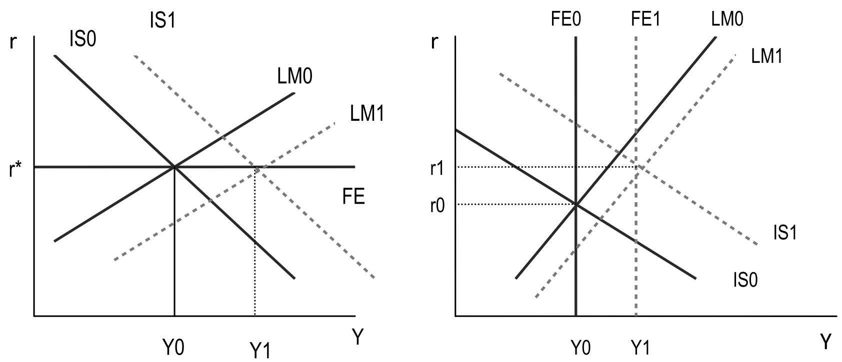
(a) Balance of payments[6]: BOP $= CA + KA$ (3)

(b) Balance of trade: $CA = NX = PX - SP * Z$ (4)

(c) Capital account: $KA = K\left(r - r^* - E(\Delta S)\right)$ (5)

where $r*$ = international interest rate, and $E[\Delta S]$ = expected nominal exchange rate depreciation.

Analysis per the Mundell–Fleming model focuses on fundamental factors of exchange rate determination, and the extent of monetary and fiscal policy effectiveness in terms of output stabilization (not price stabilization), which was the overarching concern of stabilization policy in the 1970s. The exchange rate and foreign exchange systems will influence the analysis results of the model. In this regard, the exchange rate is determined by various factors that influence balance on the domestic goods market and, pursuant to PPP theory, exchange rate adjustments will occur through changes to imports and exports in the trade balance. In more detail, Fig. 4.2 presents the Mundell–Fleming model for two specific cases. The first case looks at the influence of expansive monetary policy, assuming a flexible exchange rate regime and perfect capital



Case 1: Expansive monetary policy under a flexible exchange rate regime and perfect capital mobility.

Case 2: Expansive fiscal policy under a fixed exchange rate regime and perfect capital control system

Fig. 4.2: Policy Analysis in the Mundell–Fleming Model.
Source: Gartner (1993, p. 10 and 14).

[6]In practice, the balance of payments (BOP) consists of: (1) balance of trade (*CA*), (2) capital account (*CP*), and (3) changes in reserve assets (*OR*), thus BOP = *CA* + *CP* + *OR*. OR is omitted from this model.

mobility and the second case refers to the influence of expansive fiscal policy, assuming a fixed exchange rate regime and system of foreign exchange control (imperfect capital mobility). The IS curve has a negative slope, contrasting the positive slopes of the LM and FE curves. The FE slope depends on cross-border capital mobility.

In the first case, with perfect capital mobility and assumed exchange rate flexibility, the FE curve is horizontal. In this example, expansive monetary policy will effectively drive output. An increase in money supply would shift the LM curve to the right, thereby lowering domestic interest rates and, with foreign capital outflows, trigger exchange rate depreciation. The trade balance would increase, with rising exports and declining imports due to exchange rate depreciation, thus output would also increase. The IS curve would move to the right, hence the domestic and international interest rates would return to parity (UIRP assumption). Meanwhile, fiscal stimuli would not influence output. Increased government spending would move the IS curve to the right and edge up domestic interest rates, thus prompting exchange rate appreciation due to foreign capital inflows. The trade balance would deteriorate and return the IS curve to its original position without affecting the level of output.

In the second case, however, under a system of perfect capital control, the FE curve is vertical at a level of output that offsets the BOP. Under such conditions, expansive fiscal policy would effectively raise output.[7] Increased government expenditure would push the IS curve to the right, causing higher interest rates and appreciatory pressures on the exchange rates. To maintain the desired exchange rate, the central bank would purchase foreign exchange using domestic currency, thus moving the LM curve to the right to achieve a new balance. In contrast, monetary policy would not effectively drive output under a fixed exchange rate regime. In the event of depreciatory pressures, for instance if money supply outstripped the requirement, the central bank would have to intervene by selling enough foreign exchange for the exchange rate to return to the predetermined level.

Using the above analysis framework, another interesting feature of the Mundell–Fleming model is the presence of the policy trilemma in an open economy. Differing from the Tinbergen rule for a closed economy, namely that there should be at least the same number of instruments as there are targets, in an open economy, focusing one policy on one target must also take into consideration the exchange rate regime and foreign exchange system applied. Monetary policy would only be effective in terms of influencing output if the country in question adheres to a flexible exchange rate regime and perfect capital mobility. Oppositely, fiscal policy as an output stabilization policy instrument would be very effective under a fixed exchange rate regime and perfect capital control.

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[7] Fisher (1994) stated that under a fixed exchange rate regime, only fiscal policy is effective, thus limiting central bank independence in terms of implementing monetary policy to achieve the domestic economic goals, such as inflation and economic growth.

4.2.2. Exchange Rate Model with Sticky Prices: Dornbusch's Overshooting Model

Increasing exchange rate volatility as globalization accelerated along with international financial markets at the beginning of the 1980s attracted the attention of economists and policymakers alike. Consequently, the analysis of monetary models shifted from output stabilization policy to the phenomenon of exchange rate volatility and its implications on domestic price stability. The overshooting model proposed by Dornbusch (1976) explained that short-term exchange rate volatility was caused more by price rigidity (sticky prices) on the domestic goods market, while the exchange rate quickly reacts to economic shocks on the foreign exchange market. In the long term, however, prevailing monetary opinion is that PPP theory is valid and that inflation represents a monetary phenomenon.

Dornbusch's overshooting model differs from the Mundell–Fleming model by assuming sticky prices that cause a trade-off between inflation and economic growth in the near term since the inception of the Philips curve. In this regard, price corrections to non-traded goods occur with a lag to a new equilibrium level in response to a shock in the economy, while the prices of trade goods rise proportionally with changes to money supply. Therefore, prices increase less than money supply, causing currency supply to outstrip demand. Under such conditions, economic shocks would trigger exchange rate overshooting in the near term, implying greater exchange rate volatility than would fundamentally be expected, before returning to a new balance in the long term.

A simple version of Dornbusch's overshooting model can be found in Gartner (1993) in the log-linear equation, which mathematically illustrates balance on the goods market, money market, and foreign exchange market. Balance on the goods market is indicated by the Philips curve and aggregate demand curve as follows:

Philips curve: $\pi = \beta(y^d - y)$ (6)

Aggregate demand: $y^d = \delta(s - p) + \sigma y + g$ (7)

Meanwhile, balance on the money market is indicated by demand for currency and equilibrium as follows:

Currency demand: $m^d = p + \varphi y + \lambda r$ (8)

Money market equilibrium: $m^d = m^s = m$ (9)

On the other hand, equilibrium on the foreign exchange market is indicated through uncovered interest parity (UIP) with the formation of exchange rate expectations as follows:

Foreign exchange market: $r = r^* + E[\Delta s]$ (10)

Adaptive expectations formation: $E[\Delta s] = \theta(\hat{s} - s)$ (11)

where π = inflation, ρ = price, y^d = aggregate demand, y = output (exogenous), s = short-term exchange rate, s = long-term exchange rate, Δs = exchange rate depreciation, r, r^* = domestic and international interest rates, m^d = demand for currency, m^s = supply of currency, g = government spending, and $E[\cdot]$ = operator expectations.

The exchange rate overshooting model can be analyzed by expressing the model in two equations, namely an IS equation as goods market equilibrium and an MS equation that represents money market and foreign exchange market equilibrium. Therefore, because $y_d = y$ and $\pi = o$, the IS curve is as follows:

$$p = s + g/\delta - (1-\sigma)y/\delta \tag{12}$$

with a positive 45° slope, which demonstrates price and exchange rate homogeneity as contained in PPP theory. Meanwhile, the MS curve is as follows:

$$p = m - \varphi y + \lambda r^* + \lambda\theta(\hat{s} - s) \tag{13}$$

with a negative 45° slope that is consistent with the quantity theory of money (QTM).

Simply analyzing the influence of monetary policy in this model is as follows: In the long term, $s = s$ and $p = p'$, therefore price equilibrium is:

$$p' = m - \varphi y + \lambda r^* \tag{14}$$

with a (nominal) exchange rate equilibrium at:

$$\hat{s} = p' - g/\delta + (1-\sigma)y/\delta = m + \lambda r^* - g/\delta + (1-\sigma-\varphi\delta)y/\delta \tag{15}$$

and (real) exchange rate equilibrium level of:

$$(\hat{s} - p') = -g/\delta + (1-\sigma)y/\delta \tag{16}$$

Therefore, the long-term influence of monetary policy is as predicted by PPP theory and the QTM, namely that an increase of money supply only impacts (homogenously) the exchange rate $(\partial\hat{s}/\partial m = 1)$ and price level $(\partial p'/\partial m = 1)$, while the real exchange rate is not affected $(\partial\hat{s} - p')/\partial m = 0)$.

In the near term, however, the behavior of the (nominal) exchange rate is shown by the following equation:

$$s = \hat{s} + (m - p - \varphi y)/\lambda\theta + r^*/\theta \tag{17}$$

Therefore, an increase of money supply would cause the exchange rate to overshoot in the short term, namely excessive fluctuations beyond the long-term balance: $[\partial s/\partial m]^{ST} = \partial\hat{s}/\partial m + 1/\lambda\theta = (1 + 1/\lambda\theta) > 1 = [\partial\hat{s}/\partial m]^{LT}$. The magnitude of

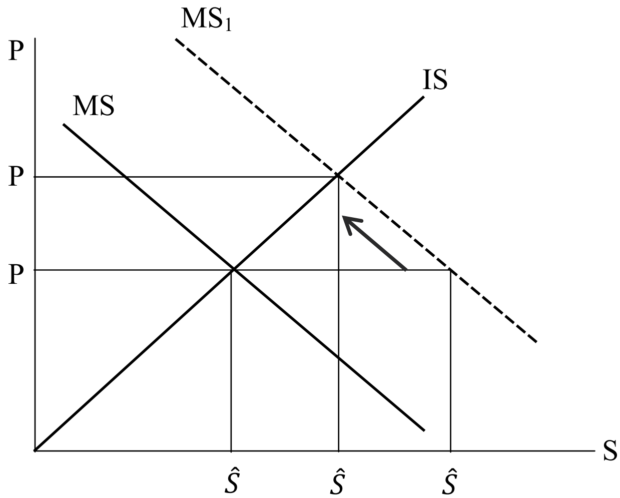


Fig. 4.3: Policy Analysis According to Dornbusch's Overshooting Model.

short-term exchange rate misalignment is more attributable to sticky prices in terms of adjusting to economic shocks as follows:

$$s - \hat{s} = [(m - \varphi y - \lambda r^{*}) - p] / \lambda\theta = [p' - p] / \lambda\theta \tag{18}$$

In other words, the preliminary results of Dornbusch's overshooting model confirmed the impact of price rigidity on heightened short-term exchange rate volatility before returning to a new equilibrium in the long term. These movements are presented graphically in Fig. 4.3 as follows:

The monetary policy trilemma in an open economy can also be analyzed using Dornbusch's overshooting model. When exchange rate stability close to the long-term equilibrium is desired, monetary policy will have less of an influence on exchange rates if cross-border capital flows are restricted though capital controls. In this case, the goal of cross-border capital mobility, such as through the application of perfect capital mobility, will not be achieved and the impact on prices will also be more pronounced (lack of monetary policy autonomy). On the other hand, when price stability close to the long-term equilibrium is desired, monetary policy autonomy to minimize price fluctuations could be achieved if capital mobility and a flexible exchange rate regime are applied. In such as case, exchange rate fluctuations would be of greater magnitude to absorb (shock absorber) the impact of cross-border capital mobility on domestic money supply.

4.2.3. Portfolio Balance Model and Central Bank Monetary Operations

Kouri (1976) was the first to expand the PBA in terms of exchange rate determination and BOP analysis. The model analyzed the exchange rate based on public behavior in terms of financial asset portfolio diversification to the domestic currency (M), domestic bonds (B), and foreign bonds (F) taking into consideration the returns (price and interest rate) of each respective asset type. Trade balance conditions will influence the impact of the exchange rate on other economic variables. Therefore, the influence of monetary policy will be determined by how the central bank conducts monetary operations, whether through the purchase of domestic bonds or intervention on the foreign exchange market.

In the near term, the total of domestic bonds is assumed to be fixed and held by the public and central bank, thus $B = B_P + B_A$. Similarly, foreign bonds are also held by the public and central bank, hence $F = F_P + F_A$, but the total will fluctuate in line with the current account surplus or deficit as follows:

$$CA = \partial F / \partial t = NX(S / P, Y) + r^*(F_P + F_A) \tag{19}$$

Primary funds represent the total of domestic and foreign bonds held by the central bank, thus $M = B_A + SF_A$. Therefore, economic worth can be expressed as follows:

$$W = M + B_P + SF_P = B + SF \tag{20}$$

In the model, balance on the money market, which is usually expressed in the following form: $M = m(r, E(\Delta S), Y, W)$, can be expressed in the relationship between the exchange rate and interest rate with the ME curve as follows:

$$S^{ME} = m(r, M, Y, W) \tag{21}$$

with a positive slope: $\partial S / \partial r = -(\partial m / \partial r) / (\partial m / \partial W)F_P > 0$. Implying that an increase in the interest rate on the money market would reduce demand for currency and trigger exchange rate depreciation.

Balance on the domestic bond market can be expressed in the following form: $B_P = b(r, E(\Delta S), Y, W)$, which can also be expressed in the relationship between the exchange rate and interest rate with the BE curve as follows:

$$S^{BE} = b(r, B_P, Y, W) \tag{22}$$

with a negative slope: $\partial S / \partial r = -(\partial b / \partial r) / (\partial b / \partial W)F_P < 0$. Implying that an increase in the interest rate on the domestic bond market would reduce central bank demand for domestic bonds and trigger exchange rate appreciation.

Similarly, balance on the foreign bond market can be expressed in the following form: $SF_P = f(r, E(\Delta S), Y, W)$, which can also be expressed in the relationship between the exchange rate and interest rate with the FE curve as follows:

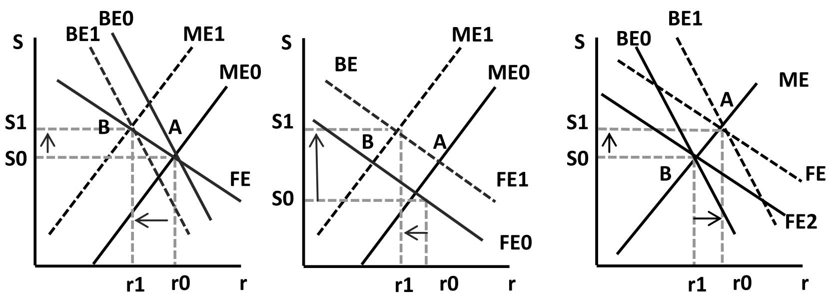
$$S^{FE} = f(r, F_P, Y, W) \tag{23}$$

with a negative slope: $\partial S / \partial r = (\partial f / \partial r) / (1 - \partial f / \partial W) F_P < 0$. Implying that an increase in the interest rate on the foreign bond market would reduce demand for foreign bonds and trigger exchange rate appreciation.

Short-term analysis of monetary policy's impact is determined by the monetary operations implemented by the central bank. Three cases serve as examples. *First,* open market operations (OMO), namely when the central bank increases money supply through the purchase of domestic bonds held by the public, thus: $\Delta M = -\Delta B_P = \Delta B_A$. *Second,* non-sterilized foreign exchange intervention, namely when the central bank increases money supply through the purchase of foreign bonds, thus: $\Delta M = -S\Delta F_P = S\Delta F_A$. *Third,* sterilized foreign exchange intervention, namely when a central bank purchases foreign bonds held by the public, while simultaneously selling domestic bonds, thus the amount of money supply remains unchanged and: $-S\Delta F_P = \Delta B_A$ and $\Delta M = 0$.

Fig. 4.4 presents an impact analysis of the three types of monetary operations available to the central bank in terms of the exchange rate and interest rate. In the first case, OMO would increase money supply, thus moving the ME curve to the left, while the purchase of domestic bonds would move the BE curve downwards, which would lower interest rates and cause exchange rate depreciation. The FE curve would not move because OMO merely create an exchange between currency and domestic bonds. In the second case, central bank monetary operations through non-sterilized foreign exchange intervention would create excess currency supply, thus moving the ME curve to the left, while the purchase of foreign bonds would move the FE curve to the right, which would lower interest rates and lead to exchange rate depreciation.

Meanwhile, sterilized foreign exchange intervention is a combination of OMO and non-sterilized foreign exchange intervention, which would not affect money supply and, therefore, the ME curve would not move. The purchase of foreign bonds would move the FE curve to the right, while the sale of domestic bonds would also move the BE curve to the right. Consequently, interest rates would increase and the exchange rate would depreciate. Exchange rate depreciation



Case 1: Open Market Operations (OPT)

Case 2: Non-Sterilized Foreign Exchange Intervention

Case 3: Sterilized Foreign Exchange Intervention

Fig. 4.4: Analysis of Monetary Operations in the Portfolio Balance Model.

would occur because excessive demand for foreign bonds would depreciate the exchange rate back to equilibrium.

The above analysis of monetary operations could be considered when formulating monetary policy to achieve the short-term economic targets required. For example, if monetary policy is required to overcome a trade deficit and stimulate economic growth, monetary operations through OMO and non-sterilized foreign exchange intervention would be the first choice because both trigger exchange rate depreciation and lower interest rates, while domestic inflation remains under control. Meanwhile, monetary operations through sterilized foreign exchange intervention would be the optimal choice to improve the trade balance and control inflation due to the resultant exchange rate depreciation and higher interest rates, while also considering the impact on growth.

4.2.4. Market Microstructure Models: Order Flows and Heterogeneous Information

The models presented above explain exchange rate theory and the impact on the economy from the perspective of macroeconomic fundamentals. The question, therefore, is how far the theoretical models can explain exchange rate fluctuations on the market? A review of the literature concluded that the predictive power of the models is no better than the random walk estimation of exchange rate fluctuations on the market (Meese & Rogoff, 1983). In other words, the theoretical models can better explain exchange rate fluctuations in the long term but not in the short term. There is some confusion when explaining short-term exchange rate fluctuations on the market, a phenomenon known as the exchange rate puzzle.

The exchange rate theories developed thereafter were based more on a market perspective, known as market microstructure analysis, namely to study how trade and foreign exchange market transactions determine exchange rate fluctuations on the market in the short term (O'Hara, 1995). The foreign exchange market microstructure consists of two segments.[8] The first segment is between the customers and dealers, where transactions are quote driven, while the second segment is between dealers, where the bid-ask mechanism is used to achieve transaction agreement. In general, the dealer is from the treasury division of a commercial bank and conducts foreign exchange transactions to serve the customer directly or as required by the bank itself. Meanwhile, the customers can be divided into two groups, namely financial customers in the form of a financial manager, another bank, the central bank or multilateral financial institutions, as well as corporate customers that buy or sell foreign exchange in line with their economic activities, such as exports-imports, external debt, or dividend proceeds. The transactions are based on an exchange rate quotation by a certain dealer, through bid-ask spreads, where the consumer or another dealer/broker can place an order of a specific amount. Order flows and transactions on the foreign exchange market

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[8]Refer to Osler (2008), for instance, for further explanation about foreign exchange market microstructure.

can occur at any time and continue indefinitely, thus influencing exchange rate fluctuations on the market.

The magnitude of impact of market microstructure on exchange rate fluctuations is in line with the results of direct surveys of foreign exchange market players (Cheung & Chinn, 2001; Taylor & Allen, 1992). The surveys showed that market players hold three beliefs that are completely different from the assumptions used in the standard exchange rate models. *First,* dealers believe that the exchange rate moves in response to trading flows (orders and transactions) on the market. This implies that order and transaction flows developing on the market will influence the behavior of dealers and brokers when setting their daily trade strategy and exchange rates. Such conditions are vastly different from the standard model that focuses on FX holdings, not trading flows, and assumes that PPP and UIRP effectively explain exchange rate fluctuations on the market.

Second, the surveys also showed that dealers consider private information an important aspect of the foreign exchange market. For example, Cheung and Chinn (2001) reported that dealers viewed large banks as having excess information because of the number of customers and extent of the networks in place. Furthermore, many big banks conduct macroeconomic and financial market research and analysis when setting their portfolio strategy and determining foreign exchange market transactions. Such conditions are very far removed from the assumptions of the standard model that all information is in the public domain and, therefore, accessible to all market players.

Third, market players consider that trading flows are a medium to transmit the impact of private information to exchange rate fluctuations. This implies that without trading flows, private information would not influence exchange rates on the market. With transactions available at any time, each order is based on the latest private information and then determines the exchange rate transacted. Thus, order process, private information and the exchange rate transacted move quickly and continuously. Such conditions are very different from the standard market efficiency model, which assumes that news causes immediate exchange rate fluctuations without considering whether there are transactions on the foreign exchange market that could influence exchange rates.

The influence of order flows on exchange rate fluctuations has been the object of several theoretical and empirical studies (Bionne & Rime, 2005; Evans & Lyons, 2005; Lyons, 1995; Payne, 2003). Based on UIRP theory, the model is as follows:

$$\Delta S_{t+1} = \beta_i (r_t - r_t^*) + \beta_z O_t \tag{24}$$

where ΔS = exchange rate fluctuations (appreciation or depreciation), also known as the exchange rate return, $(r - r^*)$ = disparity between domestic and foreign exchange rates as a fundamental indicator, and O = order flows indicator as a measure of the number of purchases minus the number of sales of foreign exchange on the market. In this context, the time interval $(t, t + 1)$ does not denote calendar days but transactions on the market.

The previous equation shows the extent of imbalances on the foreign exchange market per the order flows indicator, which can explain whether excessive exchange rate fluctuations on the market are attributable to fundamental factors, as measured by differences in the interest rate indicator. A study by Evans and Lyons (2002) found that the equation could explain around 70% of exchange rate fluctuations on the market. Furthermore, the empirical results demonstrated that an aggressive increase in purchases between dealers of US dollar (USD) 1 billion would cause 0.5% appreciation in the USD against the Deutsche eMark (DEM). Similarly, Froot and Ramadorai (2002) confirmed those results, adding that future short-term interest rate fluctuations also influence order flows and exchange rate fluctuations.

The promising empirical findings for order flows prompted further development of several exchange rate models based on market microstructure (Bacchetta & van Wincoop, 2003; Breedon & Vitale, 2004; Evans & Lyons, 2005; Froot & Ramadorai, 2005). In addition to order flows, several models were developed with the inclusion of behavioral factors and information heterogeneity regarding the market players.

Such models differed greatly from the previous exchange rate theories, which assumed that all market players have access to identical information, understand the models well and utilize all information in the decision-making process. In reality, although foreign exchange market players constantly monitor the news concerning economic fundamentals, their understanding or perception is heterogeneous. Different behaviors and reactions to information also have an influence, for instance carry traders generally base their decisions on differences between interest rates and currency risks. Similarly, despite involvement in market transactions, market players may not have access to complete information regarding order flows on the market.

Such factors precipitated model development to include interconnectedness between indicators of economic fundamentals, the magnitude of order flows as well as the behavior of market players and information heterogeneity to explain short-term exchange rate fluctuations on the market. Evans and Lyons (2005), for example, began their analysis by referring to the determination of exchange rate fundamentals in the monetary model for two countries as follows:

$$s_t = (1-\lambda)\sum_{j=0}^{\infty} \lambda^j E_t f_{t+j} \tag{25}$$

where $f_t = (m_t - m_t^*) - \varphi(y_t - y_t^*) + q_t - (r_t - r_t^* + \alpha\rho_t)$, namely the exchange rate is determined by disparity in terms of money supply, economic growth and interest rates, while paying due consideration to the risk premium. The equation can be used to predict the influence of fundamental factors on exchange rate fluctuations on the market as follows:

$$\Delta s_{t+1} = \frac{1-\lambda}{\lambda}(s_t - E_t f_t) + \varepsilon_{t+1}$$

where

$$\varepsilon_{t+1} = (1-\lambda)\sum_{j=0}^{\infty}\lambda^{j}(E_{t+1} - E_{t})f_{t+j+1} \tag{26}$$

In this case, if $s_t = E_t f_t$, then exchange rate fluctuations will follow the random walk. Alternatively, if fundamental changes follow AR(1):

$$\Delta f_t = \phi\Delta f_{t-1} + u_t \tag{27}$$

then exchange rate deviation from the fundamental value will also follow AR(1).

Differing from the macromonetary model above, the market microstructure model determines market player expectations using fundamental information available at the beginning of the transaction period:

$$s_t = (1-\lambda)\sum_{j=0}^{\infty}\lambda^{j}E_t^{m}f_{t+j} \tag{28}$$

therefore, exchange rate fluctuations on the market can be represented by the following equation:

$$\Delta s_{t+1} = \frac{1-\lambda}{\lambda}(s_t - E_t^{m}f_t) + \varepsilon_{t+1}^{m}$$

where

$$\varepsilon_{t+1}^{m} = (1-\lambda)\sum_{j=0}^{\infty}\lambda^{j}(E_{t+1}^{m} - E_t^{m})f_{t+j+1} \tag{29}$$

In this case, exchange rate fluctuations will be influenced by two factors: (1) public information that creates exchange rate deviation from the fundamental value at the beginning of the transaction; and (2) private information, namely the perceptions of each respective market player concerning future information that will influence the exchange rate.

The empirical findings relating to the influence of order flows on exchange rate fluctuations, as found previously, show that market players form private information from transaction flows occurring on the market, in addition to the public information available from the authorities, data, analysis, and market news. This can happen due to two factors. *First,* market players base their transaction behavior on the fundamental information available and, therefore, will be contained in the order flows on the market. Furthermore, there is a lag between the time when information becomes available from order flows and the time when market players process and understand that information. This is because market players only know some of the transactions when transacting during the first session, and only fully understand all order flows in the next transaction session.

Ordering to such conditions, changes in fundamental factors that influence exchange rate fluctuations on the market not only limit existing public information but also the perceptions of market players in terms of new public information and the information gleaned from order flows.

$$\begin{aligned} \Delta f_t &= \phi \Delta f_{t-1} + u_t + \delta v_t \\ O_t &= \gamma O_{t-1} + v_t \end{aligned} \tag{30}$$

Assuming news is public information, μ_t, that is known at the time of transaction and order flows information, v_t, is only understood by market players with a time lag of one period, then $E_t^m f_{t-1} = f_{t-1}$ and $E_t^m f_t = (1+\phi) f_{t-1} - \phi f_{t-2} + u_t$ thus $(f_t - E_t^m f_t) = \delta v_t$. Therefore, the exchange rate equation becomes as follows:

$$s_t = E_t^m f_t + \frac{\lambda\phi}{1-\lambda\phi} E_t^m \Delta f_t = \frac{1}{1-\lambda\phi} f_t - \frac{\lambda\phi}{1-\lambda\phi} f_{t-1} - \frac{\delta}{1-\lambda\phi} v_t \tag{31}$$

Such that spot exchange rate innovation, $\varepsilon_{t+1}^m \equiv (s_{t+1} - E_t^m s_{t+1})$, is:

$$\begin{aligned} \varepsilon_{t+1}^m &= \frac{1}{1-\lambda\phi}(f_{t+1} - E_t^m f_{t+1}) - \frac{\lambda\phi}{1-\lambda\phi}(f_t - E_t^m f_t) - \frac{\delta}{1-\lambda\phi} v_{t+1} \\ &= \frac{1}{1-\lambda\phi} u_{t+1} + \frac{[1+\phi(1-\lambda)]\delta}{1-\lambda\phi} v_t \end{aligned} \tag{32}$$

Substituting the order flows equation produces the following equation:

$$\Delta s_{t+1} = \frac{1-\lambda}{\lambda}(s_t - E_t^m f_t) + \frac{1}{1-\lambda\phi} u_t + \frac{[1+\phi(1-\lambda)]\delta}{1-\lambda\phi}(O_t - \gamma O_{t-1}) \tag{33}$$

This equation explains that there are three factors influencing exchange rate fluctuations on the market, namely: (a) the perception of market players to exchange rate misalignment from its fundamental value; (b) news appears from fundamental information; and (c) information is derived from order flows.

In their empirical study, Evans and Lyons (2005) used the following two equations:

$$\begin{aligned} \Delta s_{t+1} &= \alpha_0 + \alpha_1 O_t^A + e_{t+1} \\ \Delta s_{t+1} &= \alpha_0 + \sum\nolimits_{j+1}^{6} \alpha_j O_{j,t}^S + e_{t+1} \end{aligned} \tag{34}$$

where the first equation was estimated for order flows as an aggregate of all customer segments, while the second equation was estimated for each respective customer segment. In general, the empirical results showed that the market microstructure model has a higher prediction power than the macromodel or random walk. Moreover, exchange rate fluctuations on the market are not only influenced by public information relating to economic fundamentals but also by the private information of market players concerning current news on economic fundamentals and from studying order flows on the market.

4.3. Empirical Findings for Exchange Rates and the Economy

The various theoretical reviews presented in previous sections provide a rationale to analyze the exchange rate and its impact on the economy. How far can these theories explain empirical economic conditions in various countries? Specifically, can the exchange rate improve the trade balance and, therefore, stimulate economic growth? Also, to what extent does the exchange rate influence inflation? This kind of empirical review represents an important input for the central bank to formulate monetary policy to achieve price stability and also stimulate economic growth.

4.3.1. Exchange Rates and Trade

Theoretically, the Marshall–Lerner condition states that exchange rate depreciation will only cause a balance of trade improvement if export elasticity to the exchange rate is greater than import elasticity to the exchange rate. In general, such conditions are found in countries where the composition of manufactured commodities in exports is larger than in imports. Dynamic analysis of the elasticity concept in the Marshall–Lerner condition reveals a J-curve phenomenon concerning the exchange rate's influence on the trade balance. In the near term, exchange rate depreciation would undermine trade balance performance due to the negligible decline of imports to meet domestic demand. As time goes by, however, commodities to substitute the imports would emerge in the country and, therefore, exchange rate depreciation would reduce imports. The trade balance would improve as exports increased and imports declined in response to the exchange rate depreciation.

Numerous empirical studies have corroborated the positive impact of exchange rate depreciation on the trade balance. From a panel data study of around 100 countries for a period of 10 years (2000–2009), for instance, Nicita (2013) showed that exchange rate misalignment from the fundamental value (measured using the REER) triggers a 1% increase in world trade. Auboin and Rut (2011) found that the influence of the exchange rate on the trade balance was more pronounced in highly competitive countries, along with countries that are part of global/regional supply chains or countries experiencing domestic price rigidity, including those using foreign currencies in the production process. Meanwhile, the empirical evidence remains inconclusive about whether exchange rate volatility would adversely affect the trade balance (Clark, Laxton, & Rose, 2004). This might be due to weak predictive power of the theoretical analysis because, in general, exchange rate volatility is linked to economic shocks, such as technology, preferences, and exchange rate systems. Empirically, the ambiguous influence of exchange rate volatility can be associated with hedging and other risk mitigation practices undertaken in the corporate sector, the large amount of raw materials imports as well as the proliferation of affiliated corporations operating internationally.

Concerning developing countries, Rodrik (2008) showed that undervaluation of the nominal exchange rate (or a high real exchange rate) could stimulate economic growth. From a study of seven countries in the period of 1950–2004, the empirical results showed that a 20% undervaluation of the real exchange rate could accelerate economic growth by around 0.4%. Exchange rate undervaluation was shown to be transmitted primarily through the tradable sector, especially the manufacturing industry, which is generally competitive and profitable, thereby avoiding the market failures and institutional default that can plague developing countries. How can real exchange rate undervaluation be achieved through various policies, such as fiscal policy (large structural surplus), income policy (redistributing income to large savers through the compression of real wages), saving policy (mandatory saving schemes and pension reforms), capital account management (tax foreign capital inflows, liberalize capital outflows), or exchange rate intervention (with adequate reserve assets)? Experience in East Asia showed that countries targeting real exchange rates to enhance competitiveness successfully increased exports, which must be supported by deregulation and other structural reforms to boost economic competitiveness.

The positive influence of real exchange rates on economic growth was also empirically proven in ASEAN5 countries (Abudalu, Ahmed, Almasaied, & Elgazoli, 2014). Using quarterly data for the period of 1996–2006, the study showed the respective coefficients of real exchange rate elasticity to GDP growth in Thailand (0.11), Indonesia (0.14), Malaysia (0.16), the Philippines (0.49), and Singapore (0.88). Meanwhile, Onafowora (2003) empirically demonstrated the Marshall–Lerner condition with a J-curve in the near term regarding trade between ASEAN5 with Japan and the United States for the period from 1980 to 2001. For Indonesia, for example, exchange rate depreciation would reduce the real trade balance for a duration of around three quarters and then subsequently increase with an elasticity of 0.35 for Japan and 0.24 for the United States. Similarly, in Thailand's case, the real trade balance would reduce for around four quarters before increasing with an elasticity of 1.08 for Japan and 1.65 for the United States. Nonetheless, Hadad and Pancaro (2010) advised against enhancing competitiveness through real exchange rates in the long term because such policy would raise inflation and undermine structural reforms to boost real sector competitiveness, which would be more important to foster long-term growth.

4.3.2. Exchange Rate and Inflation

The influence of the exchange rate on inflation (exchange rate pass-through – ERPT) is an important aspect of monetary policy formulation. Thus far, observations have shown that developing countries experience a greater decline of ERPT than advanced countries. Frankel, Parsley, and Wei (2005) conducted an empirical study concerning this phenomenon and its determinants using a sample of 76 countries in the period of 1990–2001. Price indicators were differentiated between port prices, retail prices, and CPI. The results showed that countries with a history of high inflation experienced an ERPT reduction from around 0.79 during the period of 1990–1996 to just 0.20 in 1996–2001. In contrast, countries with historically low inflation recorded a decline from 0.50 to 0.12 over the same period. For all price indicators, ERPT was around 0.63 for port prices, 0.41 for import prices and 0.22 for CPI. The proliferation of post-globalization international trade was cited as the main cause of lower transportation costs between countries. Meanwhile, exchange rate volatility was shown to have an asymmetrical impact on ERPT, meaning that depreciation exceeding 25% could raise ERPT but exchange rate appreciation may not necessarily reduce ERPT. The empirical study also stressed the role of monetary policy in terms of lowering long-term inflation as an important aspect of reducing ERPT.

How monetary policy framework performance in terms of price stabilization (Inflation Targeting Framework – ITF) reduced ERPT was the focus of an empirical study by Edwards (2006), who revealed several interesting findings. *First,* countries applying ITF experienced a decline in the exchange rate influence on inflation, most significantly in Brazil, decreasing from 0.719 to 0.056, followed by Chile, Israel, and México. Meanwhile, the decline in South Korea was not significant due to the relatively low level of ERPT (0.020). *Second,* ITF application did not exacerbate exchange rate volatility (nominal or real). Nonetheless, the three countries that applied a floating exchange rate regime, namely Brazil, Chile, and Israel, experienced an uptick in exchange rate volatility. *Third,* ITF countries, especially those facing comparatively high inflation, included nominal exchange rate factors into their monetary policy. Estimating monetary policy rate determination using the Taylor rule revealed the significance of the exchange rate variables (depreciation and volatility), with coefficients varying from 0.79 for México and 0 for Chile.

Regarding ASEAN5 countries, Corinthas (2007) conducted an empirical ERPT study bilaterally between two countries, linked to the possibility of currency unification using data for the period from 1968 to 2001. The results showed that ERPT to inflation was significant and relatively large (0.10) for Indonesia–Philippines, but not significant for Thailand–Singapore or significant but close to zero for Thailand–Malaysia and Malaysia–Singapore. In terms of ERPT to import prices, the study found significant ERPT for Indonesia–Malaysia and Indonesia–Singapore. The study concluded that currency unification was only feasible between Malaysia and Singapore. For Indonesia, currency unification was shown to benefit monetary policy effectiveness in terms of lowering inflation to the regional level.

4.3.3. Empirical Findings in Indonesia

The influence of the exchange rate on the economy has also been the object of empirical studies in Indonesia. Winarno (2014), for instance, studied the Marshall–Lerner condition and J-curve effect in Indonesia using monthly data from 2008 to 2012. The results proved that the Marshall–Lerner condition was present in Indonesia, namely that each 1% depreciation of the rupiah exchange rate would raise the current account, total trade balance and non-oil and gas trade balance by 0.51%, 0.72%, and 0.77%, respectively, in the long term. Furthermore, the study also proved the J-curve phenomenon in Indonesia, namely that the exchange rate influence would reduce the current account, total trade balance, and non-oil and gas trade balance for around 3–4 months before increasing to its long-term trajectory. The J-curve effect was also demonstrated by Adiningsih, Siregar, and Hasanah (2013), using quarterly data from 1996 to 2011, for bilateral trade between Indonesia–China and Indonesia–Japan but not for Indonesia–United States. In another study, Affandi and Mochtar (2015) showed that the effect on the current account since 2000 has been more due to temporary real exchange rate shocks rather than permanent.

The ERPT analysis has also been conducted by BI and through other empirical studies. Kuncoro (2015), using monthly data from 2003 to 2013, found that ERPT has tailed off since ITF application in Indonesia. The study reported that ERPT to CPI and import prices for each 10% of exchange rate depreciation was 6% and 3% prior to ITF implementation, dropping thereafter to around 3% and 1.5%. That level of ERPT exceeded those proposed in the empirical studies conducted by BI, namely around 0.7%–0.3% for each 10% of exchange rate depreciation. The empirical results point to the need to rupiah exchange rate stabilization policy as an integral part of monetary policy to achieve the inflation target set.

4.4. Exchange Rate System and Policy

The discussion on exchange rate determination in the previous section highlighted an interesting problem for further review that the relationship between macroeconomic variables and monetary policy effectiveness in an open economy will depend on the exchange rate regime adopted. For example, under a flexible exchange rate system, Mundell–Fleming model analysis showed that monetary policy could very effectively influence output. Meanwhile, Dornbusch's overshooting model showed that the impact of monetary expansion on exchange rate depreciation is larger in the near term before returning to equilibrium in the long term. Similarly, in the context of the monetary policy trilemma, exchange rate stability may be achieved under a fixed exchange rate system with capital controls, while price stability is more achievable under a flexible exchange rate regime with capital mobility.

4.4.1. Choice of Exchange Rate System: Theoretical Review

Observations of the international monetary system after the collapse of Bretton Woods in 1942 showed that exchange rate systems vary from one country to another, from a fixed (pegged) exchange rate system at one end of the spectrum to a free-floating exchange rate regime on the other. As stated by Frankel (1999), "No single currency regime is right for all countries or at all times."

There are numerous considerations when determining the exchange rate system in accordance with prevailing economic and financial conditions in the country (Later, 1999). The factors include: (1) the relative size of the economy compared to other countries; (2) trade integration between countries or groups of countries; (3) flexibility of the economic structure; (4) production factor mobility, including labor and capital; (5) capacity to absorb or confront internal and external economic shocks; (6) economic homogeneity with other countries or groups of countries; (7) availability of external sources of financing to overcome BOP shortfalls; (8) availability of monetary policy that collectively supports exchange rate system stability; and (9) exchange rate credibility to anchor or standardize expectations.

In general, it can be said that for countries with a comparatively small economy and high factors listed from points 2 to 9, a pegged exchange rate regime should be considered. Such a system is even more important in countries with historically high inflation and low central bank credibility in terms of maintaining price stability. The application of a fixed exchange rate system under such economic conditions would help the country to lower inflation toward the country or group of countries used as the standard for exchange rate determination. Nonetheless, economic developments in the affected country would also mirror the other countries, with output fluctuations and higher unemployment, while monetary policy autonomy to achieve the domestic goals would be forfeited.

Theoretically, the selection of the exchange rate system is generally based on three salient criteria (Ghosh et al., 2001), namely (1) insulation; (2) economic integration; and (3) monetary policy credibility. The first approach emphasizes the ability of the exchange rate system to insulate the domestic economy from external shocks. In this regard, Freidman (1953) suggested that a flexible exchange rate regime is generally better because it supports the role of automatic adjustments from the nominal exchange rate to imbalances in the BOP. Similarly, the exchange rate system can also be selected based on the sources of economic shocks, in which case a floating system is typically better to protect domestic output if shocks tend to originate from the real economy, while a pegged exchange rate regime would be better if the nominal side is more dominant.

The second approach emphasizes exchange rate system selection based on the level of economic integration with another country or group of countries. In this context, optimum currency area theory (McKinnon, 1963; Mundell, 1961) states that economic integration between two countries facing a similar and closely correlated output shock could opt for an exchange rate system that applies a common peg. Under such conditions, the function of the nominal exchange rate as an adjustment instrument to shocks will be lost due to economic integration and application of the common peg. The adjustment function would have to be replaced by flexible prices and wages, production factor mobility and the availability of a fiscal transfer system contained in the economic integration. If the requirements are met, application of a common peg with a common currency would be optimal. In contrast, a flexible exchange rate regime would better suit countries with disparate economic characteristics.

The third approach stresses monetary policy credibility to achieve the domestic economic goals. Analysis is focused on how far the applied exchange rate regime can nurture central bank credibility in terms of controlling inflation. This theoretical approach refers to the monetary policy credibility and optimal inflation target theory developed by Barro and Gordon (1983) for an open economy.[9] Analysis emphasizes the ability of a fixed exchange rate regime as a pre-commitment to low inflation, thus limiting the propensity of the central bank to create surprise inflation to stimulate output. In this context, a fixed exchange rate system would be preferable if central bank credibility is low so that the costs incurred by surprise inflation would be higher than the cost of losing the exchange rate as an adjustment instrument in the economy in the form of more prominent output fluctuations and higher unemployment. Conversely, a flexible exchange rate system should be applied if central bank credibility is high in terms of controlling inflation, hence the exchange rate can function as an adjustment instrument to shocks in the economy.

In practice, exchange rate system classification is a country is also a problem in and of itself. One approach, the *de jure* approach, classifies the exchange rate system based on the official statement of the central bank submitted annually to the IMF. Nevertheless, the official statement of the central bank cannot always be used as a basis to classify the exchange rate system in accordance with theoretical analysis of exchange rate fluctuations. The second approach, therefore, known as the *de facto* approach, classifies the exchange rate regime based on observations of nominal exchange rate behavior and several monetary indicators that are in line with the theoretical concept. One such approach is exchange rate system classification based on the magnitude of nominal exchange rate volatility, fluctuations in reserve assets, nominal exchange rates, and other monetary indicators. Another technique, known as natural classification, is based on survey results, the official statement of the central bank, interviews, and so on that are subsequently compiled into an exchange rate system index (Rogoff et al., 2003).

4.4.2. Exchange Rate Systems in Practice: Bipolarization and Fear of Floating

Applying different approaches has implications on the results of assessments into the classification of exchange rate systems adopted around the world. Using a *de jure* approach, Ghosh et al. (2003), for instance, evaluated exchange rate system evolution in 165 IMF member countries for the period of 1970–1999. The exchange rate systems were classified into three main groups, namely pegged, intermediate, and floating, with varying policy practices contained within. The

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[9]In brief, the model of Barro and Gordon (1983) characterized an economy where nominal wages are determined prior to the inflation target and the central bank seeks to create surprise inflation to stimulate output. The model shows that such central bank behavior would not be optimal to achieve the inflation target and would therefore undermine central bank credibility and output. The optimal solution for the inflation target is achieved if the central bank remains committed to maintaining low inflation so that wage determination can be trusted in the decision-making process. Central bank credibility can therefore be achieved by: (1) setting an optimal inflation target, namely one that minimizes real shocks; (2) central bank commitment to the inflation target; and (3) achievement of the inflation target set.

first group consisted of the currency board system, or a hard peg to a certain currency or basket of currencies. The second group included floating systems with intervention pursuant to certain rules or masked (intermediate). The third group included free floating systems with little or no intervention.

A number of salient observations can be delivered. *First,* fewer countries are applying pegged systems, dropping from 84.8% in 1970–1979 to just 46.6% in 1990–1999, while countries favoring more flexible exchange rate systems are increasing, with intermediate systems accounting for 26.4% and floating systems for 27.0% in 1990–1999. *Second,* among the pegged systems, there has been a shift away from a peg to a single currency toward a hard peg to a basket of currencies. Similarly, flexible exchange rate systems have trended toward intermediate systems with masked intervention or floating systems with little or no intervention.

Differing from Ghosh et al. (2003), Rogoff et al. (2003) used *de facto* classifications to evaluate exchange rate systems in various countries from 1973 to 1999. The classification was differentiated into six groups, namely hard peg, other peg, limited flexibility, managed floating, free floating, and free-falling. The results revealed significant differences between the official stance and the actual system applied, with the results for only around half of the countries the same using the two approaches. The differences became even more stark when classifying the floating system, with only 20% of the sample countries actually applying a free-floating system, while 60% adopted hard-peg to managed floating systems, and the other 20% were in a state of free fall.[10] On the other hand, all *de jure* hard peg exchange rate systems were indeed hard peg systems and around 60% of the countries applied limited flexibility and managed floating systems as per the official stance.

The study also analyzed exchange rate system bipolarization, which is often found in the literature. The study showed that since the middle of the 1990s, numerous EMEs have been expected to shift to either end of the spectrum, either hard peg or floating. The growth of hard peg and free-floating exchange rate systems in the study supported the prevailing view of exchange rate system bipolarization. Nonetheless, the trend has not been accompanied by a reduction in the number of intermediate systems, a phenomenon known as "hollowing of the middle." Meanwhile, several countries opted for free-floating systems (South Korea, Thailand, Indonesia, and South Africa) or pegged systems (Argentina and Malaysia), while many others preferred intermediate systems (Brazil, Peru, Poland, Russia, and Venezuela). In addition, the transition from a *de facto* pegged system in the 1990s tended to favor a soft peg (China, Egypt, and Jordan) rather than a hard peg.

The *de jure* and *de facto* classifications raised an interesting question about whether countries that officially embraced a floating exchange rate system were actually applying a floating system. To that end, Calvo and Reinhart (2002) analyzed the behavior of exchange rates, reserve assets, monetary policy, interest rates, and goods prices between 39 countries using monthly data from January

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[10]The free-falling exchange rate regime is used to denote periods when an exchange rate experiences large-scale depreciation due to unfavorable macroeconomic dynamics, similar to what occurs prior to a crisis episode.

1970 to November 1999, which illustrated around 154 different exchange rate systems. The results implied that countries officially adopting a floating system actually had a fear of floating. Relative to the countries more committed to floating systems – such as the United States, Australia, and Japan – exchange rate volatility observed in other countries was very low. In fact, nominal and real economic shocks in such countries were larger than those found in the United States and Japan. In other words, low exchange rate volatility was more attributable to the stabilization policies instituted by the relevant authorities.

The analysis also looked at the magnitude of reserve assets, which should be smaller under a flexible exchange rate regime. Greater interest rate variability (real and nominal) showed that the authorities were not entirely reliant on foreign exchange market intervention to stabilize exchange rates, a problem also caused by a lack of central bank policy credibility. Monetary policy was also more variable, such as interest rate variability, indicating procyclical monetary policy to stabilize exchange rate fluctuations. The empirical evidence also showed that prices (in the domestic currency) fluctuated more, reflecting that authorities only allowed limited exchange rate adjustments to absorb real sector shocks. In fact, many countries showed no strong correlation between commodity prices and the exchange rate, congruous with the view that the exchange rate does not permit adjustments in response to terms of trade (ToT) shocks.

Another factor that drove monetary authorities to stabilize exchange rates was the fact that comparatively large exchange rate depreciation could be linked to recessions in EME due to the rapid increase in exposure to external obligations accompanied by the depreciation. The adjustment process in the current account to exchange rate fluctuations was also shown to be more significant in EME as a result of the relatively large import content in their export industry. Access to the international credit market is also affected by exchange rate volatility. Similarly, exchange rate stabilization is also driven by the desire for monetary policy to control inflation.

Another question raised in the classification study was whether economic performance in one country was affected by the choice of exchange rate regime adopted. This question was addressed by Rogoff et al. (2003) by investigating several aspects, especially inflation, economic growth, and developmental level (developing, emerging, or advanced) of the country involved. The study concluded that exchange rate flexibility would be more beneficial to countries with greater access to international capital markets and could also help build healthy financial systems. Furthermore, the economic improvements would be larger in countries that applied sound and consistent macroeconomic policy.

4.4.3. Exchange Rate Crises: Causes and Impacts

As more countries have adopted more flexible exchange rate regimes, exchange rate volatility has increased along with vulnerability to exchange rate crises. Consequently, crises in several EME have attracted the attention of economists offering a number of theories regarding the causes and impacts of such crises. The first-generation crisis models (Flood & Garber, 1980; Krugman, 1979) explained exchange rate crises from the perspective of differences between government behavior, assuming inconsistent policy, and private investor behavior, with access to adequate information and acting rationally. For example, under a fixed exchange rate regime (and assuming money supply is controlled), excessive fiscal expansion would deplete reserve assets and encourage investors to launch a speculative attack and, as foreign exchange reserves became more depleted, could trigger a crisis. On the other hand, under a flexible exchange rate system (and assuming no change in reserve assets), excessive fiscal expansion would increase money supply and lead to uncontrolled inflation. Such conditions, if allowed to persist, would exacerbate exchange rate depreciation and, as investor confidence declined, could trigger a crisis. Although simple, the model reminds us that rational investor behavior and confidence in economic fundamentals can change into a speculative attack on the exchange rate due to imprudent policy instituted by the government.

The second-generation models tried to explain crises in terms of changes in market perception to government policy, which subsequently becomes a self-fulfilling crisis. The conditions that cause self-fulfilling crises depend on the specific conditions found in a particular country. Referring to the attack on the exchange rate mechanism (ERM) at its inception when currencies in Europe were unified, Flood and Garber (1984) as well as Obstfeld (1986), explained that crises can occur if the markets perceive (perhaps erroneously) the government to persevere with expansive monetary policy after a speculative attack on a fixed exchange rate system. Similarly, Calvo and Mendoza (1995) showed that the 1994 crisis in México was consistent with the idea that issuing short-term securities and anticipating a government bail-out of a weak banking system would be vulnerable to changes of market perception. Similar dynamics also explained the 1997 Asian crisis, which was triggered by market panic stoked by high levels of short-term private external debt, with the risk of a currency mismatch and maturity mismatch likely, thus undermining confidence in the government maintaining exchange rate stability.

Both crisis models above emphasized government policy credibility that was not aligned with market perception. Nonetheless, several crisis models that have been developed since have tended to highlight private sector behavior on the international financial markets. The first approach maintained the rational behavior of investors, but stressed the reality that capital markets in EME cannot offer comprehensive financial products and that investors do not have access to the same information. Limited financial products on the market cause suboptimal investment portfolios due to inefficient price setting, inadequate liquidity, and a lack of diversity. Limited information on financial products, economic dynamics, and government policy can cause exposure to international investor perception. Such product and information distortion could spark a crisis if unexpected changes in conditions occurred and/or if investor confidence in the government's ability to stabilize the financial markets was lost.

The second approach emphasizes differences in the behavior of various investor types and attempts to interpret the data to illustrate such behavioral changes. This analysis is often linked to the types of capital flows in the BOP, namely direct capital investment, bank loans, and portfolio investment, which is then broken down as long or short term to observe the differences in investor behavior. The goal is to analyze how different types of investors react to changes in financial product yields and risks as well as domestic and international economic developments, and then sorts the investors into groups, including real money and noise traders. In other words, this approach focuses on the reactions of noise traders to securities that oftentimes spurs high volatility in short-term capital flows and tends to be speculative. Such investors and private capital flows more commonly spark off a crisis than other types of investors.

Crisis analyses according to the models above are important in terms of understanding not only the contributing factors but also the policy alternatives required. In that context, it is important to note that previous crises in EME, particularly in terms of BOP crises and banking crises, had a deleterious effect on economic performance, while incurring huge costs for the government and public. The cost of restructuring the banking sector can exceed 20% of GDP and the loss of output after a crisis can reach around 14%.

Deeper analysis of the contributing factors and crisis fallout such as this is crucial to formulate a policy response that can prevent and/or minimize the impact of similar crises. Referring to the crisis models above, in addition to acknowledging the importance of strengthening economic fundamentals and the financial system, the various policies required for crisis prevention include the significance of government policy consistency, information transparency to avoid self-fulfilling investor expectations, domestic financial market deepening or development, the importance of understanding the difference between short-term noise traders and long-term real money investors, external debt controls, and foreign exchange flow management to control foreign capital flow volatility. Furthermore, the development of systems and mechanisms to prevent and mitigate crises that can detect existing fragilities in the economy early as well as the availability of a financial system safety net are crucial in order to remain prepared and able to institute the policy measures required.

4.4.4. Exchange Rate Stabilization Policy: Foreign Exchange Intervention

The discussions in previous sections showed that exchange rate fluctuations in the near term tend to create misalignment from the fundamental value, accompanied by higher volatility. Consequently, the authorities in many EME have adopted exchange rate stabilization measures for numerous reasons, a phenomenon known as fear of floating in countries that apply a floating exchange rate regime. In other words, foreign exchange market intervention is not something that can be eliminated from emerging economies. Specifically, foreign exchange market intervention can be used, as part of monetary policy, to stabilize market expectations, alleviate market shocks, and control undesired exchange rate fluctuations from short-term economic shocks. Intervention is also available in line with policy to unwind macroeconomic imbalances, such as the current account deficit. To that end, intervention can support efforts to maintain macroeconomic stability by minimizing excessive exchange rate fluctuations, when there is credible commitment to the macroadjustments required.

Theoretically, the impact of foreign exchange market intervention on exchange rate stabilization can be explained using the portfolio balance model or the impact of policy signals on the market (Sarno & Taylor, 2001). In accordance with the portfolio balance model, foreign exchange intervention will influence the investor composition between domestic and foreign financial asset portfolios. Therefore, the exchange rate will be affected because changes in the portfolio will influence the yield and value of financial asset portfolios denominated in the home currency held by foreign investors. Foreign exchange intervention can also signal the market regarding the central bank's stance to the exchange rate fundamental value and the policy direction pursued. Such signals, coupled with order flows per the market microstructure above, will also influence exchange rate fluctuations on the market in the near term.

Chutasripanich and Yetman (2015) studied foreign exchange intervention strategies and their effectiveness. In general, intervention can correct exchange rate misalignment from the fundamental value or help to lean against the wind. Intervention effectiveness was evaluated based on five criteria: exchange rate stabilization, fewer current account imbalances, reducing speculation, minimizing reserve assets, and curtailing intervention costs. This study showed that intervention that can reduce exchange rate volatility can also lower the risk of speculation. Nonetheless, uncertainty concerning the fundamental value will undermine the effectiveness of intervention. Leaning against the wind that is not based on an assessment of exchange rate fundamentals could lower exchange rate volatility but would also exacerbate exchange rate misalignment. Furthermore, the cost of foreign exchange intervention will be higher is exchange rate fluctuations are caused by inconsistent interest rate policy. Regarding which approach is most effective, whether transparent or not, will depend on the desired objectives. If exchange rate stability is the goal, leaning against the wind would be optimal. If the aim is to reduce misalignment, however, the most effective approach would be transparent targeted policy. Nonetheless, both approaches would reduce speculation. Furthermore, more aggressive foreign exchange intervention by the central bank would erode the reserve assets and, therefore, reduce the effectiveness of the intervention itself.

Mohanty and Berger (2015) conducted a survey of foreign exchange intervention practices by central banks in Asia and the Pacific. Although the goals of monetary and financial system stability remained the primary consideration, a change has occurred since the GFC, namely to mitigate the economic risks. Most intervention aims to prevent currency speculation, while alleviating the impact on inflation and capital flow volatility. The survey also showed that the target of intervention is to curb exchange rate volatility rather than achieve a specific target. The basic strategy of intervention has not changed much, namely: monitoring information on the position of international investors, focusing on the most liquid market segments, and the proclivity for masked intervention to maximize the effect. Most central banks believe that intervention is effective in terms of achieving the desired exchange rate goals, although differences emerge concerning the magnitude and duration. In terms of which channel is effective, most central banks believe that intervention works best through the signaling channel. The success of intervention is also supported by macroprudential policy and capital controls.

Does intervention, in practice, support exchange rate stabilization in the wake of the GFC? Blanchard, Adler, and Filho (2015) conducted an empirical study using a sample of 35 EME for the period from 1990 to 2013. The results showed that, consistent with the predictions of the portfolio balance channel, intervention in the form of buying foreign exchange created less exchange rate appreciation during influxes of foreign capital flows after the GFC. Furthermore, the disparity between interveners and floaters was found to be large, namely approaching 1% of quarterly GDP (0.25% of annual GDP) would result in 1.5% lower exchange rate appreciation. The positive outcome would persist for three to four quarters before fading. Similarly, gross capital inflows reacted the same or more notably to intervention by interveners rather than floaters, showing the effect of exchange rate appreciation on lower capital inflows. No evidence was presented, however, for differences in interest rate behavior between interveners and floaters, indicating that central banks did not use the interest rate to mitigate capital flows.

Fratzscher, Gloede, Menkhoff, Sarno, and Stöhr (2015) presented further empirical evidence using daily data for intervention conducted in 33 countries from 1995 to 2011. Effectiveness was measured based on two motives for the intervention: to achieve the desired exchange rate direction (moving) or just to reduce exchange rate volatility (smoothing). The assessment distinguished between flexible and narrow band exchange rate system. The results showed that, for flexible exchange rate systems, the success rate was around 60% for moving and 80% for smoothing. Nonetheless, the success rate of moving could be improved to 80% if the intervention was large. Meanwhile, for narrow band exchange rate systems, the success rate was higher at 84% for moving. The effectiveness of intervention could be improved for both groups, however, if accompanied by oral intervention, especially during periods of excessive exchange rate pressures. In general, the empirical evidence showed that central banks in various countries have successfully improved the effectiveness of their foreign exchange intervention to influence exchange rate fluctuations on the market.

Nevertheless, it is important to note that intervention is not itself an instrument because the effectiveness depends on the consistency of the exchange rate desired through general macroeconomic policy (Canales-Klijenko, 2003). The policy trilemma mentioned above provides a valuable lesson about how perfect capital mobility prevents exchange rate policy and monetary policy from being instituted independently. Intervention will be ineffective if exchange rate fluctuations are due to persistent macroimbalances. Large-scale and long-term intervention is clearly unsustainable and such conditions require a change of exchange rate policy or other macroeconomic policies to improve the external balance. Similarly, a financial sector crisis, particularly a banking sector crisis, or capital flight would necessitate adjustments to several policy aspects, including the exchange rate, banking, monetary, and even fiscal policy to recover the external balance.

4.4.5. Exchange Rate System and Policy in Indonesia

As mentioned previously, the most appropriate exchange rate system is not only determined by the economic and financial dynamics specific to a particular country, but also the implications on monetary policy and foreign exchange flow management (foreign capital flows). Consistent with the monetary policy trilemma concept of an open economy, if a country applies a fixed exchange rate system, foreign capital inflows/outflows will occur that influence monetary policy effectiveness, while the reverse is true for a floating exchange rate system. If capital controls are applied, however, capital mobility to and from the country will be less and, thus, support more effective monetary policy, and vice versa for a free-floating exchange rate system. The monetary policy trilemma also implies that monetary policy effectiveness depends on the exchange rate and foreign exchange systems in place. Exchange rate policy in Indonesia, in addition to supporting sustainable economic development, also supports monetary policy effectiveness.

In the history of the Indonesian economy, a fixed exchange rate system, managed exchange rate system and floating exchange rate system have all been used. The fixed exchange rate system was applied from 1973 to March 1983.[11] A tightly managed exchange rate system was applied from March 1983 to September 1986. During that period, the Government introduced rupiah devaluation policy as follows:

(1) devaluation in November 1978 from Rp425 per USD to Rp625 per USD;
(2) devaluation in March 1983 from Rp625 per USD to Rp825 per USD; and
(3) devaluation in September 1986 from Rp1,134 per USD to Rp1,644 per USD.

Thereafter, a more flexible managed floating exchange rate system was implemented in Indonesia from September 1986 to January 1994 and with intervention bands from January 1994 to August 1997. During this period, the following exchange rate policies were introduced in the country:

(1) BI released the exchange rates each day at midday.
(2) The intervention bands were expanded eight times, namely from Rp6 (0.25%) to Rp10 (0.5%) in September 1992, to Rp20 (1%) in January 1994, to Rp30 (1.5%) in September 1994, to Rp44 (2%) in May 1995, to Rp66 (3%) in December 1995, to Rp118 (5%) in June 1996, to Rp192 (8%) in September 1996 and to Rp304 (12%) in July 1997.
(3) BI intervened on the foreign exchange market to ensure the rupiah exchange rate remained within the intervention bands set, through the purchase of USD when the exchange rate approached the floor of the intervention band and through the sale of USD when the exchange rate approached the ceiling of the intervention band.

[11]Prior to 1973, Indonesia applied a multiple exchange rate system through the Proof of Export System from 1957. The associated regulations contained trade and foreign exchange flow restrictions. Then, in 1967, the Export Bonus System was introduced that aimed to stimulate exports by permitting the public to freely transfer or trade export proceeds on the market at constantly changing prices (floating exchange rate system).

A floating exchange rate system was introduced in Indonesia on August 14, 1997, which remains in place to the present day. The new exchange rate system was introduced in response to government policy to confront speculative attacks in the foreign exchange market from July to August 1997. Demand for foreign exchange was high to meet international obligations, due to large private sector external debt in Indonesia, and for speculative attacks from certain domestic and foreign parties seeking to exploit exchange rate shocks for personal gain. Consequently, BI could no longer keep pace with such high demand for foreign exchange, after eroding the reserve assets held, to sustain the managed exchange rate system. If the managed exchange rate system was maintained, the already lower position of reserve assets stoked concerns that foreign exchange reserves could dry up. Several neighboring countries, including South Korea and Thailand, also introduced a floating exchange rate system.

By applying a floating exchange rate system, the rupiah began to move in line with market supply and demand and was relatively more stable (less sharp fluctuations) in accordance with economic fundamentals. Nonetheless, the exchange rate was often influenced by volatility non-economic domestic factors, particularly during the early part of the recovery after the devastating Asian financial crisis of 1997/1998. Attempting to stabilize the rupiah, BI intervened at specific moments on the foreign exchange market, namely when faced with excessive currency shocks and/or when exchange rate fluctuations were expected to pass through to inflation. It was not the intention of BI, however, to use foreign exchange market intervention to achieve a specific exchange rate target on the market.

The exchange rate regime in Indonesia is regulated by Act No. 24 of 1999 concerning foreign exchange flows and the exchange rate system. Pursuant to that law, the exchange rate system in Indonesia is determined based on the recommendations put forward by BI, considering that any change to the regime would entail considerable ramifications, not only in terms of monetary policy and the financial sector, but also real economic activities, such as investment and international trade. Consequently, any change to the exchange rate system must be based on established ways of thinking and backed up by robust research that pays due consideration to the economic, political, and social aspects. In this context, BI is required to submit its recommendations concerning any planned changes to the existing exchange rate regime, due to its experience and knowledge in this area as well as the influence on monetary, banking, and payment system policy.

BI's implementation of exchange rate policy is not only stipulated in Act No. 24 of 1999, but also in more detail through the BI Act No. 23 of 1999, which has been amended several times, most recently by Act No. 6 of 2009. According to Article 7 of the BI Act, the mandate of overarching purpose of BI is unequivocally specified, namely to maintain rupiah (price) stability, which also contains provisions for exchange rate policy. Therefore, pursuant to Article 12, BI is required to implement exchange rate policy in line with the exchange rate system adopted as follows:

(a) under a fixed exchange rate system through devaluation or revaluation of foreign currencies;
(b) under a floating exchange rate system through market intervention; and
(c) under a managed floating exchange rate system through daily exchange rate determination and adjusting the intervention bands.

When maintaining rupiah stability, Bank Indonesia sets exchange rate policy based on the floating exchange rate system by implementing various strategies to manage the exchange rate, including, but not limited to, OMO. The elucidation of Article 10, paragraph (1), letter b, point 1 of the BI Act also states that OMO include foreign exchange market intervention by Bank Indonesia to stabilize the rupiah.

4.5. Concluding Remarks

This chapter has discussed, in depth, monetary policy and theory in an open economy, particularly through the exchange rate and its impact on the economy. *First,* the theoretical and empirical studies demonstrate that exchange rate fluctuations are influenced by fundamental factors, such as money supply, the interest rate, BOP and economic growth, as well as foreign exchange market microstructure dynamics, specifically order flows and information heterogeneity. Although long-term exchange rate dynamics can be explained by fundamental factors, short-term developments are extremely volatile and difficult to predict. Therefore, the exchange rate not only influences inflation and growth, but the volatility can trigger monetary and financial system instability.

Second, the exchange rate correlates positively with the trade balance and, therefore, economic growth. Theoretically, this occurs if the Marshall–Lerner condition is met, such as if export elasticity to the real exchange rate exceeds the corresponding import elasticity. Empirically, however, that positive exchange rate effect can be observed to increase in countries where the export structure is dominated by competitive and hi-tech commodities, as part of the international trade chain, and the domestic industry structure can reduce dependence on imports. The use of exchange rates to boost competitiveness and, thus, improve the trade balance and stimulate growth is often a polemic in international economic politics, which has manifested in the ongoing debate on currency wars. Nonetheless, solid competitiveness to support trade and growth is required, through improvements to the investment climate, infrastructure and structural reforms in the real sector.

Third, exchange rate dynamics directly and indirectly impact price stability and, therefore, domestic economic growth. The empirical findings in many countries have evidenced a dwindling exchange rate effect on inflation as a positive consequence of post-globalization international trade as well as monetary policy credibility. A more significant decline has been found in countries that apply an ITF but the central bank must still pay due consideration to the influence of the exchange rate on inflation, especially the asymmetric nature of the impact, meaning that the influence is stronger during periods of high depreciation.

Consequently, the central bank must consider the impact of the exchange rate when formulating monetary policy to achieve low inflation, while also considering economic growth.

Fourth, exchange rate volatility in many countries increased in the wake of the GFC, with capital flow volatility the inauspicious outcome of expansive monetary policies to stimulate economic recovery in advanced countries. In fact, more volatile exchange rates and capital flows have complicated policy response formulation in EME to maintain domestic monetary and financial system stability. The policy of foreign exchange market intervention to stabilize exchange rates has been widely adopted in EME. Although the empirical data points to a relatively high level of effectiveness, exchange rate volatility is still a risk that demands the attention of central banks. Intervention also becomes more effective if supported by policy to manage capital flows and macroprudential policy to maintain financial system stability.

Understanding the role of the exchange rate and monetary policy, as explained in this chapter, is crucial for the discussions in subsequent chapters. Monetary policy transmission effectiveness and the mechanisms to influence the economy through the exchange rate channel, money supply, the interest rate channel and bank credit channel are discussed in Chapter 5. Thereafter, the strategic and operational framework of monetary policy set by central banks is discussed in Chapters 6 and 7. The two chapters underlie the discussion in Chapters 8 and 9 on the ITF, as applied by many central banks, Indonesia included. Finally, foreign intervention supported by capital flow controls to address the policy trilemma in an open economy, as well as the linkages with monetary policy in the central bank policy mix, is discussed in Chapters 14 and 15.

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Chapter 5

Monetary Policy Transmission Mechanism

5.1. Introduction

The Monetary Policy Transmission Mechanism (MPTM) is always an important and interesting topic in terms of the theory of monetary economics and its practice at central banks. Like a roadmap, MPTM illustrates the process of how the monetary policy implemented by the central bank influences various economic and financial activities to achieve the desired target, namely price stability and economic growth (Taylor, 1995; Warjiyo, 2004).[1] For the central bank, understanding MPTM is crucial when determining the monetary policy stance, the instruments as well as the timing and magnitude of the measured and apposite response. For the financial community, corporate sector, and public, understanding MPTM is required to anticipate and consider the effect of monetary policy on their economic decisions. For academia, MPTM issue poses a challenge to the underlying theoretical thinking and empirical studies.

MPTM represents a relatively complex process involving interactions between the central bank, financial sector and economic players as well as the policies instituted by the government and other relevant domestic and international authorities. During the two decades leading up to the GFC, for instance, monetary policy credibility in various jurisdictions contributed to lower inflation and stimulated economic growth (Berg et al., 2013; Taylor, 2014). The era of Great Moderation in the US, for example, was characterized by low inflation and robust economic growth for nearly two decades. Nevertheless, the US economic boom also facilitated product innovation in the financial sector, which became increasingly complex and disparate from the traditional intermediation function through deposits and credit, with a proliferation of derivatives that contained risks and for

[1]A detailed explanation of the monetary policy transmission mechanism and the empirical findings in Indonesia is presented in Perry Warjiyo, *The Monetary Policy Transmission Mechanism in Indonesia*, Central Banking Series No. 11, Centre for Education and Central Banking Studies, May 2004. Those unfamiliar with the topic are recommended to read this book before proceeding.

Central Bank Policy: Theory and Practice, 115–158

doi:10.1108/978-1-78973-751-620191008

which the transmission mechanisms were difficult to comprehend. Global financial integration has also accelerated and, hence, tightened the linkages in terms of financial transmission and international capital flows. The ongoing developments demand greater attention, particularly in understanding MPTM, to ensure greater monetary policy effectiveness.

This chapter explores MPTM, focusing on the periods before and after the GFC. In many ways, this chapter updates the description of MPTM in Indonesia presented by Warjiyo (2004) and its linkages with monetary policy (Warjiyo & Solikin, 2003) for the periods before and after the Asian Financial Crisis in 1997/98. The discussion is divided into six main sections. After the introduction, Section 5.2 presents the MPTM map as well as its relevance to monetary policy, interest rate policy, and monetary operations as the starting point of MPTM. Section 5.3 discusses the transmission channels based on the money view, while the credit view is examined in Section 5.4, including the influence of imperfections or financial frictions with the emergence of the risk-taking channel. Section 5.5 outlines the empirical studies of MPTM in several countries, including advanced countries, developing countries, and Indonesia, with an emphasis on the various theoretical factors discussed in the previous two sections, together with developing the economic and financial dynamics. Section 5.6 concludes the chapter with some important lessons from MPTM theory and empirical studies, while the implications for monetary policy at the central bank are presented in the subsequent chapters.

5.2. Monetary Policy and the Transmission Mechanisms

MPTM mapping strives to provide theoretical and empirical answers to two salient questions asked by Bernanke and Blinder (1992), namely: (1) how can monetary policy influence the real economy, in addition to the impact on prices; and (2) through which mechanisms is monetary policy transmitted to the economy? Both questions represent an important issue in the monetary policymaking at the central bank as well as in the theoretical discussions on monetary economics by economists.

5.2.1. MPTM Map

Initially, MPTM theory referred to the role of money in the economy, as explained by the quantity theory of money (Fisher, 1911). After further development, consistent with the advancement of non-bank financial sectors combined with increasing global financial integration, six monetary policy transmission channels emerged in the literature (Cecchetti, 1995; De Bondt, 2000; Kakes, 2000; Mishkin, 1996). The seven monetary policy transmission channels are the direct monetary channel, interest rate channel, exchange rate channel, asset price channel, credit channel, balance sheet channel, and expectations channel. Furthermore, observations of imperfections and financial frictions during the decade before and after the GFC compelled several economists (Allen & Carletti, 2008; Borio & Zhu, 2008) to view the risk-taking channel as another monetary policy transmission channel.

The workings of MPTM begin with monetary policymaking at the central bank through interest rate policy and other monetary instruments, such as monetary operations, foreign exchange intervention, reserve requirement, and other instruments (Fig. 5.1). Subsequently, the mechanism is influenced by activities in the financial sector and real economy through the various monetary policy transmission channels, namely the interest rate channel, exchange rate channel, asset price channel, credit channel, balance sheet channel, and expectations channel. Monetary policy is transmitted through two stages of interaction in the economy, namely: (1) interaction between the central bank and the banks as well as other financial institutions in terms of various financial sector transactions; and (2) interaction between the banks and other financial institutions with economic players in the real sector through financial intermediation concerning various domestic and international economic activities. The uppermost panel in Fig. 5.1 presents the various MPTM stages from the central bank, to the financial sector, real economic activities, and the ultimate target.

In the financial sector, monetary policy affects interest rates, exchange rates, bond yields, and stock prices, as well as the volume of private funds deposited in the banking industry, bank lending and placements in bonds, stocks, and other securities. Meanwhile, monetary policy influences the real economy through aggregate demand, including domestic demand (consumption and investment) as well as external demand (exports and imports). Furthermore, monetary policy also affects aggregate supply through the cost of production capital as well as the decisions on wages and the duration of work contracts. Ultimately, the magnitude of the gap between aggregate demand and aggregate supply determines the pace of economic growth and rate of inflation as the final targets of monetary policy. Additionally, monetary policy also influences inflation directly through the effect of the exchange rate on the prices of imported goods and services and indirectly through the current account as well as the capital and financial account in the balance of payments (BOP). The central panel of Fig. 5.1 illustrates the interactions between various financial and economic variables in the financial sector and real economy.

In reality, MPTM is a complex process and, therefore, is often known as the "black box" by monetary economists (Bernanke & Gertler, 1995). MPTM is predominantly influenced by three factors, namely: (1) behavioral changes in the central bank, government, and banking industry or among economic agents in terms of their economic and financial activities; (2) the lag from when the monetary policy is implemented by the central bank until the time when the effect manifests in terms of economic growth and the final policy target is achieved; and (3) a change occurs in the monetary policy transmission channels due to a change in (1) above in accordance with economic and financial developments in the country involved. The lower panel in Fig. 5.1 shows the determinants of MPTM beyond the control of the central bank, including changes in the risk premium, bank capital, global economy, fiscal policy, and commodity prices.

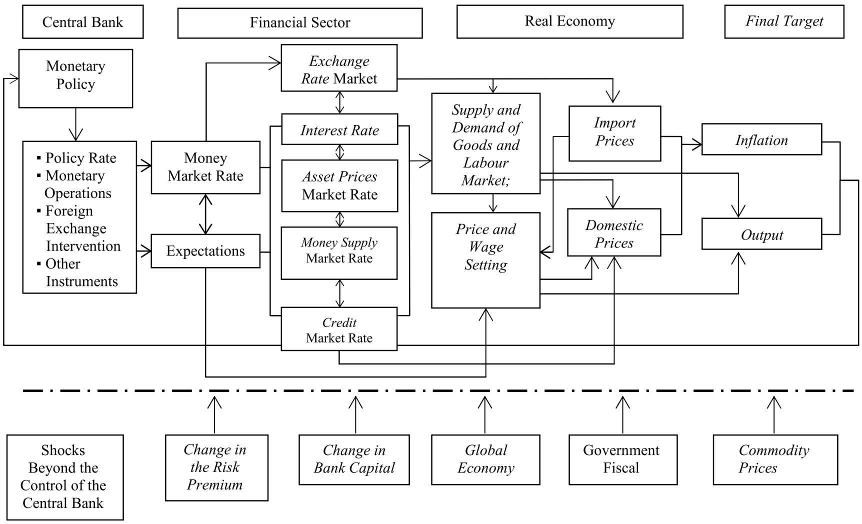

Fig. 5.1: Monetary Policy and the Transmission Mechanism.

5.2.2. The Importance of Monetary Policy Transmission

Behavioral changes in the central bank, government and banking industry or among economic agents will clearly affect the interactions of various economic and financial activities and, therefore, prompt a change in MPTM. Furthermore, due to changes in behavior and expectations, MPTM is replete with uncertainty and is relatively difficult to predict. If the central bank lowers the interest rate, for instance, bank behavior and balance sheet dynamics will conspire to lower the interest rates and adjust the volume of deposits and loans. Such interest rate policy would also influence financial market, exchange rate, and stock price expectations. Additionally, household behavior and balance sheet dynamics also influence how much interest rate policy affects deposits and demand for new bank loans. Similarly, a behavioral change in the banking industry, concerning financial operations and product innovation, such as a reluctance to lend or a proliferation of derivatives in foreign exchange transactions, would also determine how the monetary policy of the central bank influences various economic activities.

In general, the effect of MPTM on economic growth and inflation has a relatively long and varied lag (Friedman & Schwartz, 1963). The impact of monetary policy on economic growth and inflation typically takes six to eight quarters. During the preliminary stage in the financial sector, the impact of lowering policy rate on lending and deposit rates usually takes around three to six months because the banking industry must wait for existing term deposits and loan agreements with customers to mature. On the other hand, foreign exchange market and stock market respond more quickly through exchange rate depreciation and elevated stock prices, resulting in changes in the composition of the investment portfolio. Meanwhile, during the second stage, from the financial sector to the real economy, household, and corporate demand for new bank loans takes time to manifest. The corporate sector requires assurance that the business outlook is promising enough to pay loan installments to the banking industry.

The complexity of MPTM also stems from changes in the role and processes of monetary policy transmission channels in the economy. For economies which are still dominated by the banking industry, MPTM through the interest rate, money supply, and credit channels will play an important role. As financial markets become more developed, however, the role of the interest rate channel and asset prices, such as stocks and bond yields, become more important. Likewise, open economies with perfect capital mobility and a floating exchange rate regime facilitate a faster exchange rate channel than countries applying foreign exchange controls (imperfect capital mobility) and a fixed exchange rate regime. In an open economy, MPTM is also influenced by economic and financial developments in other countries, among others, through changes in the exchange rate, export and import volume, as well as the magnitude of capital inflows and outflows. The speed and scale of MPTM also differ during periods of economic boom and bust (asymmetric). Interest rates rise faster than they fall. Similarly, credit is more expansive (contractive) during an economic upswing (downswing), a phenomenon known as the financial accelerator.

MPTM complexity demands analysis and empirical studies to map the various existing transmission channels, including the money supply, credit, interest rate, exchange rate, asset price, or expectations channels, in line with economic and financial development. In terms of monetary policy and theory, MPTM studies generally review two salient aspects. *First,* to investigate the most dominant transmission channel in the economy to from the underlying base for monetary policy-making strategy. If the money supply channel is dominant, for example, a monetary policy strategy targeting money supply is still relevant. On the other hand, if the interest rate channel or exchange rate channel were dominant, the central bank would have to consider targeting the interest rate or exchange rate as the monetary policy strategy. In general, the exchange rate channel is important, specifically in the case of EME.

Second, to investigate the magnitude and duration of the lag of each respective transmission channel, from when the monetary policy decision is taken by the central bank, the effect through each respective transmission channel, to inflation and economic growth. Such analysis is important to determine the strongest economic and financial variables as leading indicators for inflation targeting and future economic growth, as well as to determine the operational target of monetary policy. Consequently, if money supply, credit, interest rate, and exchange rate have strong influences over future economic dynamics and inflation, those variables should be used as leading indicators. Of those leading variables, the most appropriate one is selected as the operational target, for instance the monetary base or interest rates. Furthermore, analysis of the lag profile of each transmission channel is also required to formulate a forward-looking monetary policy strategy. By understanding the lag of the MPTM, the central bank could better formulate forward-looking and pre-emptive monetary policy in the current period to steer future inflation and economic growth in line with the desired targets.

Comprehensive and in-depth understanding of the MPTM is vital in terms of reviewing the monetary policy-making practices of the central bank as well as to understand and develop monetary economics theory in academia. For the central bank, MPTM mapping facilitates monetary policy formulation in terms of setting the operational target (policy rate and monetary operations), the intermediate target (interest rate, money supply, credit, exchange rate, asset prices, and expectations) and the effectiveness of achieving the final target (inflation and economic growth). For academia, MPTM is an interesting topic for empirical and theoretical study, linked to the debate between monetary economists in line with the monetarists' viewpoint, credit, and even imperfections and financial frictions.

5.2.3. Policy Rate and Monetary Operations

The processes of MPTM begin with the monetary policy decision taken by the central bank. When formulating monetary policy, the central bank generally sets the interest rate to influence macroeconomic projections in line with the desired target, namely price stability and economic growth. Interest rate policy is implemented through monetary operations by the central bank to influence liquidity and, therefore, interest rates in the money market. In addition, the central bank also conducts monetary operations in the foreign exchange market when intervention is required to stabilize exchange rate fluctuations. Other monetary instruments include the deposit facility (DF), lending facility (LF), or discount window, reserve requirement and lender of last resort function using high-quality and liquid securities as collateral.

5.2.3.1. Interest Rate Policy. The interest rate policy adopted by the central bank will influence short-term interest rates on the interbank money market as well as market expectations of macroeconomic projections and the future direction of the central bank's interest rate policy. Such developments will subsequently affect bank deposit and lending rates, asset prices on financial markets, including stock prices and bond yields, the exchange rate and long-term interest rates. Using a simple mathematical approach, interaction between the policy rate and bank interest rates can be expressed as follows:

(1) transmission from the central bank interest rate (r_{CB}) to the interbank rate (r_P) is expressed as the following function: $r_P = f(r_{CB}$, other factors[2]);
(2) transmission from the interbank rate (r_P) to the deposit rate (r_D) is expressed as the following function: $r_D = g\ (r_P$, other factors); and
(3) transmission from the deposit rate (r_D) to the lending rate (r_K) is expressed as the following function: $r_K = h\ (r_D$, other factors).

Confirming previous findings, the transmission process typically contains a lag, primarily due to the maturity structure as well as the loan and deposit agreements between the banks and their customers. Consequently, the reduced form

[2]Other factors generally relate to internal conditions in the banking sector.

equation to investigate the transmission of the policy rate (r_{CB}) to lending rates can be expressed as follows[3]:

$$r_k = \alpha + \sum \beta_j r_{CB,t-j} + \sum \gamma_j Z_{j-t} + \varepsilon_t \tag{1}$$

where Z = internal bank conditions, such as deposit composition, liquidity (loan-to-deposit ratio – LDR), capital (capital adequacy ratio – CAR), non-performing loans (NPL), and so on, and ε_t = random shocks in the econometric equation.

5.2.3.2. Monetary Operations. The central bank performs monetary operations to influence liquidity conditions and, therefore, interest rates on the interbank money market in line with the policy rate set. To that end, the central bank projects bank liquidity conditions (bank reserves) and base money on a weekly basis as a guideline for monetary operations. Liquidity on the interbank money market is monitored, while base money projections are made using a central bank balance sheet equation as follows:

$$\Delta B = \Delta \mathrm{NFA} + \Delta \mathrm{NCG} + \Delta \mathrm{NCP} - \Delta \mathrm{NOI} \pm \Delta \mathrm{OPT} \tag{2}$$

where ΔB = change in base money, ΔNFA = change in net foreign assets, ΔNCG = net claims to government, ΔNCP = net claims to private sector,[4] and ΔNOI = net other items. In Equation (2), the magnitude of weekly monetary operations (ΔOPT) represents the difference between the expected position of base money and the policy rate set.

When projecting base money, the central bank must consider whether the total money supply in the economy is adequate. In general, the procedure is as follows. *First,* total money supply (M1 and M2) is projected annually in line with the requirement of the real economy based on the inflation target set, economic growth predictions, and the estimated income velocity in the economy. This can be achieved simply by forecasting the demand for money by the public, for instance, using the following function:

$$M / P = h(y, r, \Delta s) \tag{3}$$

where y = economic growth, r = interest rate and Δs = change in the exchange rate. *Second,* using the projection of money supply and estimation of the money multiplier, the position, and growth of base money can be predicted on an annual, quarterly, monthly, and weekly basis using the following equation:

$$m = M / B \tag{4}$$

[3]The time lag is a characteristic that must be monitored to understand monetary policy transmission. This is applicable to various equations mentioned in this chapter. To simplify understanding, the lag is not always mentioned in the equations.

[4]If applicable.

The weekly projection of base money is subsequently used as a guideline for monetary operations at the central bank to control liquidity and, therefore, interbank rates in line with the policy rate determined by Equation (2).

It is important to note that differing from monetary policy using the interest rate as the operational target, as outlined above, monetary policy targeting money supply can also be achieved through monetary operations with Equation (2) based on the projections of base money Equation (4) and money supply Equation (3). In practice, MPTM through the money channel such as this is generally applied by central banks in low-income countries with an underdeveloped financial sector. In this case, base money is controlled through Equation (2) in line with projection Equation (4), which is subsequently transmitted to total money supply (M1 and M2) congruent with public demand Equation (3) and, ultimately, money supply influences inflation and economic growth.

The effectiveness of monetary policy in targeting the monetary base depends on several factors. *First,* there is a stable and predictable correlation between total money supply and inflation and economic growth, reflected in the predictability of income velocity or money demand function stability. *Second,* there is a causal relationship from total money supply to economic growth and inflation but not vice versa. And *third,* there is a stable and predictable relationship between base money and total money supply, reflected in the predictability of the money multiplier. As the financial sector develops, however, not only do income velocity and the money multiplier become less stable and more difficult to predict, but an inverse causal relationship often emerges, namely from economic growth and inflation to total money supply. Such conditions impede the control of money supply as the basis of implementing monetary policy based on the money channel in many countries, including EME and advanced countries.[5]

In addition to the implications for monetary policy, the MPTM through the money channel is also one reason for restrictions on central bank lending to the government. As found in Equation (2), central bank lending to the government to finance the fiscal deficit would increase total base money and liquidity in the economy. On top of influencing interest rate policy effectiveness, this would also increase total money supply in the economy beyond demand, leading to uncontrolled inflation.[6]

5.2.3.2. Foreign Exchange Intervention. Exchange rate stability is very important for the economy, especially in EME and developing countries, therefore many central banks, including BI, pay a lot of attention to MPTM through the exchange rate channel. In this case, the exchange rate affects international trade and investment, such as exports, imports, foreign capital investment, portfolio

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[5]This is evidenced by the growing number of central banks that have switched from targeting money supply to interest rate targeting as the operational target of monetary policy.

[6]If the central bank reabsorbed its loans to the government, interbank rates would rise and, therefore, exacerbate investment, and economic growth. In this case, investment distortions would appear from the private sector to the government, which is often known as the crowding out effect in the literature.

investment, and external debt. Furthermore, the exchange rate and non-resident capital flows influence inflation and economic growth. The exchange rate also affects financial (balance sheet) conditions in the banking industry, household sector, and corporate sector, namely through foreign currency assets and liabilities. Moreover, wide currency fluctuations create economic uncertainty and could even threaten political stability. With greater economic openness, accompanied by a floating exchange rate system and perfect capital mobility, comes a larger influence of exchange rates and non-resident capital flows.

In addition to interest rate policy, the central bank also implements monetary policy through foreign exchange market intervention to stabilize exchange rates. Central bank intervention has a direct impact on the supply of foreign exchange and, therefore, exchange rate developments on the market. In this regard, there are several motives and targets for foreign exchange intervention, namely to control exchange rate volatility, prevent misalignment from the currency's fundamental value and to anchor future exchange rate expectations (Moreno, 2005; Neely, 2001). An empirical study concerning the targets of foreign exchange intervention could be conducted using the following equation (Adler & Tovar, 2014; Neely, 2005):

$$\mathrm{INT}_t = \alpha_0 + \alpha_1 \Delta s_{t-1} + \alpha_2 (s_{t-1} - s^*_{t-1}) + \alpha_3 \mathrm{VOL}_{t-1} + \alpha_4 \mathrm{INT}_{t-1} + \varepsilon_t \qquad (5)$$

where INT = foreign exchange intervention, Δs = changes to the exchange rate (appreciation or depreciation), ($s - s$*) = exchange rate misalignment from the fundamental value, and VOL = exchange rate volatility.

A change in the monetary policy rate would affect the interest rate differential and then the exchange rate due to the size of foreign capital flows beyond supply and demand in the foreign exchange market. As explored in Chapter 4, exchange rate transmission is shown through international interest rate parity, namely: (1) based on uncovered interest parity (UIP), $r - r^* = (s^e - s)/s$, or (2) based on covered interest parity (CIP), $r - r^* = (F - s)/s = (s^e - s)/s + \rho$, where {$r$, r^*, s, F, ρ} = {domestic interest rate, international interest rate, spot exchange rate, forward exchange rate, and risk premium}.

Several studies have also been conducted to measure the effectiveness of interest rate policy and foreign exchange market intervention in terms of exchange rate stability (Galati & Disyatat, 2005; Gerl, 2006; Miyajima & Montoro, 2013; Sarno & Taylor, 2001). For example, the following model can be estimated:

$$\Delta s_t = \beta_0 + \beta_1 (r_t - r_t^*) + \beta_2 \rho_t + \beta_3 \mathrm{INT}_t + \varepsilon_t \qquad (6)$$

where Δs = a change in the exchange rate, ($r - r$*) = interest rate differential, p = risk premium, and INT = foreign exchange market intervention.[7] In this case, β_3 measures the effectiveness of foreign exchange market intervention in terms of exchange rate stability.

[7]Empirically, intervention and changes in the exchange rate are mutually reinforcing and, therefore, Equations (5) and (6) contain heteroscedasticity. In general, GARCH is the estimation technique for the instrumental variables used in empirical studies.

5.3. Monetary Transmission Channels: The Money View

In monetary economics theory, a debate rages about how the MPTM affects the financial sector and real economy. Per the former theory, known as the money view (Cecchetti, 1995) or neoclassical channels (Boivin, Kiley, & Mishkin, 2010), monetary policy through short-term interest rates and/or base money control influences consumption and investment through long-term interest rates, asset prices, and the exchange rate. A financial system is assumed to be efficient and, therefore, there is no need to discuss the banks' role in the MPTM. Per the latter theory, however, which is often referred to as the credit view or the non-neoclassical channels, financial system imperfections affect the MPTM in the economy through the credit channel and balance sheet channel. Furthermore, the implications for monetary policy are also different. The assumption of financial market efficiency per the money view implies that monetary policy does not have a distributive effect on the various segments of the real economy along with price setting. In contrast, financial frictions per the credit view cause the impact of monetary policy to vary by segment in the financial sector and real economy, depending on bank behavior and internal conditions, the lending relationship between lender and borrower, as well as risk-taking behavior among economic players.

The workings of the MPTM per the money view on investment, consumption and international trade were developed based on the theories that emerged in the middle of the twentieth century, such as the neoclassical investment theory (Jorgenson, 19963; Tobin, 1969), the lifecycle and permanent income models (Ando & Modigliani, 1963; Brumberg & Modigliani, 1954; Friedman, 1957), as well as the international IS/LM analysis models (Fleming, 1962; Mundell, 1963). The MPTM works through the interest rate channel, asset price channel, and exchange rate channel. In this case, monetary policy affects investment via the interest rate channel in terms of the cost of capital and the asset price channel in terms of the value of the firms, which is also known as "Tobin's *q* channel." The influence on consumption occurs through the interest rate channel due to the intertemporal substitution effect and through the asset price channel because of the wealth effect. Meanwhile, the exchange rate channel facilitates direct and indirect exchange rate pass-through to inflation and the international trade channel through changes in the relative prices of exports and imports.

5.3.1. Interest Rate Channel

The most commonly cited MPTM channel in macroeconomic modeling involves the influence of the interest rate on corporate and household propensity to invest, including investment in fixed assets, such as land and buildings, as well as durable assets, such as motor vehicles, machinery, and equipment. Therefore, macroeconomic models take into account the influence of the interest rate on private consumption.

Transmission through the cost of capital on investment. The standard neoclassical investment model shows that the user cost of capital is a key determinant that affects demand for investment. The user cost of capital, μ_c, is usually expressed by the following equation:

$$\mu_c = p_c[i-(E(\pi_c)-\delta)] \qquad \text{with} \qquad E(\pi_c)=\sum\nolimits_{i=0}^{\alpha}\beta_t E_t(\Pi_{t+i}) \tag{7}$$

where p_c = relative price of new capital asset (compared to the replacement), i = nominal interest rate, $E(\pi_c)$ = expected capital asset price hike, and δ = rate of depreciation. The equation shows that the external finance premium represents the interest rate cost to obtain new capital minus the increase in the value of the capital asset after depreciation.

In terms of corporate investment, the expected capital asset price hike may represent the current market value of the asset or profit rate (II) or current and future cash flow. In this case, firms or households will only take on new investments using bank loans if the user cost of capital is negative. Consequently, expected capital asset price hike, after depreciation, would exceed the interest payments of the loan received. Therefore, the investment demand function can be expressed as follows:

$$I^d = g(E(Y),\mu_c,Z) \tag{8}$$

where I = investment, $E(Y)$ = economic growth outlook, μ_c = user cost of capital in Equation (7), and Z = vector of other variables that affect investment, such as business risk, ease of licensing, level of competition, and so on. Demand for investment would increase if the cost of capital was low or decreasing and the business outlook was promising.

Several factors demand attention when analyzing the effect of monetary policy on investment through the interest rate channel. *First,* the long-term interest rate has a greater effect on investment, considering the decision to invest is long term and, therefore, firms will consider interest rate fluctuations and expected increases in asset prices over the same horizon as the investment. *Second,* monetary policy is determined by and works through short-term interest rates, therefore, MPTM effectiveness through the interest rate channel is also affected by the term structure of interest rates, which is linked to short- and long-term interest rates. When the policy rate is reduced, for instance, the long-term interest rates also decrease due to price rigidity in the near term. Nevertheless, this only occurs if the financial markets can form the term structure of interest rates and if the central bank's monetary policy is credible and, hence, able to anchor the expectations of economic players in terms of their business decisions.

Transmission through the substitution effect on consumption. Interest rates affect consumption through the intertemporal substitution effect. In other words, interest rates influence household preferences or decisions about whether it would be better to increase consumption in the current period or postpone the increase in consumption until a future period. This can be expressed using the following equation:

$$C = c(Y^d,r_D,r_K) \tag{9}$$

where C = consumption, Y^d = disposable income, r_D = term deposit interest rate, and r_k = consumer loan lending rate. If the policy rate was lowered, for instance, lending rates would also be reduced, which would therefore drive household consumption. This would depend on the respective preferences and financial conditions of each household, namely as savers or borrowers. In Equation (9), in addition to the impact on lending rates, the central bank's interest rate policy would also affect deposit rates and, therefore, interest income for the savers. The income effect and substitution effect determine the influence of the interest rate channel on consumption.

5.3.2. Asset Price Channel

Monetary policy, implemented through interest rates, monetary operations, and foreign exchange intervention or other instruments, also affects other asset prices, including financial asset prices, such as bond yields and stock prices, as well as physical asset prices, particularly property prices and gold. This transmission occurs because funds in the investment portfolio are not only in the form of term deposits held at a bank and other investment instruments on the rupiah money market and foreign exchange market, but also as bonds, stocks and physical assets. Therefore, changes in the interest rate and exchange rate as well as the magnitude of investment on the rupiah money market and foreign exchange market would affect the volume and prices of bonds, stocks and physical assets.

For financial assets, the basic theory applied is investment portfolio formation by the investor, à la Markowitz model, as an alternative investment in central bank instruments with a risk-free interest rate of r_{CB}, bank term deposits, bonds and stocks as follows:

$$\text{Max}\, U(C, A) \text{ with constraint}: \qquad \sum p_i C_i = W_0 + \sum r_i / \sigma_i A_i \tag{10}$$

where $\{U, C, A, W, p, r, \sigma\}$ = {utility, consumption, assets, wealth, consumer goods prices, asset yield, asset risk}. From the investment portfolio formation, the demand solution for each respective financial asset in equilibrium produces a yield relationship for each respective financial asset after calculating the risk factor as follows:

$$r_B / \sigma_B = f(r_{CB}, r_D / \sigma_D, d_S / \sigma_S) \tag{11a}$$

$$d_S / \sigma_S = g(r_{CB}, r_D / \sigma_D, r_B / \sigma_B) \tag{11b}$$

where r_D and σ_D = term deposit rate and risk, r_B and σ_B = bond yield and risk, d_S and σ_S = stock yield and risk. Equation (11) shows that the central bank interest rate affects the yields of bonds and stocks.

Transmission through Tobin's q on investment. The investment decisions of firms and households can be analyzed using Tobin's q ratio. For business investment, Tobin's q is defined as the ratio of the market value of a company's assets divided by the replacement cost of the company's assets. A high Tobin-*q* value, for instance during an economic upswing, when the company's market value is high and/or the policy rate is lowered, thereby reducing the replacement cost, implies that corporate demand for investment would increase. Companies would, therefore, expand bank borrowing or issue stocks or bonds to finance the investment. A similar analysis could be applied to household investments, for instance property, because of rising asset prices as lending rates fall. Tobin-*q* analysis, which emphasizes transmission through the asset price channel, may be linked to the user cost of capital approach, which emphasizes transmission through the interest rate channel as explained previously.

Transmission through the wealth effect on investment. Rising asset prices could also drive consumption due to an increase in net worth. The lifecycle hypothesis of saving and consumption shows that current spending on consumption is determined by the resources at hand to the consumer over his/her lifetime, encompassing not only future income but also net worth, such as property, stocks, and other assets. A reduction to the policy rate would stoke demand for assets, such as property and stocks, thereby edging up asset prices. Similarly, a reduction in the interest rate means a lower discount rate, hence elevating the current value of income flows and returns on investments in stocks, property and other assets received at a future date. An increase in net worth would drive household consumption and aggregate demand. Therefore, the wealth effect through asset prices represents an important MPTM channel, especially in countries where household investment in financial assets, such as stocks and bonds, or property is high.

5.3.3. Exchange Rate Channel

According to the money view, monetary policy transmission through the exchange rate channel affects the real economy through the export and import component of aggregate demand as well as the imported prices of goods on inflation.

Exchange rate transmission on international trade. A reduction to the policy rate, consistent with the interest rate parity theory mentioned previously, precipitates exchange rate depreciation because the return on domestic assets declines compared to the return on assets available offshore. Exchange rate depreciation would stimulate exports and, simultaneously, reduce imports. Nonetheless, the impact of net exports is not always desirable, depending on export elasticity to exchange rates relative to the import elasticity, otherwise known as the Balassa-Samuelson effect:

$$x_t = \alpha + \beta\tau_t + \gamma(y_t - y_t^*) + \varepsilon_t \qquad \text{with} \qquad \tau_t = \Delta p_t - (\Delta s_t + \Delta p_t^*) \quad (12)$$

where $\{x, \tau, y, y^*, s, p, p^*)$ = {net exports, terms of trade (ToT), domestic economic growth, global economic growth, exchange rates, domestic prices, international prices}. The exchange rate channel plays an important role in small open economies, with a more favorable impact on economic growth if the export structure is dominated by manufacturing commodities rather than primary goods.

Exchange rate transmission on inflation. In addition to indirect pass-through via net exports and growth, as mentioned above, there is also direct pass-through to inflation. Direct pass-through occurs because exchange rate developments affect corporate price setting and the public's inflation expectations, particularly for imported goods as finished goods, raw materials, and capital goods. This can be explained using purchasing power parity (PPP), namely $sP^*/P = 1$, which can be estimated using the following equation:

$$\Delta p_t = \alpha \Delta S_t + \beta \Delta p_t^* + \varepsilon_t \tag{13}$$

where Δp_t = domestic inflation, Δs_t = exchange rate depreciation, Δp_t^* = global inflation, ε_t = random shock, with $\alpha = \beta = 1$ for PPP. In general, the effect of the exchange rate on inflation is lower in countries with a flexible exchange rate system and sufficient monetary policy credibility to anchor inflation expectations.

5.3.4. Expectations Channel

As uncertainty continues to accumulate in the economy and financial sector, the expectations channel is becoming increasingly important in terms of monetary policy transmission to the real sector. Economic players, such as banks and companies, will base their business decisions on the economic and financial outlook. They form certain expectations from the performance of various economic and financial indicators. How economic players form their expectations is influenced by their anticipation of central bank and government policies, as well as global economic developments. Consequently, monetary policy credibility to anchor inflation expectations plays an important role in terms of the MPTM.

Inflation expectations are often used by economic players as an indicator of the central bank's interest rate policy. The problem is that inflation expectations are unobservable. Therefore, one estimation method applies Fisher theory, namely[8]:

$$r^e_{t+j} = i^e_{t+j} + \pi^e_{t+j} \qquad \text{14a}$$

If there is a forward interest rate on the market for tenor j, then:

$$\pi^e_{t+j} = r^f_{t+j} - i^e_{t+j} \qquad \text{14b}$$

where $\{r^e, r^f, i^e, \pi^e\}$ = {nominal interest rate expectations, nominal forward interest rate, real interest rate expectations, and inflation expectations}. Mishkin (1990) used this equation to measure future inflation expectations as follows:

[8]Another method would be to use the inflation expectations of the business community and consumers contained in various surveys, such as the Business Survey and Consumer Expectations Survey published by BI.

$$[\pi_{t+j} - \pi_t] = \alpha_{t+j} + \beta_{t+j}[r_{t+j} - r_t\ ] + \varepsilon_t \tag{15}$$

where β_{t+j}†0 shows the inflation expectations information contained in the interest rate for period *j*.

Inflation expectations affect various real sector activities. The effect on aggregate demand occurs due to the impact on real interest rates, which determine the magnitude of demand for consumption and investment. Meanwhile, inflation expectations affect aggregate supply through corporate product price setting. In addition, the effect of inflation expectations on aggregate supply and demand will also determine the rate of inflation and real output in an economy. Greater monetary policy credibility will reduce the deviation in inflation expectations from the inflation target set by the central bank. Consequently, the distortions that emerge in terms of real output and monetary policy effectiveness to achieve the inflation target become smaller.

5.4. Monetary Transmission Channels: The Credit View

As explained previously, the money view theory of MPTM assumes an efficient financial system. The interest rate structure on financial markets transmits the short-term monetary policy rate to longer term interest rates, which subsequently influences relative prices, the components of aggregate demand (investment, consumption, and international trade) and inflation. The financial system also supplies financial products, thus making it possible for companies and households to build an optimal investment portfolio and/or access financing sources other than bank loans, such as issuances of stocks and bonds. Similarly, the foreign exchange market also works efficiently. Exchange rate fluctuations, therefore, are not misaligned from the currency's fundamental value, including when faced with volatile non-resident capital flows. The various aspects of financial system efficiency influence the effectiveness of monetary policy transmission through the money view channels.

Departing from the money view, the credit view of MPTM theory is based on imperfect financial markets due to asymmetric information and moral hazard in various financial transactions (Bean, Larsen, & Nikolov, 2002; Cecchetti, 1998). The effect of financial frictions on the MPTM was observed by Bernanke and Gertler (1995), who found three anomalies in the aggregate demand response to changes in the interest rate as follows: (1) composition, namely that changes in the short-term interest rate affected durable goods, such as houses, which should be more responsive to long-term interest rates; (2) propagation, namely that the real economy continues to respond after the short-term interest rate has been raised and even when the interest rate has been lowered; and (3) amplification, namely that the interest rate triggers a faster change in output even though the investment expenditure of an individual firm does not significantly respond to the cost of capital. In the context of the exchange rate channel, international interest rate parity is not really explained by the economic cycle (Meese & Rogoff, 1983). The effect of the exchange rate on inflation is also somewhat subdued, with a portion, therefore, spilling over to undermine corporate profit margins. In addition, there were indications of an asymmetric impact of monetary policy on real output and inflation between tightening and easing, which cannot be explained by the interest rate or exchange rate channels.

Such observations compelled the credit view of MPTM theory to propose three alternative transmission channels, namely the bank lending channel and bank capital channel, balance sheet channel and risk-taking channel. The bank lending and bank capital channels explain how the MPTM works through bank lending behavior in the face of asymmetric information concerning borrower conditions and bank modeling requirements. The balance sheet channel focuses on MPTM analysis in terms of the loan relationship between lender and borrower, which is influenced by the external finance premium and collateral constraints due to moral hazard. Meanwhile, the risk-taking channel analyzes the effect of monetary policy on risk management and financial product innovation in the financial system.

5.4.1. Bank Lending and Bank Capital Channels

The bank lending and bank capital channels play an important banking role in the intermediation function in an economy to overcome the problems associated with asymmetric information concerning the actual feasibility of debtor projects. To that end, the banks are selective regarding applications for new loans (demand for credit) and then monitor the borrower to avoid loan delinquency and default. The ability to select borrowers and the amount of capital required to mitigate credit risk will affect the MPTM through the supply of credit by banks.

5.4.1.1. Bank Lending Channel. The first model of the bank lending channel was developed by Stiglitz and Weiss (1981) based on the assumption that the borrower has access to private information about the feasibility of his/her businesses. Despite producing the same expected return, the borrower's business project contains different probabilities of success. With limited liability, pursuant to the loan agreement from the bank, the borrower could choose to ignore loan delinquency if the business fails. Therefore, if the central bank tightened monetary policy by applying a higher policy rate, only the debtors with a low chance of success (high risk) would opt to borrow from a bank, leading to a low rate of return on the loan. Consequently, as explained in Chapter 3, the credit market is always in a state of imbalance and the banks are inclined to ration credit. The phenomenon of credit rationing affects the transmission effectiveness of interest rate policy through the bank lending channel on investment and consumption in the real economy.

Several studies propose solutions concerning the need for a selection instrument to reveal the actual business feasibility of a prospective borrower. In the present context, the selection instrument may be in the form of a credit information system that contains the business conditions of borrowers and their credit history. This view was proposed by Rothschild and Stiglitz (1976), who found that such a selection instrument would overcome credit rationing. Meanwhile, Bester (1985) proposed the need for credit guarantees as a selection instrument, thereby only safe borrowers would apply for new loans rather than risky borrowers. Nevertheless, as sophisticated as the credit selection process may be, or even with credit guarantees in place, credit risk will always emerge after the loan has been disbursed. Business failure is always possible due to a plethora of reasons, including management negligence, moral hazard, or general economic conditions. Banks continue to face the problem of adverse selection when extending loans. In a bank's credit portfolio, there are always borrowers whose businesses succeed and those who contain the risk of default.

Several later studies linked credit supply to the bank funding structure. Bernanke and Blinder (1988), for instance, showed that the funding structure of retail banks would influence bank lending. Such conditions are typically found in countries dominated by the banking sector and with an underdeveloped capital market. The banks will rely on retail funding, while the corporate sector also depends on bank loans. The business scale is usually small and medium enterprises, which do not meet the requirements to issue stocks or bonds on the capital market. Therefore, the banks could face funding shortfalls if the central bank tightened its monetary policy stance because retail funding is not normally sensitive to the interest rate, while a dip in income could reduce bank funding. Consequently, the banking industry would ration more credit, with small and medium borrowers most affected.[9] Bernanke and Blinder (1988) demonstrated that monetary policy will tend to amplify the real economy under such a bank credit and funding structure, implying that credit will be too tight when the economy experiences a downswing and too loose during an economic upswing.

5.4.1.2. Bank Capital Channel. Bank capital also affects credit supply. Van den Heuvel (2002) studied the impact of the CAR and ability of banks to issue shares. Bank capital could be influenced by the ability to generate profit, the market price assessment of net assets on the balance sheet and the mandatory reserves required to mitigate credit risk and default of the securities held. The ability of a bank to issue shares on the capital market also affects bank capital. More bank capital implies greater lending capacity. Capital constraints in terms of credit supply become more binding if the level of capital approaches or falls below the capital requirements set by the central bank or relevant supervisory authority.

Such factors in the bank lending and capital channels influence the supply behavior of banks. In general, the bank supply function is determined not only by the lending rate (r_k) and economic outlook (y), but also by credit risk (ρ_k) liquidity (LDR) and capital (CAR) at each respective bank as follows:

$$K^i = f(y, r_k^i, \rho_k^i, \mathrm{LDR}, \mathrm{CAR}) \qquad \text{with} \qquad r_k^i = r_d + c^i \tag{16}$$

[9]Under such conditions, the LDR is not always constant and, therefore, monetary policy would have an asymmetric impact on bank lending if the policy stance is changed. This is very different to the money supply channel that assumes no funding constraints, thus changes in lending will always mirror changes in money supply.

In the credit supply function (16), banks may set different lending rates for different borrowers depending on the cost of monitoring (c^i) and credit risk ($\rho_k^{\,i}$), despite the same funding rate (r_d). Such conditions show that monetary policy has a distributive effect on loan allocation by borrower group. Similarly, the internal bank conditions, such as LDR and CAR, also affect the credit supply function and, therefore, monetary policy transmission at each respective bank.

Furthermore, the workings of the MPTM through the bank lending and capital channels are more complex and tend to be asymmetrical. During a period of monetary policy easing, lower interest rates may drive credit supply in several ways. *First,* a lower policy rate would increase the net interest margin (NIM) and increase profits, thereby improving the balance sheet over time. *Second,* monetary policy easing also pushes up asset prices and directly boosts capital. *Third,* credit risk would also dissipate due to sounder business conditions among the borrowers during an economic upswing and, therefore, reduce the need for reserves and increased bank capital. In general, through the bank capital channel, monetary policy easing could increase bank capital and lending beyond that possible by lowering the interest rate and, therefore, drive aggregate demand, especially by facilitating bank-dependent borrowers to increase spending on investment and consumption.

Nonetheless, an economic downswing, or financial crisis, would exacerbate conditions. A decline in asset value, increase of NPL and lower profits would undermine bank capital, thus compelling the bank to reduce lending. Issuing stocks to maintain bank capital would not be a feasible option because assets prices would track a downward trend and there would be fewer investors to purchase the stocks as the economy moderates. Less credit availability would exacerbate economic conditions and also, therefore, the business conditions and balance sheets of the debtors, with a higher intensity than the effect of the interest rate, which would subsequently compound bank capital due to escalating credit risk and lower profits. Such financial sector deleveraging would undermine economic conditions further, particularly due to a decrease of corporate spending that depends on bank loans. According to such conditions, the central bank's interest rate policy would also become less effective.

5.4.2. Balance Sheet Channel

Differing from the bank lending and capital channels, which stress adverse selection and credit rationing in the face of (*ex ante*) asymmetric information, the bank balance sheet channel emphasizes the issue of moral hazard because of (*ex post*) asymmetric information in terms of lending transactions between lender and borrower. The borrower, after receiving the loan from the lender, may be unable to fulfill their obligations because of moral hazard. Due to the costly state of verification of the borrower's business feasibility and/or limited liabilities in terms of enforcing the loan agreement, this channel proposes two solutions to overcome moral hazard. *First,* is the external finance premium and the *second* is to demand additional collateral (collateral constraints) in the loan agreement.

5.4.2.1. External Finance Premium. The ideas of Bernanke, Gertler, and Gilchrist (1999) as well as Clastorm and Fuerst (2000) represent two salient references in the external finance premium model. For each business transaction, there is always the possibility of default, where it would be more efficient for the borrower not to honor the loan agreement. Nevertheless, there is also moral hazard because of the limited liability contained in the loan agreement. Due to the costly state of verification of a project, the borrower must pay a cost to ensure actual business feasibly. Therefore, the lender imposes an external finance premium on the borrower to compensate the cost of verification.

The external finance premium creates imperfections, or financial frictions, in the lending transaction between the lender and borrower. For safe businesses, with a feasible project, the external finance premium implies it is better to use internal funding rather than external finance. A highly profitable project would encourage the business to increase investment, thus driving aggregate demand and real output. Meanwhile, for higher-risk businesses and those facing balance sheet constraints to internal finance, investment funding in the form of lending is an alternative option that incurs the cost of the external finance premium. In other words, the external finance premium is a veil that creates financial frictions between capital finance and investment. The issue is that the external finance premium lowers the profitability of the borrower and, therefore, increases the risk of default.

Furthermore, Bernanke, Gertler, and Gilchrist (1999) showed that the eternal finance premium also triggers the financial accelerator, which amplifies procyclicality between the financial cycle and economic cycle. Such analysis is significantly different from the real business cycle because of the productivity effect without including financial aspects (Kydland & Prescott, 1982; Long & Plosser, 1983; Lucas, 1980). The financial accelerator phenomenon fundamentally changes MPTM analysis in the financial system and real economy. In the case of monetary policy easing, for instance, a reduction to the interest rate would spur increased investment at feasible and profitable businesses, thus strengthening the economic cycle. Increased corporate investment by less feasible or unfeasible businesses, however, would exacerbate the risk of default due to the external finance premium or business failure. Consequently, not only does the financial accelerator affect the financial and economic cycles, the risk that an economic upswing could reverse into a recession, or boom become bust, also increases, which could potentially spur a crisis.

The reverse is true concerning the effect of tighter monetary policy on investment and real output. A hike to the interest rate would undermine cash flow and, therefore, the profitability of the borrower's business. In addition, a higher interest rate would also prompt the lender to raise the external finance premium imposed on the borrower, which would further compound the risk of business default and loan delinquency. A similar analysis was also applied to lending transactions in the form of loans between firms and the banks. Amalgamating MPTM analysis through the corporate balance sheet channel with the bank lending and bank capital channels would better explain the financial accelerator that occurs due to the interest rate policy of the central bank (Bernanke & Gertler, 1995). Analysis of how the financial accelerator exacerbates the impact of monetary policy on the financial and economic cycles is an integral part of understanding the boom-bust cycle, which often culminates in a crisis, including the GFC.

5.4.2.2. Collateral Constraints. Financial accelerators can be analyzed using the collateral for loan transactions between lender and borrower. The seminal views of Kiyotaki and Moore (1997) represent an important reference in the analysis based on difficulties enforcing the loan agreement due to limited liability, amounting to the nominal value of the loan. Therefore, to prevent moral hazard by the borrower, the lender typically requests collateral in the form of physical assets in addition to collateral to ensure business feasibility. The size of the loan offered by the lender is, therefore, based on an assessment of assets to be used as collateral by the borrower and will hence fluctuate in line with the market value of the collateral.

Subsequent MPTM analysis, using collateral constraints, was similar to the financial accelerator analysis conducted by Bernanke, Gertler, and Gilchrist (1999) mentioned above. A reduction to the central bank's policy rate would improve economic performance and raise asset prices. A higher market value of the assets used as collateral would raise the credit ceiling from the borrower for the lender to offset the cost of investment. The financial accelerator would then occur between the higher asset prices, size of loan, demand for investment, economic growth, and so on. The reverse is true if the monetary policy rate is raised. Less economic activity would lower the market price of assets as well as the value of collateral and investment, and further undermine economic performance. Consequently, the financial and economic cycles could change from boom to bust and, therefore, exacerbate the risk of a crisis occurring.

A simple analysis of the financial accelerator is presented in the following section, including the external finance premium and collateral constraints. Supposing a business requires an input, x, for the upcoming production cycle with a production function, $y_1 = z_1 F(x_1)$, where y = output and z = productivity. To pay for the production input, the business could use internal finance from the sales of production in the previous period, $p_0 y_0$, or borrowed capital with instalments $(1+r)L$. Therefore, the financing constraints to procure the production input can be expressed as follows:

$$x_1 \leq [p_0 y_0 - (1+r_0)L_0] + L_1 \tag{17}$$

To receive the loan, the business must provide additional collateral, for instance in the form of buildings (W), with a market value, q_1 and α as the (inverse) of the loan-to-value (LTV) ratio. Therefore, the loan received by the business, with the collateral constraints can be written as follows:

$$L_1 \leq \frac{q_1 W}{\alpha(1+r)} \tag{18}$$

Under such conditions, the optimal level of credit for the business is selected to maximize profit to ensure production can continue even if the internal capital is insufficient as follows:

$$\text{Max}_{(L_1)}.\Pi = p_1 z_1 F(x_1) - (1 - r_1) L_1 \tag{19}$$

with constraints (17) and (18). If the market value of the collateral is adequately large that the business no longer faces any collateral constraints, profit maximization is, therefore, as follows:

$$\text{Max}_{(L_1)}.\Pi = p_1 z_1 F(p_0 y_0 - (1 + r_0) L_0 + L_1) - (1 - r_1) L_1 \tag{20}$$

Therefore, the first-order condition of optimization requires the same marginal product value of the input as the cost of borrowed capital as follows:

$$p_1 z_1 F_x(x^*) = (1 + r_1) \tag{21}$$

with the optimal loan size as follows:

$$L_1 = x_1^* - [p_0 y_0 - (1 + r_0) L_0] \tag{22}$$

Therefore, if the value of collateral is adequate, the business will be able to overcome a lack of funding from the bank loan if production declines. In other words, a temporary shock, for instance from the effect of monetary policy, would not disrupt production, merely erode some of the income between the business and bank.

Conditions would be different if the market value of the collateral was insufficient for the business to secure a loan of the desired amount, thus the business would face collateral constraints per Equation (18). In this case, the scale of the production inputs purchased by the business would be determined by the maximum loan feasible, namely:

$$x_1 = \overline{x_1} = [p_0 y_0 - (1 + r_0) L_0] + \frac{q_1 W}{\alpha(1 + r_1)} < x_1^* \tag{23}$$

In this regard, production would be suboptimal due to less corporate investment. In other words, monetary policy would lower the value of collateral, leaving the business unable to secure an adequate loan, which would not only affect the size of the loan but also production and real output. Therefore, the impact of monetary policy would exceed the effect of the interest rate.

Both credit conditions and the impact on real output are illustrated in Fig. 5.2 if the collateral constraints are not binding and in Fig. 5.3 if they are.

The previous description shows that collateral is required to protect the lender if the borrower defaults on their loan. Through collateral, the banks can offer a

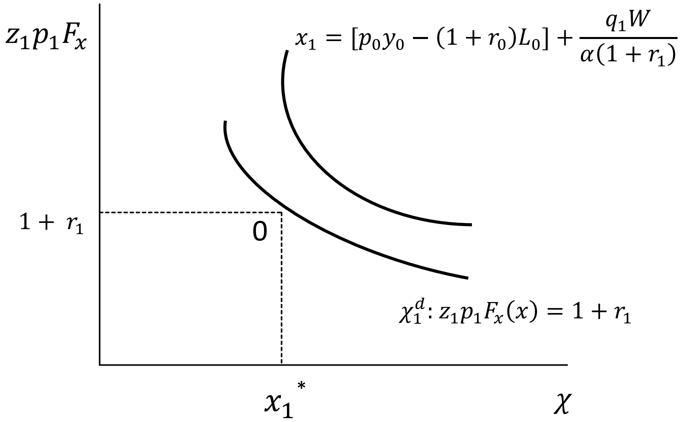

Fig. 5.2: Business Decisions without Collateral Constraints.

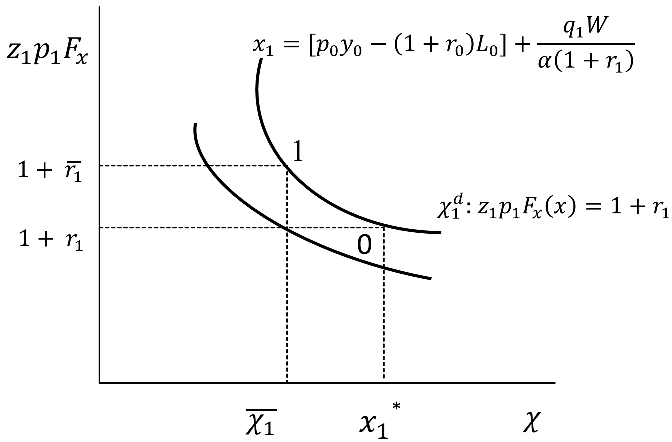

Fig. 5.3: Business Decisions with Collateral Constraints.

lending rate approaching the risk-free rate. This does not mean, however, that the lending rate will be the same as the risk-free rate. While credit rationing is present due to asymmetric information, the marginal value of each additional loan is generally larger than the market value, namely the lending rate. This is congruent with the external finance premium explained by Bernanke and Gertler (1999). In Fig. 5.2, if the collateral constraints increase, conditions are determined by the business that can only procure $\overline{x_1} < x_1^*$ input units. In this case, there is an external finance premium that must be borne by the borrower, which exceeds the cost of using internal funds (or equivalent to the net return on production) equal to $\overline{r_1} < r_1$. The external finance premium will become smaller as corporate balance sheet conditions improve, measured by net capital and liquidity.

Collateral constraints and the external finance premium create financial accelerators in the economy. During an economic recession and financial crisis, for instance, corporate balance sheets tend to decline, while the banks tighten lending standards and raise the external finance premium because of several factors, including less production and lower prices and, thus, less internal business capital, a lower market value of property and other financial assets and, therefore, the value of collateral also declines and the banks offer fewer loans (LTV ratio is elevated) due to the higher-risk business outlook, while the costs of monitoring and resolving bad loans increase. In contrast, during an economic upswing, the uptick of credit expansion tends to outstrip demand due to more profitable businesses, higher market value of collateral and more loans offered by the banking industry due to the promising business outlook and less stringent lending standards.

MPTM analysis through the balance sheet and collateral constraints channels can also be applied to borrowing transactions in the household sector. For example, rising house or property prices would raise the value of collateral, thus facilitating larger loans and more lenient lending standards. In other words, higher property asset prices could lower the external finance premium and unwind the collateral constraints to the amount of credit available to households. Financial accelerators occur in the financial and economic cycles, namely the chain of rising property prices, credit, demand for investment and property development, and increasing economic activity, which subsequently edge up property prices and so on. A housing sector boom, accompanied by a low monetary policy rate, was prevalent during the era of Great Moderation in the United States, which subsequently culminated in the worst crisis since the Great Depression in the 1930s.

5.4.3. Risk-taking Channel

The periods before and after the GFC showed that the risk-taking channel in the financial system contributes to fragilities, contagion and asset price bubbles. Risk amplification by the financial system could occur due to several reasons, including financial product innovation, the capital and accounting valuation methods, ease of funding, risk-tolerance as well as global financial integration. Financial shocks leading up to and after the GFC proved that the financial system is not necessarily a shock absorber but can also perform as a shock amplifier (Allen & Carleti, 2008).

From many perspectives, risk amplification in the financial system is inextricably linked to monetary policy. Monetary stability and readily available liquidity during an economic boom (upswing) period facilitates greater financial product innovation, while exacerbating risk-taking behavior and financial fragilities. In contrast, during an economic downswing, financial amplifiers due to risk-taking behavior compound a financial crisis and the contagion, thereby complicating the appropriate monetary policy response required. Specifically, the risk-taking channel is defined as

> the impact of changes in policy rates on either risk perceptions or risk-tolerance and, hence, on the degree of risk in the portfolios, on the pricing of assets and on the price and non-price terms of the extension of funding.

5.4.3.1. Risk-taking Dynamics. Financial theory in terms of investment teaches return optimization in line with the levels of risk of each respective alternative investment. This is achieved through risk diversification by forming an optimal investment portfolio, congruent with the return and risk profiles. Financial product innovation was created to be return enhancing in terms of investment portfolio optimization, such as through housing loan securitization with mortgage-backed securities and asset-backed securities. Financial derivatives, including forward, swap, option and credit default swaps (CDS), were also created to mitigate and hedge the risks. Various measurement techniques and techniques to set prices per the risks were also developed, such as value at risk (VaR) and the probability of default (PoD). Similarly, risk-management techniques were also strengthened, which led to the innovation of financial products as if they were risk free.

From the investors' perspective, financial product innovation and forming the investment portfolio could boost yields and enhance risk-management, which would subsequently drive further innovation of financial derivatives that become more developed and complex. In addition, financial product innovation has also changed business models in the financial system, including the banking business model, by facilitating the proliferation of credit securitization strategies and wholesale funding, thereby facilitating the further creation of liquidity and lending, which has become known as the originate to distribute strategy in the United States and United Kingdom. Increasingly complex financial product innovation, coupled with a lack of investor understanding, flourished during the decades of Great Moderation prior to the GFC in the United States, supported by low inflation and monetary policy rates.

Omitted from the previous analysis is the fact that financial product innovation and monetary stability have increasingly encouraged risk-taking behavior in the financial system in the pursuit of higher yields during an economic boom. In fact, such observations were also made by Minsky (1982), namely that the capitalist economic system, based on the trade of capital as investment funding in the economy, tends to create financial instability, with uncontrolled debt accumulation during a boom period. Expectations of further profit gains and increasing net worth compared to the burden of loan installments during an economic boom causes more investors to speculate and become involved in Ponzi schemes rather than hedge. This is also in line with the observations of three MPTM analysis anomalies in the economy by Bernanke and Gertler (1995), which were explained through the bank lending and capital channels as well as the balance sheet channel in two subsections above.

How does the risk-taking channel strengthen financial accelerators and the amplification of financial and economic cycle dynamics? Borio and Zhu (2008) alluded to three ways in which the risk-taking channel affects the MPTM in the financial system and the economy. *First,* the effect of the interest rate in terms of valuations, revenues, and cash flow from the investments. The mechanisms of the risk-taking channel are nearly the same as the financial accelerator through the balance sheet channel explained previously, and the two channels also tend to be mutually reinforcing. A reduction to the interest rate, for instance, would edge up the valuation of assets prices and collateral, cash flow and profits, thus strengthening the financial accelerator in bank lending and other financial transactions. Similarly, the perception of bad loans would also decline during an economic upswing, hence lowering the external finance premium and easing tolerance of lending standards.

Second, the link between the interest rate and target yield is generally used as a benchmark to evaluate the performance of financial investments (Rajan, 2005). A higher yield compared to the benchmark implies a larger bonus for the investment manager. The nominal yield target does not often change due to the nature of the contracts with investment managers, namely in pension funds and insurance. Consequently, a reduction to the policy rate would create a large disparity with the target yield and encourage investors to seek alternative investments with a higher yield (search for yield). Investor perception, which tends to overestimate asset prices above the fundamental value during an economic boom, also coined as irrational exuberance by Alan Greenspan in 1996, is yet another reason for financial accelerators.

Third, the effect of monetary policy communication and transparency by the central bank. As an integral part of the monetary policy framework, many central banks are becoming increasingly transparent in their communications on future inflation and macroeconomic projections as well as the interest rate policy and underlying policy considerations. The direction of monetary policy is also communicated, including several key indicators underlying future interest rate policy-making decisions, often known as forward guidance, or statements from central bank officials. Monetary policy transparency can anchor the expectations of economic players and lower the risk premium on the financial markets. Economic stability and a clear monetary policy direction strengthen business assurance and, therefore, nurtures investment activity and various other economic activities, especially by prudent and productive economic players.

5.4.3.2. The Role of Liquidity. Financial amplification through the risk-taking channel is also inextricably linked to liquidity, as an important element of financial transactions. There are two types of liquidity, namely funding (cash) liquidity and market liquidity. Funding liquidity represents the investors' ability to obtain cash for the assets held, through selling or use, to receive external finance, as loan collateral. Meanwhile, market liquidity indicates conditions where the volume and price setting on the market work efficiently, thereby each investor can transact at market prices. From those two perspectives, liquidity conditions reflect an important dimension of the effect of financial conditions on the real economy. Easy access to funding liquidity and market liquidity removes the constraints to investment and other real economic activities. Such analysis is similar to collateral constraints from the perspective of Kiyotaki and Moore (2001) above.

Liquidity and risk-taking behavior are closely interconnected and mutually reinforcing. When the central bank eases monetary policy, for example, the lower interest rate also loosens funding liquidity and market liquidity conditions. Simultaneously, the lower interest rate also improves the risk perception and eases risk-tolerance, thus encouraging economic players to engage in higher-risk investments. The interaction between risk-taking behavior and liquidity during an economic upswing can create financial amplifiers.

The opposite is true when the economy is moderating or in recession, when risk perception and risk-tolerance tighten. Lower asset prices trigger sales in order to realize a profit or, at least, prevent the risk of further losses. Consequently, more investors are looking to sell assets rather than buy, which further tighten funding liquidity and market liquidity. Asset prices subsequently tumble, thus encouraging more investors to sell. Fire sales begin to appear as the majority of investors prefer to hold cash rather than assets, a phenomenon known as cash is king. The declines are amplified, which creates financial system instability and could spur a crisis. A clear example of this is the case of bad housing loans that led to the subprime mortgage debacle in the United States and culminated in the GFC.

The explanation above demonstrates the importance of risk-taking behavior in the financial system in MPTM analysis of the economy. The presence of and interaction with other MPTM channels creates financial accelerators and amplifiers, driven by the increasing rapidity and complexity of financial product innovation, which is not easily understood by risk-taking behavior and the implications on the financial system and real economy. The risk-taking channel strengthens the bank lending and bank capital channels as well as the balance sheet channel in terms of financial accelerators and amplifiers. Similarly, interaction between the risk-taking channel and funding liquidity and market liquidity could create a liquidity multiplier. The boom-bust financial and economic cycles resulting from interactions between the various MPTM channels often cause fragilities and crisis risks in the financial system and economy. Theoretical and empirical understanding of this is crucial in terms of monetary policy analysis and formulation at the central bank.

5.5. Monetary Policy Transmission in Various Countries

Numerous MPTM studies have been performed by central banks and academia in a range of advanced countries and EME. This corroborates the importance of understanding MPTM as well as how the transmission channels interact and are affected by the specific economic and financial conditions of a country. In terms of monetary policy formulation at the central bank, empirical MPTM studies are vital to deepen our understanding of how the monetary policy response, including the interest rate policy, exchange rate policy, or other instruments, is transmitted to the financial system and its influence on the final targets of inflation and economic growth. For academia, many empirical MPTM studies have been conducted to find empirical evidence for a particular theory, or even propose a new theoretical idea from the empirical evidence observed in terms of financial system and economic behaviors.

The approach used in empirical studies depends on the objectives, methodology and availability of data. In terms of the objectives, empirical MPTM studies are typically conducted in the form of mapping and investigating the relative effectiveness of each respective MPTM channel, with a focus on MPTM channels and specific cases, such as the workings of the bank lending and capital channels or the corporate balance sheet channel, or a comparison between countries, for example, monetary policy transmission in EME. In general, the estimation method applies a vector autoregressive (VAR) model to map the transmission channels or single equation model using panel microdata for specific channel case studies (Amato & Gerlach, 2001). The latest empirical MPTM studies have also applied dynamic stochastic general equilibrium (DSGE) models to investigate monetary policy transmission on financial accelerators in the financial and economic cycles (Boivin et al., 2010). Data availability is also very important, not only time series data but also disaggregated data at the micro level for banks, corporations, and households. The simultaneous effect between monetary policy and financial and real economic variables demands attention in the estimation model selected.

In general, several empirical MPTM studies have produced interesting findings. *First,* the traditional transmission channels, namely the interest rate, asset price and exchange rate channels, are still the core monetary policy transmission channels. The interest rate and asset price channels play a greater role in advanced countries, while monetary policy in EME is generally transmitted through the interest rate and exchange rate channels. *Second,* in a more developed financial system, monetary policy transmission through the bank lending and capital channels, corporate and household balance sheet channels, as well as the risk-taking channel tend to amplify the effect of monetary policy on the financial and economic cycles. Furthermore, the effect tends to be asymmetric, with a stronger effect felt when the financial system is under pressure. *Third,* global financial integration facilitates the stronger transmission of monetary policy from advanced countries to EME through global spillovers from the interest rate, exchange rate and asset prices. Liquidity conditions and risk-taking behavior on global financial markets also amplify the effect of monetary policy spillovers.

5.5.1. Monetary Policy Transmission in Advanced Countries

In advanced countries, most empirical studies have been conducted for the European Union (EU), with a focus on changes in monetary policy transmission related to the monetary union launched in 1999, increasing financial innovation and the impact of the GFC (Angeloni, Anil, Benoit, & Daniele, 2002; ECB, 2010). Meanwhile, studies of financial transmission in the United States have primarily focused on the economic boom and low interest rates during the 20 years of Great Moderation and the impact thereafter of the GFC. Various studies have tried to capture and reveal financial system behavior in terms of amplifying and accelerating the financial and economic cycles, which often culminates in a crisis.

5.5.1.1. Monetary Policy Transmission in the EU. Unification of the euro currency had an immediate impact on monetary policy transmission, with the loss of the exchange rate channel between countries in the EU. The economic and monetary union led to a uniform exchange rate channel between EU member states (Boivin, Giannoni, & Mojon, 2008). Furthermore, eliminating currency risk in the euro area reduced the cost of trade transactions and increased capital market integration. Nevertheless, although the benefits of the economic and monetary union depended on country-specific conditions, several studies demonstrated a 5–10% increase in trade after unification of the euro currency (Baldwin, DiNino, Fontagné, De Santis, & Taglioni, 2008). Bank holdings and cross-border transactions also increased significantly after currency unification (Kalemli-Ozcan, Papaioannou, & Peydró, 2009).

The centralization of monetary policy at the European Central Bank (ECB) brought stronger credibility to the new regime and a clear mandate on price stability. The immediate positive effects included controlled inflation and anchored expectations to the targets set. This was evidenced by long-term bond yields and surveys concerning inflation expectations (Beechey, Johannsen, & Levin, 2007; Ehrmann, Fratzscher, Gurkaynak, & Swanson, 2007). In addition, evidence was put forward that the Phillips curve was becoming flatter, indicating a weaker correlation between inflation and the output gap (Calza, 2008). This was also linked, however, to more credible monetary policy under the ECB in terms of anchoring expectations. The effects on wages and prices were also more controlled. In general, central bank credibility benefits monetary policy transmission in terms of maintained price stability, which drives real economic activities (Erceg & Levin, 2003; Paries & Moyen, 2009).

Several empirical studies have produced differing results concerning the effect of monetary unification under the ECB on output and inflation. Meanwhile, the VAR model developed by Gerke, Weber, and Worms (2009) did not find significant differences between the periods before and after currency unification. Cecioni and Neri (2010), using a DSGE model, showed that monetary policy was more effective in terms of economic stabilization by reducing nominal rigidity and anchoring inflation. Angeloni, Kashyap, Mojon, and Terlizzese (2003) as well as Angeloni and Ehrmann (2003) showed that monetary policy transmission through the interest rate channel was still relevant, with a significant impact on output after around one year. Nonetheless, monetary policy transmission through the financial system began to play a more important role through the bank lending channel and innovation on financial markets. The importance of financial intermediation in terms of monetary policy transmission was also stressed by Beck, Colciago, and Pfajfar (2014).

Monetary policy transmission in the EU has also been influenced by financial innovation since currency unification. Asset securitization increased, including bank loans, and banks could easily obtain non-term deposit funds by issuing bonds. The ease of funding drove robust credit expansion in the EU, which was also facilitated by working around the capital constraints through derivative instruments such as CDS to contain the interest rate and credit risks. Consequently, financial innovation tended to strengthen non-loan financing in the economy, while bank lending was influenced more by the bank balance sheet channel rather than the interest rate channel (Altunbas, Gambacorta, & Marquéz-Ibañez, 2009; Loutskina & Straham, 2006). This phenomenon was also bolstered by monetary policy credibility, with low inflation and interest rates, coupled with increasing financial integration in the euro area.

On the other hand, however, greater dependence on wholesale funding from the capital market with financial innovation rather than traditional retail funding also exposed the banks to financial market dynamics. This occurred, specifically, when the markets were experiencing financial distress. Bank difficulties securing wholesale funding exacerbated liquidity condition, and then lending, and even spurred a crisis in the financial system (Hempell & Sorensen, 2009). Conditions were then compounded by capital constraints, which could no longer be overcome under financial distress because of difficulties issuing bonds and/or hedging against the risks. Several studies have shown banks with limited capital undertaking less lending when the economy moderates (Altunbas, de Bondt, & Marquéz-Ibañez, 2004; Maddaloni & Peydro, 2010).

Financial innovation also expanded the role of the risk-taking channel in the banking sector in response to monetary policy, but the empirical evidence for this transmission channel is difficult to prove in the EU and the United States (Altunbas, Gambacorta, & Marquéz-Ibañez, 2010). As mentioned previously, low short-term interest rates would encourage risk-taking behavior in the banking industry in terms of lending quantity (volume and value of approved loan applications) as well as prices (low lending rates). Jimenez, Ongena, Peydro, and Saurina (2012), for example, using credit registry data from Spain showed that banks with lower capital and liquidity ratios engaged in riskier lending during normal periods and were, therefore, the hardest hit during a crisis episode. The risk-taking channel has been proven to be stronger during normal periods because of low short-term interest rates and the proliferation of financial innovation (Maddaloni & Peydro, 2010). The rapid increase of asset securitization and ease at which risk was transferred through derivative instruments during the decade before the GFC, exacerbated risk-taking behavior in the banking sector with looser credit standards and oversight.

5.5.1.2. Monetary Policy Transmission in the United States. Bernanke and Blinder (1992) were among the first to apply a VAR model to map monetary policy transmission channels in the United States, revealing three important findings. *First,* the federal funds rate (FFR) is a solid measure of the effect of monetary policy. *Second,* the effect of the FFR on real economic variables (inflation and GDP) is more dominant than money supply, T-Bill rates, and bond yields. *Third,* monetary policy affects bank asset composition and, thus, lending to the real economy. Follow-up empirical studies showed that the effectiveness of interest rate transmission fades with financial innovation and monetary policy credibility to form expectations in the real economy (Kuttner & Mosser, 2002). The bank lending and balance sheet channels become more important, however, in terms of monetary policy transmission (Boivin et al., 2010; Endut, Morley, & Tien, 2013).

Boivin et al. (2010) also stressed that the core channels of monetary policy transmission through short-term interest rates to asset prices and the exchange rate remain an important element of structural macroeconomic models through to the latest DSGE models. The impact of the interest rate on investment is generally modeled through the user cost of capital in the macroeconomic structural model or through Tobin's q in a DSGE model. Nonetheless, the empirical findings indicate that investment elasticity to the user cost of capital is relatively small, at around minus 2–10%, thus several economists (such as Bernanke & Gertler, 1995) questioned the importance of this transmission channel. The effect of monetary policy on consumption in a macroeconomic model is generally modeled through the wealth effect with a comparatively small elasticity at around 3–4% for property and stocks. Meanwhile, Case and Shiller (2003) showed that rising asset prices, such as property, had a significant impact on investment and consumption. The importance of the housing market, in terms of transmitting monetary policy to the real sector, including during the US housing crisis in 2007/08, was also shown by Mishkin (2007).

The latest empirical studies have tended to focus on the role of the banking industry in terms of the MPTM by separating the influence of supply and demand for loans. On the supply side, the bank funding structure responds differently to monetary policy. For example, using US bank balance sheet microdata, Kashyap and Stein (2000) showed that the credit portfolios of small and illiquid banks were most affected by changes in monetary policy. Similarly, banks with a small ratio of capital to assets and banks that have not yet gone public were also more affected by monetary policy (Kishan & Opiele, 2000). On the supply side, several studies have shown diverse impacts of changes in monetary policy on the corporate sector. Gertler and Gilchrist (1994) found that investment at small corporations was more susceptible to monetary policy due to greater dependence on bank loans.

Using US corporate microdata for the period from 2003 to 2008, Ippolito, Ozdagliy, and Perez (2015) showed monetary policy transmission to corporate balance sheets through bank lending using floating interest rates. Companies more dependent on bank loans and not engaged in hedging activity, and were the most sensitive to the response of stock prices, selling, liquidity, inventory, and investment to changes in monetary policy. Meanwhile, Beck et al. (2012) demonstrated that monetary policy also influences aggregate demand through household balance sheets, especially those reliant on bank loans. These studies serve to confirm the findings of Ciccarelli, Maddaloni, and Peydro (2010, 2013) for the case of the EU, who found a stronger influence in the balance sheet channel during normal periods, while the bank lending channel was more influential during periods of financial system distress.

Like conditions in the EU, financial innovation also plays an important role in the MPTM through the risk-taking channel in the United States (Smets, 2013). Angeloni et al. (2011) showed that expansive monetary policy increases leverage in the banking sector and, therefore, credit exposure. Securitization tends to increase incentives for risk-taking at the banks, facilitated by low interest rates and the proliferation of derivative instruments. Norden, Buston, and Wagner (2012) showed that banks transfer risk with derivative instruments to borrower institutions. Meanwhile, Buch, Eickmeier, and Prieto (2013), using data from the Fed's credit survey for the period from 1997 to 2008 showed that small banks increase credit exposure after monetary policy easing, while big banks extend riskier new loans without adjusting the composition of their credit portfolios. These empirical findings are consistent with the study by Dell'Ariccia, Laeven, and Suarez (2016), who also showed that a lower policy rate increased risk-taking behavior in terms of lending. Nevertheless, differing from the findings of Jimenez, Ongena, Peydro, and Saurina (2012) for the case of Spain, this study showed that securitization and derivatives allowed banks with a large capital base to transfer the risk to other financial players, while banks with a small capital base tended to face credit exposure.

Several studies also investigated the effectiveness of monetary policy during a crisis period. Theoretically, monetary policy can be effective during a financial crisis if able to overcome financial market failures, for instance, by easing the credit constraints or restoring confidence. If not, deleveraging and uncertainty would undermine monetary policy effectiveness in terms of driving the economic recovery. Using data from the 1980s, Jannsen, Potjagailo, and Wolters (2014) proved that expansive monetary policy could drive output during a recession due to a financial crisis. In contrast, the effect of monetary policy on output fades during the crisis recovery period. Dalhaus (2014) also showed that monetary policy is more effective and persistent during a period of financial distress in terms of the macroeconomic impact on output, consumption, and investment compared to normal periods. The impact is transmitted through the bank lending channel, as confirmed by other studies (Gambacorta & Marques-Ibanez, 2011).

5.5.2. Monetary Policy Transmission in EMEs

Studies have also been conducted concerning the MPTM in EME. Mohanty and Turner (2008), for example, reviewed MPTM in EME based on previous research by Kamin et al. (1998), considering several fundamental changes that had occurred, namely greater monetary policy independence and credibility with a strong emphasis on inflation control, more developed financial markets and an economic structure that had changed toward freer international trade and investment. Thereafter, Mohanty and Rishah (2016) analyzed the impact of the GFC on MPTM in EME.

5.5.2.1. Interest Rate Channel, Asset Price Channel, Exchange Rate Channel, and Expectations Channel. A study by Moreno (2008) showed that, in many EME, the effect of the policy rate on deposit and lending rates is generally stronger and more persistent than that of bond yields. The effect of the monetary policy rate on long-term bond yields tends to be temporary. In EMEs with strong demand for external financing, such as in Latin America, monetary policy could influence the inflation risk premium on debt denominated in the domestic currency and even the country's risk premium.

The asset price channel is also becoming more significant in the Asian region, particularly in terms of property prices. In several countries, including China, Hong Kong and Singapore, property prices have generally mirrored bank lending since 2005. In fact, in Hong Kong, the interest rate channel through property prices has a stronger impact on inflation than household consumption and net worth. In Southeast Asia, asset price sensitivity to the interest rate has also changed. In South Korea, house prices have become more sensitive to changes in monetary policy and bank lending since the Asian crisis. In Singapore, the effect of the interest rate on the property market cycle plays a significant role in terms of the consumption cycle, while stock prices do not. Oppositely, in Thailand, the interest rate has more of an impact on stock prices than property prices.

The influence of the interest rate on the exchange rate has also evolved. A low and stable risk premium due to sound macroeconomic dynamics would elicit a more predictable exchange rate response to the monetary policy rate. In general, an unexpected 25 bps hike to the policy rate would trigger immediate exchange rate appreciation of around 0.5–1.0%. Similarly, the impact of the exchange rate on inflation has also been observed to fade in many EME (Mohanty & Klau, 2008), which is linked to more credible monetary policy in terms of anchoring inflation expectations. Nevertheless, more open trade has amplified the effect of the ToT on the real exchange rate in the medium-long term due to a change in the composition of trade from primary goods to manufacturing goods. Consequently, the central bank is required to monitor structural changes such as these when considering the real exchange rate underlying monetary policy.

The expectations channel has also become more effective. In many EME, including the Czech Republic, México, and South Africa, private Southeast inflation expectations tend to hover around the inflation target of the central bank. The financial markets also react more strongly to policy rate announcements, as is the case in Indonesia, Thailand, and the Philippines. In fact, in Thailand, the housing market and bond market respond faster to changes in monetary policy proceeded by changes in housing loan rates than by changes in the central bank's policy rate. With financial markets that are becoming better equipped to overcome changes in monetary policy and with more anchored inflation expectations, the need to adjust the policy rate has dwindled. In other words, with stronger monetary policy credibility, the financial markets often do the central bank's work.

The effect of monetary policy on output in many EME is relatively small, at 1–8%, and temporary, around one to two years (Mohanty & Turner, 2008). This provides compelling evidence for the neutrality of money in the long term. In contrast, the effect of monetary policy on inflation has increased but to a different extent in each respective country. In Indonesia, México, and South Africa, the inflation response to monetary policy has increased from around 2–5% in the period from 1998 to 2003 to around 10–30% from 2004 to 2008. On the other hand, the corresponding response in India, South Korea, Thailand, Chile, the Czech Republic, and Poland has remained relatively unchanged at around 2–3%. The empirical findings also show lower variability in terms of inflation and output, which is consistent with monetary policy credibility to control inflation and, therefore, reduce the variability in real output. More anchored expectations also play an important role.

5.5.2.2. Credit and Balance Sheet Channels. Dominant banking sectors in many EME mean that the credit channel plays an important role in terms of influencing investment. The degree of influence varies, however, namely that it is stronger in Latin America and Eastern Europe than in Asia, but the experience of each individual country also varies. In Asia, more developed financial markets, as alternative investment financing, do not always weaken the bank lending channel. In the Philippines, for instance, although the commercial paper market has flourished recently and many banks have committed a credit ceiling to corporate customers, the effect of monetary policy through the bank lending channel remains important due to the size of NPL in the banking system. In Thailand, post-Asian crisis financial diversification has dampened the response of inflation and output to bank lending. This is also the case in Singapore, where small and medium enterprises are increasingly issuing stocks and bonds for financing. Meanwhile, in China and India, the bank lending channel remains dominant. In China, monetary policy is directed toward controlling the quantity of money supply and there remain policy controls on credit, therefore, monetary policy influences demand primarily through changes in credit supply. In India, studies have shown that small banks tend to reduce credit supply more than big banks during a period of monetary policy tightening.

The household balance sheet channel is now playing a more important role in line with the proliferation of consumer loans for the nascent middle class in EME. The surge in leverage has exacerbated the balance sheet constraints in terms of household consumption decisionmaking and, therefore, the temporal substitution effect of monetary policy has gained impetus. Furthermore, changes on the household balance sheet could amplify the wealth effect of monetary policy, mainly through home-ownership. The wealth effect is reinforced when using the property as loan collateral. Another implication regarding the influence of the balance sheet is linked to the effect of cash flow from monetary policy on consumption and investment through the nominal interest rate, scale of financial assets and liabilities or household loan agreements. In general, household loans are short term and, therefore, more sensitive to the monetary policy rate, moreover when the loan applies a variable interest rate. In addition, the interest rate on housing loans in many countries is generally tied to the base lending rate or monetary policy rate.

With the dominance of bank loans in corporate financing, the corporate balance sheet channel also plays a salient role. Furthermore, corporate leverage is more significant in EME than in advanced countries. In Asia, for instance, leverage averaged around 38% during the period from 1993 to 2003, exceeding the 24% reported in the United States, Europe, and Japan. Higher leverage in Asia is due to the comparatively low market value of corporations, which compels the corporate sector to finance investment using debt rather than issuing shares. Nonetheless, in some EMEs, including Indonesia, a large portion of firms rely on internal financing, which implies low leverage. The implications of corporate financial conditions on MPTMs are difficult to gauge. Low levels of debt ease corporate cash flow constraints and, therefore, undermine the investment response to monetary policy. A sound corporate balance sheet, however, would also undermine the role of financial accelerators in the economy.

On one hand, financial market development with alternative sources of financing to bank loans tends to weaken the balance sheet channel in the transmission of monetary policy. The propagation of derivative instruments has facilitated hedging of interest rate and currency risks in the corporate sector, thus unwinding balance sheet fragilities to changes in monetary policy. Financial market liberalization also opens corporate access, especially large corporations, to international sources of financing, including debt as well as issuances of stocks and bonds. On the other hand, a better capital market function could reinforce the influence of the monetary policy rate on the prices of various financial assets and, therefore, strengthen the direct cost impact of changes in monetary policy on investment. Similarly, a more liquid market and active trading of securities strengthens the impact of changes in the interest rate on the market valuation of corporate balance sheets, while investment spending could become more responsive to the monetary policy rate. Changes in expectations of the monetary policy rate are also now playing a more important role in terms of corporate financial conditions than previously.

As mentioned above, greater banking system soundness and efficiency in many EMEs have influenced MPTMs. On one hand, such conditions have strengthened the direct influence of monetary policy. In Malaysia, for instance, the long-term elasticity of the money market rate to the lending rate has continued to rise, increasing from 0.3 in 1998 to 0.5 in 2005, in line with increasing competition and banking system efficiency (Ooi, 2008). On the other hand, less bank balance sheet vulnerability has also reduced non-price distortions to credit supply and, therefore, undermined the bank lending channel in the MPTM of many countries. A solid capital base and greater bank access to alternative sources of funding through certificates of deposit and long-term bonds, such as the case in Chile, have had a similar effect in terms of unwinding the funding constraints in the banking sector, particularly during a period of tighter monetary policy.

A change in bank balance sheet conditions could intensify market risk and the effect of monetary policy on risk-taking behavior and market exposure. In several EME, the share of variable interest rate loans exceeds the composition of term deposits. The impact of the monetary policy rate is stronger, therefore interest rate exposure on the bank balance sheet increases if the average funding rate is not adequately adjusted to the change in the monetary policy rate. In many ways, such market risk can be mitigated through an appropriate hedging strategy. In South Africa, for instance, banks tend to transfer the market risk to their customers by offering variable rate term deposits. In addition to market risk, the banking sector also faces maturity risk because the average maturity of term deposits is shorter than the maturity of loans, such is the case in Latin America and Indonesia, where the average maturity if term deposits is less than six months,

Another important source of exposure from changes in the monetary policy rate could appear on the banks' investment portfolio. During the decade prior to the global crisis, commercial bank investments in government and central bank securities increased in terms of share to total assets. In a number of countries, that trend was linked to central bank intervention in order to prevent exchange rate appreciation by selling government securities or central bank issued securities to absorb the excess liquidity. An increase of government and central bank securities in the banks' asset portfolios would increase exposure to market price valuations on the balance sheet, thus exacerbating the financial accelerators. Banks could aggressively expand lending during a period of monetary policy easing because of elevated asset prices and the profits from trading bonds. In contrast, during a period of tighter monetary policy, the banks tend to ration credit in order to compensate for the decline in the prices of their financial assets. Another implication is the impact on the central bank in terms of setting the monetary policy rate because of paying due consideration to the impact on overall financial system stability. That observation was more visible during the GFC, when global uncertainty and spillovers besieged the financial systems of many EMEs.

5.5.2.3. Monetary Policy Transmission in the Post-GFC Era. The GFC precipitated an important MPTM evolution in EMEs due to three salient factors, namely the relative role of banks on the bond market, money market globalization, and low long-term interest rates (Mohanty & Rishab, 2016). From 2004 to 2013, the ratio of total economic financing in the form of domestic and offshore bank loans and bond issuances to GDP in EMEs increased rapidly and then accelerated again in the wake of the global crisis due to low interest rates and abundant excess liquidity on global financial markets originating from the expansive monetary policies adopted in the United States and Europe. Growth was even more rapid in EME with perfect capital mobility and/or stable exchange rates compared to imperfect capital mobility and/or a flexible exchange rate regime. Hong Kong is a good example, although China has also experienced similar dynamics due to the attractiveness of its economy. Similarly, South Korea, Malaysia, and Singapore all noted an uptick in total financing after the GFC. Although bank credit has remained dominant, its share of total economic financing has decline, especially in Asia. China, for instance, experienced a 21% drop in the share of credit from 2004 to 2013, a trend that has been mirrored throughout Asia.

How do global interest rates and excess liquidity affect the interest rate, exchange rate and bank lending transmission channels in EME? Several studies have shown that low long-term interest rates on global financial markets are affecting long-term interest rates in EME rather than short-term interest rates (King & Low, 2014). Furthermore, Miyajima, Mohanty, and Chan (2015) found that the response of the benchmark 10-year bond yield in 11 EME to a 100 bps hike in the 10-year US T-Bill yield was to increase from just 31–53 bps after the GFC. During periods of global financial market turbulence, however, such as the Fed's Taper Tantrum in May and June 2013, the effect increased to more than 100 bps. Using more recent data, Sobrun and Turner (2015a) reported similar findings, namely that the weak correlation between bond yields in EME and United States yields from 2000 to 2004 strengthened and became significant after 2005. In another study, Kharroubi and Zampolli (2015) demonstrated that the long-term bond yield correlation was around 60 bps, while the response of the short-term interest rate was just 20 bps. Sobrun and Turner (2015b) also reported similar findings, noting that, like Miyajima, Mohanty, and Yetman (2014), the effect of the term premium on bond yields was larger than the effect of future expectations of short-term interest rates.

How do long-term global interest rates influence bank lending rates in EME? Differing from the impact on bond yields, bank lending rates are not particularly responsive to long-term interest rates. The average response of interest rates on household loans and corporate loans was just 16 and 34 bps, respectively, due to the funding structure of the banking industry, which is typically short term, with a median of just four months in the 11 EME studied at the end of 2013 (Ehlers & Villar, 2015). In addition, the share of variable interest rate term deposits was shown to be relatively high, at around 50–70%, implying that term deposit rates are effectively indexed to the policy rate of the central bank. In terms of credit, variable interest rate housing loans account for around 70−99% of total housing loans in Asia, contrasting conditions in Latin America, where fixed-rate loans account for nearly 100% in Brazil and around 70–96% in Argentina, Chile, Colombia, and México.

Changes have also occurred in the effect of global spillovers on the exchange rate channel. Prior to the global crisis, monetary policy independence and credibility, as the ITF gained popularity, reduced the impact of the exchange rate on inflation. Previous crisis experience also helped to control currency risk and maturity risk in the banking and corporate sectors in terms of their external debt. Nevertheless, the global crisis reaffirmed the need for central banks in EME to stabilize exchange rates. In addition to the effect of global spillovers, increasing exchange rate volatility was also linked to the significant spike in external debt at non-financial institutions in EME (Chui, Kuruc, & Turner, 2016). Exchange rate fluctuations had a notable impact on economic and financial conditions in many Latin American countries, including Brazil, as well as in several Asian countries, including India and Indonesia, and in Eastern European countries, such as Turkey and Hungary, but with a smaller intensity. Another factor affecting exchange rate volatility was commodity prices, particularly in net exporters (BIS, 2014). In addition, the correlation between exchange rates in EMEs and global risk perception indicators, such as the volatility index (VIX) market volatility index, increased significantly after the GFC (Rajan, 2014). In fact, Miranda-Agrippino and Rey (2013) as well as Rey (2013) found that changes in the perception of global investors had created global convergence in terms of exchange rate and asset price fluctuations.

One effect of exchange rate fluctuations, while external debt is accumulating due to excess global liquidity, is that the credit cycle in EMEs tends to mirror the dollar exchange rate. Bruno and Shin (2014) called this phenomenon "the risk-taking channel of the exchange rate" to illustrate the important effect of global financial markets on the transmission of external shocks through the distribution of liquidity on the US money market to the banking systems in EME. USD depreciation improved the balance sheets of banks indebted in USD, therefore, the credit risk faced effectively eased, which prompted further credit growth. In contrast, a period of strong USD appreciation was accompanied by a contraction of the bank balance sheet and widespread USD shortages. Evidence also pointed to the effect of the exchange rate on funding and financing conditions, although without financial imbalances. The mechanism could operate through interactions between the foreign exchange market and bond market, possible caused by a change in the portfolio strategies of speculative investors and professional asset managers (Feroli, Kashyap, Schoenholtz, & Shin, 2014). Turner (2012) showed different returns on yields in EME between those hedged and those not. The lack of UIP implied that non-resident investors affect the risk premium and cause large fluctuations in domestic monetary conditions. When the exchange rate appreciates, investors engage in carry trade on the bond market for profit taking due to future exchange rate appreciation, which reduces the risk premium and bond yields in EMEs to very low levels. In contrast, under conditions of market distress, exchange rate depreciation and uncertainty would increase, hence non-resident investors would withdraw, pushing up yields and tightening financing conditions further.

The bank lending channel is also influenced by global spillovers. One effect is through banking globalization, which has eroded the link between monetary policy and credit (Cettorelli & Goldberg, 2012). The ability of global banks to efficiently lend across borders through their networks has undermined the impact of the domestic policy rate on credit, implying that credit conditions in EME are more vulnerable, especially to operational target changes in US monetary policy. Cettorelli and Goldberg (2012) reported that a 100 bps hike in the FFR would reduce commercial bank lending in the United States to EME by around 2.2%. In addition, a larger USD debt burden in EMEs has created a tighter correlation between domestic credit conditions and the availability of dollar liquidity. USD scarcity after the Lehman Brothers collapsed in 2008 spurred intense deleveraging pressures on EME (McGuire & von Peter, 2009). Consequently, as corroborated by Borio et al. (2011), the sharp spike in USD obligations in EME caused the EME credit cycle to synchronize more with the global credit cycle. During an economic boom, global credit tends to increase more rapidly than credit overall, with banks seeking large-scale funding. In contrast, a higher US policy rate and lower USD debt burden undermined lending, which spread to EME.

Increased risk-taking behavior made the bank lending channel more important to financial system stability in many EME. Unfortunately, differing from studies in advanced countries, research into the risk-taking channel in EMEs remains limited due to a lack of individual borrower and lender data. Nonetheless, studies using aggregate data have provided some preliminary indications. Using a cross-border panel model, for example, Kohlscheen and Rungcharoenkitkul (2015) found that external factors, such as the USD exchange rate and US capital market volatility (VIX Index), were significant drivers of credit growth in EME after the GFC compared to pre-crisis conditions. Consistent with the risk-taking and exchange rate channels, this study showed that 10% exchange rate appreciation in EME against the USD would boost credit growth in EMEs by 85 bps in the near term and by 135 bps in the long term.

5.5.3. Monetary Policy Transmission in Indonesia

The first general study on MPTMs conducted by BI was in 2001 and documented by Warjiyo and Agung (2002). The study focused on empirically mapping the interest rate channel, bank lending channel, corporate balance sheet channel, exchange rate channel, asset price channel, and expectations channels. The study covered the period from 1990 to 2001 and was divided into the periods before and after the Asian Financial Crisis of 1997/98. The research methodology used a VAR model, structural equations and surveys. In general, the results provided invaluable empirical findings on MPTM in Indonesia and showed how the mechanism evolved due to the Asian Financial Crisis of 1997/98. Several empirical studies were subsequently conducted by BI and academia regarding specific transmission channels, such as the interest rate, exchange rate, lending, and balance sheet channels. In fact, several studies also investigated monetary policy transmission through the risk-taking channel and the transmission of BI's policy mix.

5.5.3.1. Interest Rate, Exchange Rate, Asset Price, and Expectations Channels. A study by Kusmiarso et al. (2002) documented the growing role of the interest rate transmission channel in Indonesia. Specifically, using a money market structural model, the study showed that the interest rate of BI Certificates (SBI) and liquidity conditions in the banking sector influence interbank rates (PUAB), term deposit rates, and lending rates. Furthermore, changes to the interest rates would also affect investment and consumption through the cost of capital as well as the substitution effect and income. Investment growth was shown to be very responsive to lending rates, while consumption growth was responsive to term deposit rates. In another study, using data from January 1988 to June 1997 and January 1999 to December 2010, Tai, Sek, and Har (2012) demonstrated an asymmetric response, namely that the term deposit rate was more responsive than the lending rate to BI's policy rate after the Asian Financial Crisis in 1997/98. A study by Natsir (2008) also concluded that transmission through the interest rate channel was effective in terms of achieving the final target of monetary policy. Using a VAR model and data for the period 1990:2 to 2007:1, Natsir showed that the SBI rate, through the interbank rate, could explain around 63.1% of the variation in inflation with a lag of 10 quarters.

Concerning the exchange rate channel, Siswanto, Kurniati, Gunawan, and Binhadi (2002) showed that, with a more flexible exchange rate system after the Asian Crisis, exchange rate transmission increased. The direct impact of the exchange rate on inflation (through changes in the import prices of goods) was stronger and instant from the first month compared to the indirect impact (through aggregate demand) that had a lag of two months. This empirical finding was linked to the magnitude of the risk premium due to high country risk and suboptimal market mechanisms due to problems in the banking industry at that time. The findings were also confirmed by surveys of banks, businesses and households, which showed that non-economic factors, limited supply compared to demand for foreign exchange and global exchange rate developments were the three main drivers of rupiah exchange rate fluctuations. Most banks regarded foreign exchange market intervention as BI's most powerful monetary policy instrument in terms of influencing the exchange rate. Only a few banks cited the SBI rate as having an influence over the exchange rate.

With monetary policy credibility at BI since implementation of ITF in 2005, coupled with increased macroeconomic stability, subsequent studies have shown that the effect of the exchange rate on inflation is fading. Kuncoro (2015), for example, using monthly data from 2003 to 2013 proved that with ITF implementation, the effect of 10% rupiah depreciation on the inflation of import prices and producer prices declined respectively from 6% and 3% to 3% and 1.5%. That estimate is slightly higher than BI's own estimates of the effect on inflation, namely around 0.7–1.2% for each 10% of rupiah depreciation. The diminishing effect of the exchange rate on inflation after ITF implementation was also evidenced by Siregar and Goo (2009).

A study on the asset price channel was conducted by Idris, Yanuarti, Iskandar, and Darsono (2001), which focused on share prices due to the lack of data on property and land prices. In general, the study concluded that stock prices were comparatively weak in terms of transmitting monetary policy to the real sector. Although monetary policy could influence share prices and the size of the financial asset portfolio, the pass-through effect to inflation was weak. In other words, changes in share prices have not displayed evidence of the wealth effect in the economy. This may be due to the relatively small share of stocks (only around 5%) in the investment portfolio compared to alternative sources of funding, especially deposits held at a bank as well as property and land assets. Meanwhile, Yuniarsa and Chang (2015) used monthly data from 2000 to 2013 to show that the correlation between the interest rate, exchange rate volatility, and stock prices had increased since the GFC.

Wuryandani, Ikram, and Handayani (2001) focused their study on the inflation expectations channel using inflation expectations data from the BI Business Survey. The study concluded that inflation expectations are affected more by exchange rate fluctuations, inflation in the previous period (inertia) and the interest rate.[10] The empirical evidence was further substantiated by a survey of banks, businesses, and households. Inflation in the previous period was again shown to influence the formation of inflation expectations, evidencing the importance of enhancing monetary policy effectiveness in terms of controlling inflation. On the other hand, the effect of the exchange rate on the formation of inflation expectations tends to be asymmetric. In the corporate sector, there is downward price rigidity when firms set prices, implying that businesses are generally reluctant to lower prices when the rupiah experiences appreciation. In contrast, exchange rate depreciation beyond a given threshold would force companies to raise prices.

Nevertheless, a subsequent empirical study on the inflation expectations channel by Tjahjono, Harmanta, and Nugroho (2012) produced some interesting findings after implementation of the ITF in Indonesia. *First,* the expectations indicator contained in the Consensus Forecast (CF) was superior to other measures of inflation expectations, including the survey results. In addition to the availability of month data, the CF indicator provided inflation expectations for the upcoming one to four quarters. *Second,* the study did not reveal heterogeneous inflation expectations between economic players and professional projections for the upcoming one month and, therefore, the inflation expectations could be considered homogeneous. *Third,* causality analysis showed that consumer inflation expectations were influenced by retailers' inflation expectations, while professional inflation expectations were more affected by wholesalers' inflation expectations.

[10]Testing was also conducted using other indicators to measure inflation expectations, for example, the Consumer Expectations Survey performed by BI as well as Fisher Theory. The results confirmed that the inflation expectations indicator contained in the Business Survey behaved consistently.

Concerning the transmission of the GFC, Raz, Indra, Artikasih, & Citra (2012) performed an empirical study comparing the impact on economic growth of the Asian Financial Crisis of 1997/98 and the GFC. In general, the study showed that East Asian countries were more resilient to the GFC than the Asian financial crisis of 1997/98, primarily due to stronger economic fundamentals, a more prudent banking sector and more credible monetary policy. Banking sector reforms instituted in the wake of the Asian financial crisis of 1997/98 have strengthened bank capital and oversight by the regulator. Corporate transparency has also played an important role in terms of strengthening economic fundamentals, while larger reserve assets have contained global crisis spillovers. Meanwhile, Nurhidayah, Kaluge, and Pratomo (2014) compared the roles of the BI Rate and exchange rate on inflation and economic growth in Indonesia prior to and after the GFC. Using monthly data from 2005:10 to 2012:12 and a Vector Error Correction Model (VECM), the study showed that the contribution of the BI rate, measured using variance decomposition, had declined from 22.3% to 13.6% before and after the global crisis. Conversely, the contribution of the exchange rate to inflation had increased from 0.5% to 1.95% after the GFC.

5.5.3.2. Bank Lending and Balance Sheet Channels. Previous research provided complete empirical evidence concerning the bank lending channel in Indonesia (Agung, Morena, Pramono, and Prastowo, 2002a, 2002b). Three methods were used in the study. *First,* a VAR model was used to analyze the impact of changes in the monetary policy rate on bank behavior concerning term deposits, loans and securities. *Second,* the demand and supply of loans was analyzed to determine whether credit market imbalances were more attributable to credit supply as indicated by the bank lending channel. *Third,* the panel data of individual banks was investigated to discover whether monetary policy had differing impacts on the different bank characteristics, reviewed from the perspective of capital strength and asset size.

The evidence from aggregate data analysis confirmed that monetary policy influences bank lending with a lag because banks tend to avoid reducing the volume of term deposits rather than selling securities to meet their credit obligations. Further empirical evidence showed that after monetary policy was adjusted, flight to quality was observed from national banks to foreign banks and joint venture banks. The phenomenon explains why loans from foreign and joint venture banks are not sensitive to changes in monetary policy compared to national private banks. Furthermore, disaggregated bank lending data to the individual bank and corporate level showed that corporate loans were relatively less sensitive to changes in monetary policy, while individual loans experienced a sharp drop off. Such developments can be explained by the phenomenon of flight to quality in terms of borrower selection by the banks if tighter monetary policy is perused, namely that only viable firms would be eligible to receive a loan, while less feasible businesses and individuals would face credit rationing.

In terms of capital-based bank groups, the study showed that the impact of monetary policy was more pronounced for banks with a relatively small capital base. Furthermore, time series data and panel data analysis in the study demonstrated a more asymmetric impact of monetary policy during a recession compared to boom conditions. Less effective monetary policy on bank lending during the period prior to the Asian Financial Crisis of 1997/98 was caused by the banks' ability to access offshore funds. After the Asian Financial Crisis of 1997/98, however, characterized by less bank capital and heightened credit risk, a higher policy rate due to a tighter monetary policy stance, increased the PoD and, therefore, the banks were more reluctant to extend loans.

Meanwhile, an empirical study on the corporate balance sheet channel focused on two salient questions. *First,* which balance sheet conditions play an important role in influencing corporate investment decisions? *Second,* how does monetary policy affect corporate balance sheets and their investment decisions? In the case of Indonesia, the previous study showed that the importance of balance sheet dynamics, especially. cash flow and leverage, in terms of influencing corporate investment decisions, while the impact was more pronounced for small business rather than large corporations (Agung et al., 2002a, 2002b). Investment sensitivity to corporate balance sheet dynamics actually increased during periods of contractionary monetary policy compared to expansionary monetary policy. The empirical evidence pointed to the importance of the corporate balance sheet channel in the transmission of monetary policy in Indonesia. Several other studies have also investigated the bank lending and balance sheet channels. Deriantino (2013), for example, researched the impact of bank competition on monetary policy transmission effectiveness using panel data for 95 banks from 2001 to 2012. The empirical results showed that more competitive banks tended to be less responsive in terms of adjusting their lending rates after a change in the monetary policy rate. Although the elasticity of term deposit rates and lending rates to the BI rate were 0.52% and 0.30%, respectively, the level of bank competition did not significantly change monetary policy transmission. Similar conditions were found concerning bank capital.

Meanwhile, Yarasevika, Tongato, and Muthia (2015) showed that the bank lending channel was influenced more by past economic growth and credit behavior rather than lending rates and reserve requirements. In a study using monthly microdata for foreign banks from 2002 to 2007, Fazzaalloh and Sasongko (2014) showed that foreign banks with comparatively smaller total assets or capital reacted more strongly in response to changes in monetary policy when lending compared to the big banks. Concerning the transmission of the Asian Financial Crisis of 1997/98 to credit, Silalahi, Wibowo, and Nurliana (2012) studied the impact of foreign bank loans in Indonesia by combining macro push and pull factors of foreign capital flows. In general, the study showed that the global crisis had an impact on foreign bank lending in Indonesia, and that the impact was more significant on foreign bank branches compared to joint venture banks. In addition, foreign bank loans were also influenced by push and pull factors of foreign capital flows, such as economic growth, risk factors as well as domestic and global liquidity dynamics. From a micro perspective, foreign bank loans were also affected by balance sheet conditions and the banks' asset portfolios.

5.5.3.3. Transmission of Risk-taking Behavior and the Central Bank Policy Mix. Research performed by Satria and Juhro (2011) was the first empirical study concerning the impact of risk-taking behavior on lending by banks in Indonesia. The econometric model used the following error correction model (ECM) specification:

$$\Delta K_t^i = \alpha(K_{t-1}^i - \beta_1 \mathrm{GDP}_{t-1} - \beta_2) + \gamma_1 \Delta \mathrm{GDP}_{t-1} + \gamma_2 \Delta K_{t-1}^i + \gamma_3 A_t + \gamma_4 \mathrm{DD}_t \\ + \gamma_5 \mathrm{STANCE}_t + \gamma_6 (\mathrm{A}_t \times \mathrm{STANCE}_t) + \gamma_7 (\mathrm{DD}_t \times \mathrm{STANCE}_t) + \varepsilon_t \tag{24}$$

where K_t = real credit disbursed by the banking sector with the market equilibrium interest rate (investment loans, working capital loans, and consumer loans), GDP_t = Real GDP, A_t = bank portfolio risk perception index, DD_t = distance to default, and STANCE_t = monetary policy stance (tight or loose). The bank risk perception index is a measure of the risks contained in the asset portfolio as follows:

$$A = \frac{E(r_p) - r_f}{y \times \sigma_p^2} \tag{25}$$

where $(E(r_p) - r_f)$ = the difference between the yield of the portfolio at risk and the yield of the risk-free portfolio, y^* = bank total assets at risk (excluding SBI and SUN), and $\sigma_p{}^2$ = asset yield variance. Meanwhile, the risk of bank default was measured using the distance to default of the assets' market values compared to the minimum capital requirements as proposed by Chan-Lau and Sy (2007) as follows:

$$\mathrm{DD}_t = \frac{\ln(\mathrm{VA}_t / X_t) + (\mu - \sigma_A^2)T}{\sigma_A \sqrt{T}} \tag{26}$$

where VA_t = market value of at-risk assets, X_t = total liabilities, $\lambda = (1 - \mathrm{CAR})$, where CAR is the minimum capital requirements pursuant to banking regulations, μ = average asset value growth, σ_A = standard deviation of asset value, and T = maturity date of corporate loans.

The study produced several interesting findings. The risk-taking channel plays a significant role in terms of monetary policy transmission through the bank lending channel in Indonesia, where credit and risk-taking behavior tend to be procyclical, while monetary policy is countercyclical. Furthermore, the risk perception index of the bank portfolio and risk of default tend to weaken the effectiveness of the monetary policy stance. If a loose monetary policy stance is adopted, the banks are more prudent when lending and, oppositely, the banks are more inclined to take risks when a tighter monetary policy stance is pursued. These findings contradict Taylor (2009) for advanced countries, where bank risk-taking behavior tends to exacerbate lending procyclicality in response to monetary policy. One explanation for the case in Indonesia is the banks perhaps detect signals that the economy would be compounded by monetary policy easing, an explanation that requires further empirical evidence.

Another interesting empirical study, by Wimanda, Maryaningsih, Nurliana, and Satyanugroho (2014), evaluated transmission of the BI policy mix. As discussed in Chapters 13–15, after the GFC, central banks, including BI, implemented a mix of monetary policy foreign capital flow management and macroprudential policy. The study reviewed how the BI policy mix is transmitted to the economy, particularly the final targets of price inflation) stability and financial system stability. In this research, monetary policy and macroprudential policy were analyzed. Empirically, monetary policy was shown to influence inflation and financial system stability with respective lags of 18 and 10 months. All monetary policy transmission channels identified with the interest rate channel were observed to be more dominant in terms of the impact on inflation, while the asset price channel was more dominant in terms of financial system stability. In general, macroprudential policy had a moderate and temporary impact on the intermediate targets but was not shown to effectively influence the final targets of inflation and financial system stability. Monetary policy was also shown to be transmitted through the corporate balance sheet channel, becoming more sensitive after the application of tighter monetary policy, which evidenced greater utilization of internal funds for investment in line with less bank loan availability. Such developments were primarily found at small businesses that typically face financial constraints.

5.6. Concluding Remarks

The theoretical and empirical studies elaborated in this chapter revealed several determinants of MPTM dynamics and evolution from one period to another. The first factor is central bank monetary policy credibility and consistency. Experience since the 1990s has shown that monetary policy at many central banks has become more credible, especially since implementation of price stability as the final target through the ITF. Monetary policy was shown to lower inflation in many advanced countries and EMEs and support robust economic growth (Beck et al., 2015). Monetary policy consistency strengthens the effectiveness of the interest rate channel, which is subsequently transmitted to the other channels. In fact, monetary policy credibility, supported by central bank transparency and communication, could successfully anchor inflation expectations and, thus, reduce the interest rate policy response required.

The second factor is the condition of the financial system in terms of banking and financial product innovation. The supply of bank loans is not only determined by interest rates and the economic outlook but also by liquidity conditions, the funding structure and strength of the capital base. Therefore, the central bank's interest rate policy will have different impacts on the economy, in line with the internal conditions of banks and the corporate sector, and an asymmetric impact in terms of looser or tighter monetary policy. Meanwhile, financial product innovation, such as credit securitization and derivative instruments, may unwind internal bank constraints to the supply of credit, especially during an economic upswing. Nevertheless, the simultaneous accumulation of risk in the financial system would also accelerate and could culminate in crisis if the economy moderated. In brief, MPTM analysis should focus on understanding the dynamics of the financial and economic cycles that tend to amplify and accelerate between economic boom and bust.

The third factor is increasing global financial integration. Foreign capital flows volatility and risk-taking behavior in the economy, and global finance impacts significantly on monetary policy transmission effectiveness in EME with an open economic system and perfect capital mobility, including Indonesia. How spillovers from the United States crisis led to the massive and protracted GFC is the latest tangible example. The impacts of global spillovers are transmitted through the trade channel and financial channel. In terms of the trade channel, economic moderation and lower world trade volume (WTV) undermine external sector performance in the form of declining exports and current account performance, thus exacerbating pressures on macroeconomic stability and economic growth in EMEs. In terms of the financial channel, global spillovers increase exchange rate and foreign capital flow volatility, which complicate monetary policy formulation in EME. Both conduits of global spillovers influence monetary policy transmission effectiveness through nearly all channels, not only the money view channels, such as the interest rate, exchange rate, asset price, and expectations channels, but also the credit view channels, namely the bank lending and balance sheet channels. In fact, global spillovers also affect monetary policy transmission through the risk-taking channel.

Those three factors have several salient implications in terms of monetary policy formulation. Monetary policy consistency and credibility are crucial to maintain macroeconomic stability and support economic growth, which not only requires a solid monetary policy framework, as explained in Chapters 6–9 on ITF-based monetary policy, but also institutions that emphasize independence, consistency, and transparency, as described later in Chapters 10–12. Furthermore, the effectiveness of monetary policy formulation and implementation requires comprehensive understanding of the financial system, including the implications of financial frictions, the proliferation of financial product innovation, and spread of risk-taking behavior. The stronger impact of increasing global financial integration on domestic economic and financial dynamics must be well understood and anticipated when formulating monetary policy. Such conditions cannot be overcome merely by strengthening economic fundamentals. The central bank's policy mix response, in the form of interest rate policy, exchange rate policy, foreign capital flow management, and macroprudential policy, is required to achieve price stability and support maintained financial system stability, as explained in Chapters 13–15.